Solidon3D

Manual

Design, generate and edit for 3D printing

Version 0.3.5

Contents

Reference — every operation with its values

What Solidon is

What the program is for, and what it is not, in four paragraphs.

Solidon builds, changes and print-readies models for the 3D printer.

Three things set it apart from what else is open:

- **Every step stays a number.** A hole two millimetres off gets moved, not drilled again.
- **Fits come from the material, not from a feeling.** Measure your clearance once and you improve parts you built before.
- **Parts instead of handwork.** Nut trap, heat-set insert, latch, living hinge, magnet pocket — one click each instead of half an hour each.

Sketches are part of it: basic shapes and free drawing with constraints — extruded, pocketed, revolved or swept along an arc, and every measure may be a project parameter.

What it is not: no slicer — the print file still comes from the slicer, Solidon only searches and judges. No cloud — no account, no telemetry. Of its own accord Solidon only reports in at start-up, to ask about a newer version — in doing so it states nothing but its own version number, and this can be switched off in the settings.

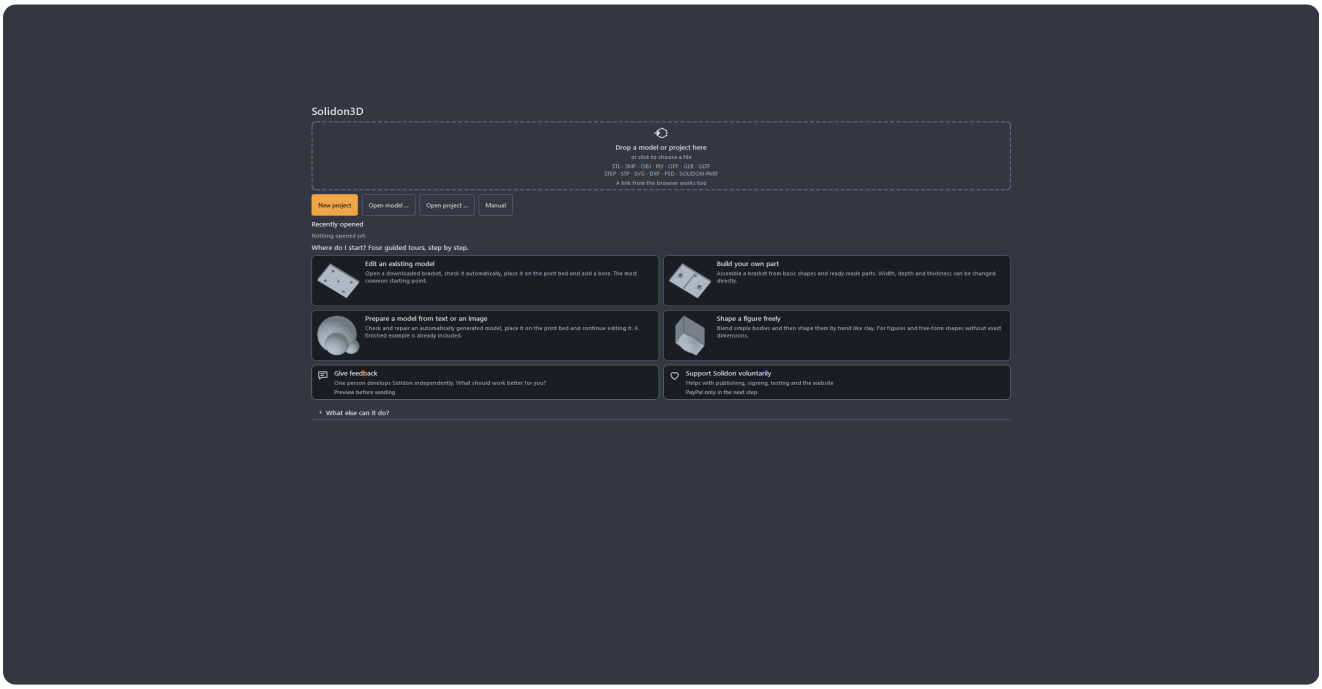
Without a network, without an account and without a language model, everything except the chat stays usable.

The first fifteen minutes

One full run, from the downloaded file to the print file.

Anyone who has never worked with a construction program starts here. A downloaded model gets a hole and then gets printed — the most common case of all, in eight steps.

1. Answer two questions at first start. Language and printer. In the filament inventory, you can explicitly import the filaments loaded in your slicer, including type and colour, as your own spools. If your own printer is not in the list, a similar one will do — the dimensions can be changed later. *Set up later* works too, and then the defaults apply.



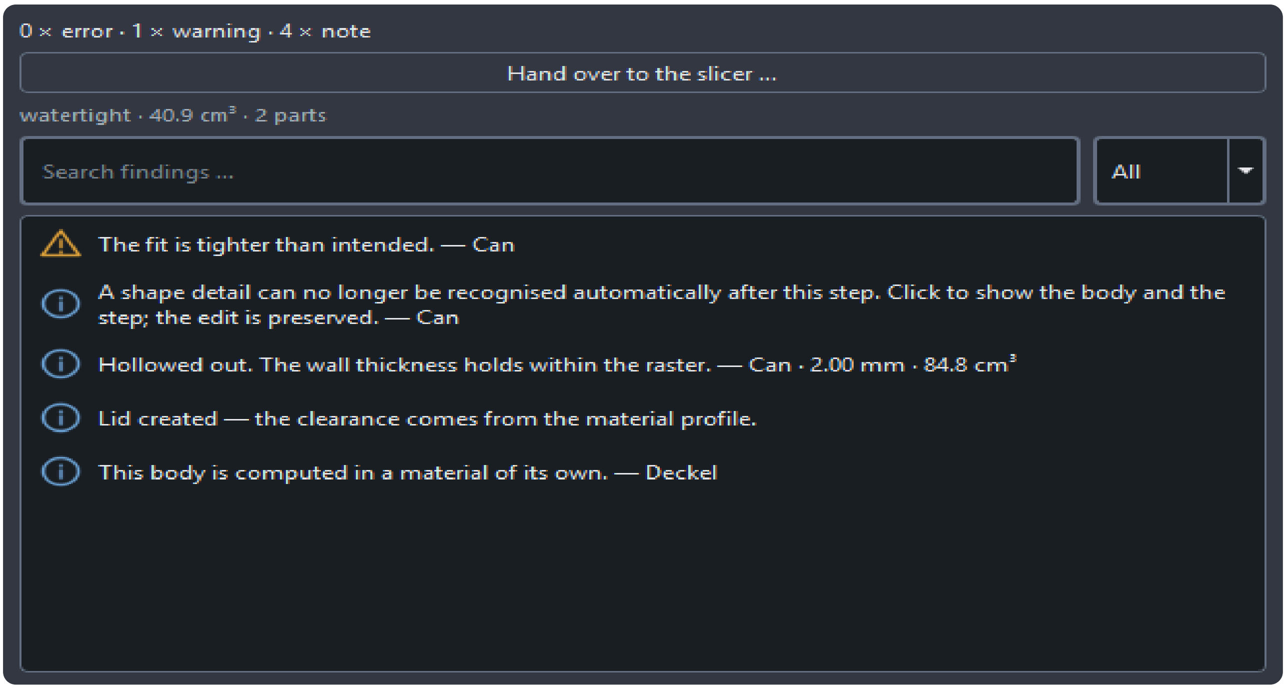
No empty start screen — there is always something to begin with.

2. Drag a file onto the window. STL, 3MF, OBJ, PLY, OFF, GLB or GLTF and STEP or STP; plus SVG and DXF for drawings meant to be extruded. If the file records no unit — and STL never does — Solidon asks instead of guessing. When in doubt, millimetres: almost everything on the internet is in millimetres.

Large files are read in the background: from about eight megabytes a loading indicator saying “Reading model” stands in the picture and the window stays usable. Smaller ones have long since loaded.

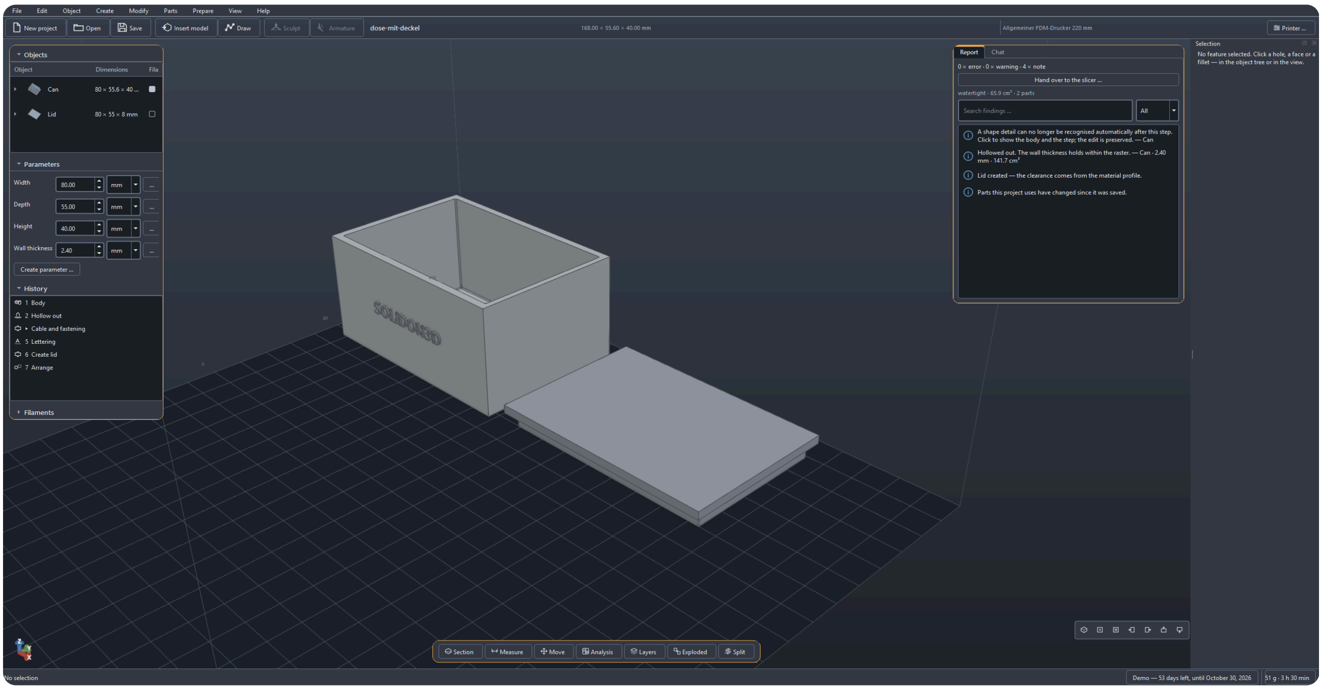
If the file is not on your disk yet, its address will do as well: drag the download link out of the browser and onto the window, or use *File → Model from the web* — the field is prefilled with whatever is on the clipboard. Catch the address of the model page instead of the file, and you will be told so.

3. Look at the report. On the right it says what is wrong with the model. Holes, duplicate faces, triangles facing the wrong way — with downloaded models that is the rule, not the exception. The most common findings carry a button that fixes them; where none is there, the report at least says what the finding means.



A finding never stops at the observation.

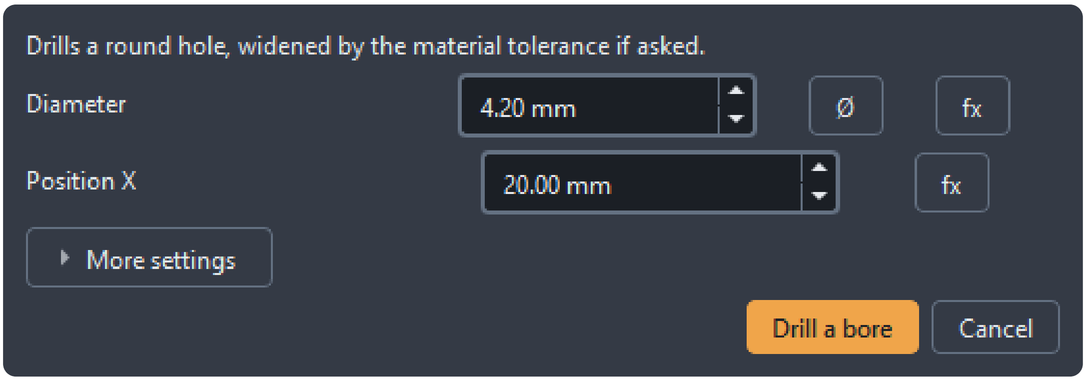
4. Repair. Usually the action suggested in the report is enough. The model stays what it was — the repair is a step in the history and can be taken back.



5. Point at the face the hole should go into and right-click. The menu names exactly those operations that suit this face — on a flat face, for instance, *drill hole*. A right-click always hits the most specific thing under the pointer, with no groundwork.

A left-click goes one step slower, and that is on purpose: the first one selects the **whole part**, the second the spot within it — a face, a hole. That is also how you get at the part itself, to move or rotate it. `Esc` steps back out again.

6. Enter the diameter and apply. Position and direction are already filled in, because the face was selected. Everything else — tolerance, resolution — sits behind *More settings* and is left alone as a rule.



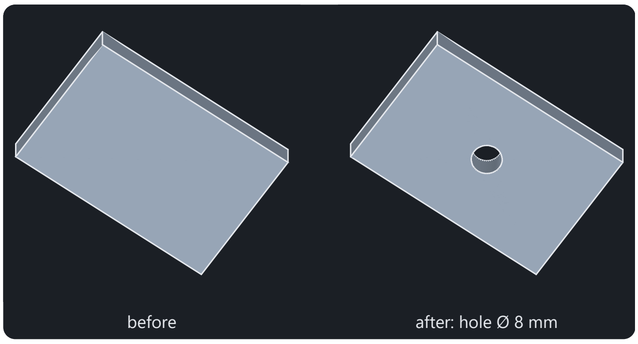
Staged depth: a good default is worth more than a good setting.

The result: the same body, one hole more.

7. If the hole sits wrong, it is moved, not drilled again. A double-click on the step in the history reopens it. Change the number, apply, done. That still works next week.

8. Print. *File → Print settings*, then *Slice*: the slicer computes the print file without ever showing itself, and *Save print file* puts it where you want it. If you would rather carry on in the slicer, use *Open in slicer* — or *File → Export* for a plain 3MF.

The first run needs no more than this. Everything else in this manual builds on it.

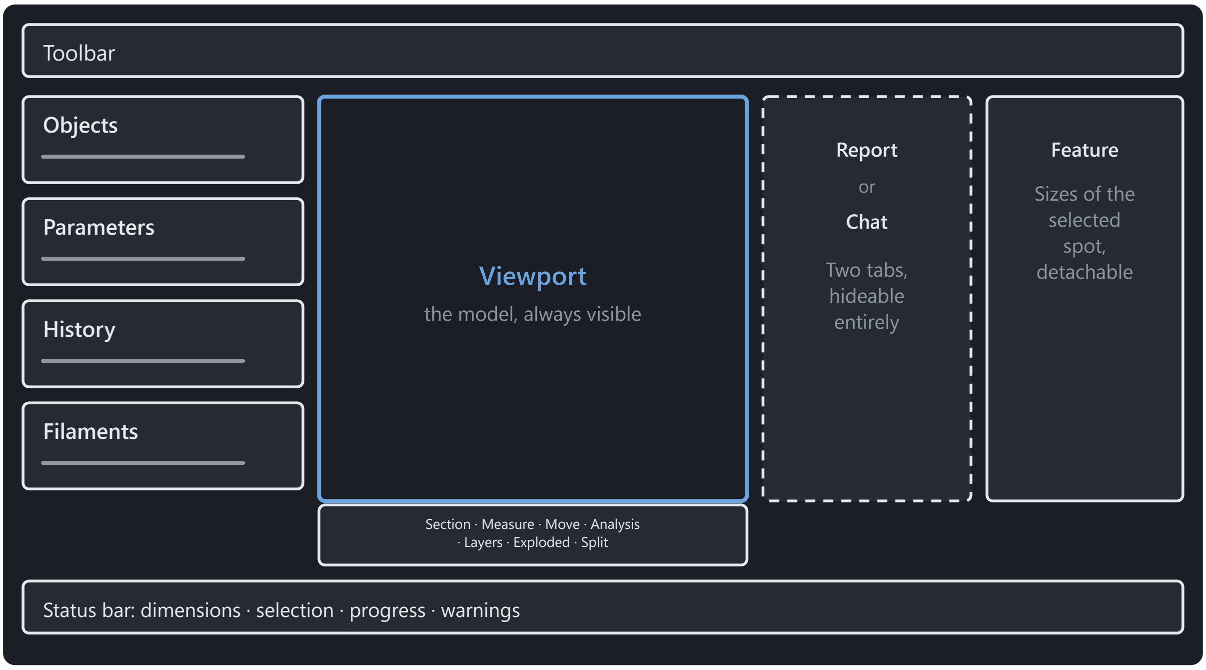


Both pictures are computed by the same operation that sits in the menu.

The window

Which part of the window answers which question.

Each area answers a distinct question.



Four areas, no modes. The right-hand side can be hidden entirely.

The project header shows the project name, printer, and filaments actually in use. Clicking the printer opens its selection for this project. Different materials and colours are assigned to individual bodies or faces.

At the top the toolbar: new, open, save, *Insert model*, *Draw* — which swings the view square onto the drawing plane, the model stays put and steps back translucent —, plus *Sculpt* and *Armature*, both of which need a selected body and say so before you click. Escape comes back out of all three like out of any other tool. The buttons carry symbol and name side by side, so no symbol has to be guessed.

Below the model the tool strip: *Section*, *Measure*, *Move*, *Analysis*, *Layers*, *Explosion*, *Split* — left to right on Alt+1 to Alt+7. Most of them only look; *Move* and *Split* really change the model, and both do it as a step in the history.

On the left sit Objects, Parameters, History and Filaments one below the other, each section collapsible. The object tree shows what is in the scene — and under every body what the recognition found inside it; the parameter list the project's named dimensions; the history every step that led to where things stand. *Filaments* sits collapsed: which spools are on the shelf is asked once at the start and once before printing.

In the middle the model. Dragging with the left button moves the view, clicking with it selects — move the mouse and you pan, tap and you select. With a body selected, pressing the left button on it moves the body; beside it, the view. Dragging with the right button orbits around the centre of the view, holding the wheel tilts up and down, and scrolling zooms to wherever the pointer is.

And the keyboard flies. W and S go forward and back, A and D sideways, Q and E tilt like the held wheel. Hold the key down — the view flies evenly for as long as it rests; hold A and D at once and you stand still, because the directions cancel each other out. The difference from zooming: zoom travels up to the part, flying goes through it. The keys only arrive once the view has the focus — one click into it is enough; anyone coming from an input field and pressing W straight away types a W. If the view loses the focus, the flight stops.

Used to something else? Switch. The settings hold four more layouts, mimicking Cura, Bambu Studio, Orca and PrusaSlicer, CAD and Blender; there the left button selects again, and the keyboard does not fly. A 3D mouse (SpaceMouse) drives the same camera as soon as it is plugged in: the cap is the part — push it and it moves, twist it and it turns, pull it towards you and it comes closer, and a device button fits everything into view. Speed and direction are in the settings as soon as a device has been seen.

Home puts the camera right. *Fit to view* frames the body you have selected — and with nothing selected the whole scene.

Next to it two tabs: *Report* and *Chat* — never both side by side, because nobody reads both at once. When a warning appears, the view jumps to the report by itself. While sketching a tab for the constraints of the sketch joins them, and during the introduction one for the tour. A key press hides the whole column, and then the model has the screen to itself.

At the far right stands the selection window, over the full height, as soon as you have selected something: at the top the dimensions of what you clicked, below them the actions for it. The ones that suit the selection come first: with two bodies *Unite*, *Subtract* and *Intersection*, with a single one *Drill a bore*, *Hollow out* and *Split*, on a bore you clicked *Change bore*, *Countersink* and *Fill a bore*. The search below finds all the others; the part catalogue stays an area of its own. The window can be pulled off and put elsewhere, and *View → Selection* hides it. What is in it is told by the chapter on features.

At the bottom the status bar: dimensions of the selection, progress of a running computation, open warnings.

There are no modes. No switching between “editing” and “building”, no workbenches — there is one state, and that is the scene.

When you are looking for something, open the command palette. It finds every operation by name and shows the keyboard shortcut right beside it. That way you learn the shortcuts in passing instead of memorising them.

Looking before printing

What the report reports, what the analysis maps show, and when to look.

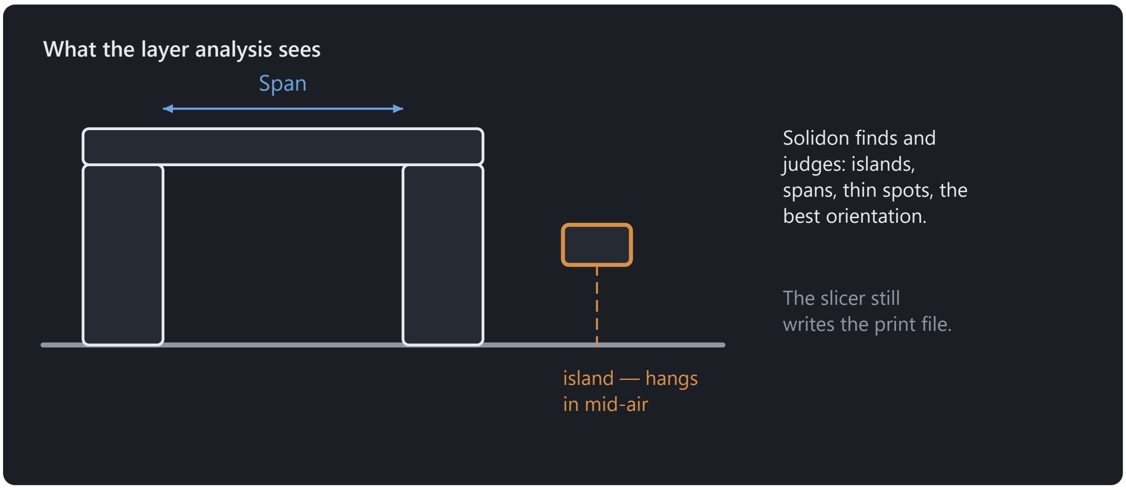
A model almost always looks fine on screen. Whether it can be printed is determined elsewhere — by wall thickness at the thinnest point, an overhang beyond the limit, or an island that starts in mid-air. Five tools in the bar below the view answer these questions, and none of them changes the model.

Measure (toolbar, *Measure*) sets two points and shows the distance. The pointer snaps to corners and edges, so you need only come close rather than hit them exactly. For **wall thickness**, one click on a face is enough: Solidon sends a ray inwards and measures where it emerges again. A measured distance remains as a **dimension** until you remove it — allowing three measurements to be compared side by side instead of remembered.

Section (*Section*) places a plane through the model and shows what lies behind it. The cut face is displayed closed rather than as an open hole, so a cavity is recognisable as a cavity instead of a display error. The plane can be drawn through the model with the slider.

Analysis maps (*Analysis*) colour the model according to a number. There are seven: **Wall thickness**, **Overhang**, **Support needed** and **Curvature** answer questions about printability. **Mesh defects** shows open edges and places where more than two faces meet — almost always the source when a boolean operation fails. **Features** colours what recognition found in the body: which face counts as a hole and which as a pocket. **Fits** shows which faces belong to a fit and which one violates it. The legend always gives both the range of values **and** their source — a map without a scale is merely a pretty picture. Clicking a warning in the report moves the camera to that location.

Layer preview (*Layers*) shows the model as it breaks down into layers. With one body it starts immediately; with several, first select the body you want to inspect. The slider moves through the layers. Beside it are the layer number and total count, Z height, cross-sectional area, island count and overhang area. In addition to colour, the legend distinguishes contour, island and overhang. What you see is Solidon's own layer analysis and **not** what the slicer computes later. Both sets of numbers remain labelled separately so that no estimate is mistaken for a measurement.



Analysed here, sliced as before in the slicer.

Exploded view (*Exploded*) pulls several bodies apart so you can see what fits into what. It changes the display only — the stack and export remain untouched.

The shadow under the model also conveys information. It falls separately for each piece, not for the body as a whole — a part that broke into several pieces during computation therefore casts several shadows before this is visible in the mesh itself.

None of these tools creates an operation. They are a lens, not a tool: you cannot damage anything with them, and there is nothing for Undo to take back here.

What Solidon recognises in a model

What Solidon recognises again in a loaded model, and what one click on it makes possible.

An STL file contains triangles and nothing else. No hole, no radius, no dimension — open a downloaded model and you have a shell of ten thousand faces in front of you, and it knows nothing of the hole somebody once designed.

Solidon reads it back. After loading, a recognition pass goes over the body and fits shapes into its faces: cylinders, cones, spheres, rings, roundings. Whatever fits becomes a **feature** — with a centre, an axis, a size and an identity that stays. From then on the hole is a hole again and no longer a ring of triangles.

Nine kinds, and the name tells you which way it faces:

- **Hole** — a cylinder going in. They are numbered: "Hole 1", "Hole 2".
- **Peg** — a cylinder standing out.
- **Countersink** or **taper** — a cone, going in or standing out.

- **Socket** or **dome** — a spherical face.
- **Groove** or **bead** — a ring running around.
- **Concave fillet** or **fillet** — a rounded edge.
- **Internal thread** or **external thread**.
- **Open edge** — not part of a body but a hole in the mesh. It stands in the same list because it sits in the same place and wants repairing.
- **Face** — named after the way it points: “Top”, “Front”, “Right”. A wall pointing inwards is called “Top inner” — otherwise the inside of a box would be called the same as its lid.

Where two names stand, direction decides: the same rounding is a concave fillet at an inner corner and a fillet at an outer edge. The list is in the object tree under the body, with alike ones gathered under one roof.

Shapes that are too small are not listed. Below half a millimetre across nothing counts as a feature — no printer can make it smaller than that. On a real customer model this dropped 296 of 1130 entries, among them roundings with a radius of seven ten-thousandths of a millimetre. Those are not features, that is the resolution of the triangle mesh, and a list made up of it by a quarter is not a list but noise.

The selection window

Click something — in the picture or in the object tree — and its dimensions are in the *Selection* window. It stands at the far right over the full height — not as a card below the report, but as a window of its own beside it. It can also be pulled off and put elsewhere, and *View → Selection* hides it entirely. It follows the selection and empties when there is none.

One place, one answer. Up to version 0.3.5 the actions for the selection sat in a card of their own above the view and the dimensions here — on a countersink you clicked you found the same three actions twice, once as buttons and once as fields. They now stand together: the dimensions on top, the actions below.

At the top is what was measured: “Hole 1 · Ø 5.19 mm”, and for a hole a line saying which standard size that is — 5.19 mm is the clearance hole for M5. Below it, each action has its own group of fields and a button.

The fields stand on the measured values. That is the whole idea behind it: a changed number is the operation. Whoever wants the hole two millimetres further right changes the one number they mean and leaves the others alone. A default that was not the measured value would change something on the first click that nobody wanted changed.

A changed number shows itself before it applies. The picture shows the preview as soon as the typing pauses; only the button turns it into a step in the history.

There are five actions, and which of them is there depends on the kind:

- *Move feature* — the new centre as X, Y, Z.
- *Change bore* or *Change feature* — the new diameter. For a hole the material tolerance is taken into account.
- *Turn feature* — axis and angle.
- *Duplicate feature* — the same hole a second time, somewhere else.
- *Remove feature* — a button, not a field.

Holes and pegs can do all five. A spherical face can do all but turning — it has no orientation that could be turned, and turned it would look as it did before. A cone can do them too, as long as it stands on its own; sitting as a countersink over a hole it cannot be moved, and why is further down. Face, open edge, ring and fillet can do none.

What does not work stands there as a sentence and not as a greyed-out field. At a ring you read: “An individual ring surface cannot be edited directly. To change its position, move the whole body. To create a new groove or bead, you can use a ring as a tool.” That ends the search. A gap would leave it open whether the action is missing or was forgotten.

And two of those sentences say “not yet” rather than “never”. Changing the radius of a recognised fillet and taking a fillet away again are both sensible actions that do not exist today; until then the only way is to round the edge anew. The other notes explain which edits are possible for the selected feature type.

Where several alike features sit on the same body, every action carries a tick box “Apply to all 4 alike”. Boring out the four holes of a hole pattern at once is one handgrip, and it becomes one: **one** transaction arises, and a single Ctrl+Z takes them all back together. The tick box only appears where there are siblings — “Apply to all 1” would be a question without a difference.

At a face the catalogue stands there instead of five refusals. None of the feature actions applies to a face; twenty-five parts, on the other hand, are placed on exactly one. The button *Insert a part …* opens the same catalogue as the way through the object tree.

With two features marked, their distance stands there: the centre-to-centre span and the three axis distances separately — “Distance 30.00 mm · in X 0.00 · in Y 30.00 · in Z 0.00”. The first number answers whether the spanner fits between them, the other three how far the template has to be shifted. It is information and not an action: no button below it, no step in the history. With three marked the window stays as it was — then there is not one span but three.

What else you can do with a feature

Del hits the feature, not the body. With a hole selected, the key takes the hole away and leaves the part standing. With no feature selected it deletes the object as it always did.

The move handle sits on the selected feature. Click a hole and reach for *Move* and you find the arrows at the hole and not in the middle of the part — for a hole at its opening, because at half depth the handle was stuck in the material. A drag there becomes *Move feature* or *Turn feature*, not a displacement of the whole part. There is no scaling cube there: a hole grows through its diameter, not through a bounding box. The status bar says for every handle what it is about to move.

While dragging you can see where it is going. A cylinder in the diameter and depth of the hole travels along; a pale one stays behind at the starting place for as long as you drag. The body itself only changes on release.

Moving is one step, not two. The feature is closed at the old place and created at the new one — and the identity travels with it. A fit pointing at this hole keeps its reference. Duplicating is the other way round: the copy gets an identity of its own, the original stays untouched, and the fit notices nothing of it.

Two warnings can meet you on the way, and both are useful. A through hole that no longer goes all the way through after being moved says so. And a countersink over a hole cannot be moved on its own — whether the hole goes through or not: there the countersink is not a cavity of its own but passes into the hole, and moved it would drag the hole along. The message names the reason and the way out: choose the feature in the middle, or close the hole and create both anew — *Countersink* is an operation of its own.

Features are also an analysis map. The *Analysis* tool below the view carries a map called *Features*: it colours the body by what the recognition made of it — that is the quickest way to see whether it found the place you mean, before you click on it.

Moving and colouring

Moving, rotating and arranging objects — and assigning faces to a filament.

Two ways change the model itself and not just the picture: *Move* from the tool strip below the view, and **Colour** from the menu. Both keep the promise of the history — every drag and every assignment is its own step, and Ctrl+Z takes it back on its own.

Move shows the handle in the picture: three arrows for moving, three rings for turning, a cube for uniform scaling — labelled X, Y, Z and S.

What moves is what is selected. With the part selected, the handle moves the part. With a hole inside it selected, the handle sits on the hole and moves the hole — the part stays put. The status bar says for every handle what it is about to move; what else is different at a feature is in the chapter *What Solidon recognises in a model.*

Anyone who would rather type than drag, types. Three buttons in the bar pick what it is about — *Move*, *Rotate*, *Scale* — and next to them stand the numbers: X, Y, Z in millimetres, an angle in degrees, a factor in per cent or the target size itself. Enter in the field applies it; there is no button for it, because dragging the handle has none either. With a face selected, the bar only offers *Move* — *Rotate* and *Scale* sit greyed out and say in the tooltip why.

On release the drag stands in the history as *Move*, *Rotate* or *Scale*, with numbers that can be changed there afterwards like any others.

While you drag, the number stands above the picture. Whoever wants it exact types it: the first digit takes over the drag, Enter applies exactly the typed value — without snapping, because whoever types means it exactly — and Esc discards the whole drag.

And you can see where it is going. The shadow follows along while you move; lift a part and you watch it run out sideways — that is the only account of height a picture without perspective can give.

Grid snap and angle snap round every drag to the step you set. Both sit behind the button with the grid, and both are zero out of the box: a snap nobody asked for swallows every small drag. When turning, a magnet takes over instead — near a multiple of 45 degrees the angle locks briefly and lets go again as you keep pulling. An arc in the picture grows with the rotation and stops while it is locked, so that the locking is something you see and not only read off the number.

With a face selected, the handle sits on the face. Dragging it offsets the face along its direction, and the neighbouring walls grow with it (*Offset face*) — the 20 mm block becomes a 25 mm block, not a displaced box. Whatever is dragged across the face is discarded: a face only knows forwards and backwards.

The cube scales uniformly — for an exact target size there is *Fit to size*, for per-axis distortion the dialog of *Scale*. And whoever wants to put two parts together does not drag pixel-perfectly but uses *Align to feature*: face onto face, hole onto hole.

Colouring comes twice, and the difference is the size: *Assign filament* takes the entire body, *Filament on a face* exactly the face you point at. For a face the shortest way is a right-click on it — the dialog then already knows it.

What you pick is a **filament**, not a number: with a name and a colour, from your own filament inventory or created on the spot. At the very top stands what the body already carries — usually that is the answer. On 3MF export this becomes the filament change for the printer; a part without its own filament keeps the colour of the body.

Open the **filament inventory** from the start screen or the *Edit* menu. Each spool has its own remaining amount, even when two have the same name. If the amount is unknown, it is shown as *Unknown*. The quick picker next to the selected bodies assigns a spool directly; with faces selected, it affects only those faces. Changes in the inventory do not overwrite saved project colours or print settings.

The history sees both exactly like a menu entry: the same operations, the same undo, the same chance to change your mind.

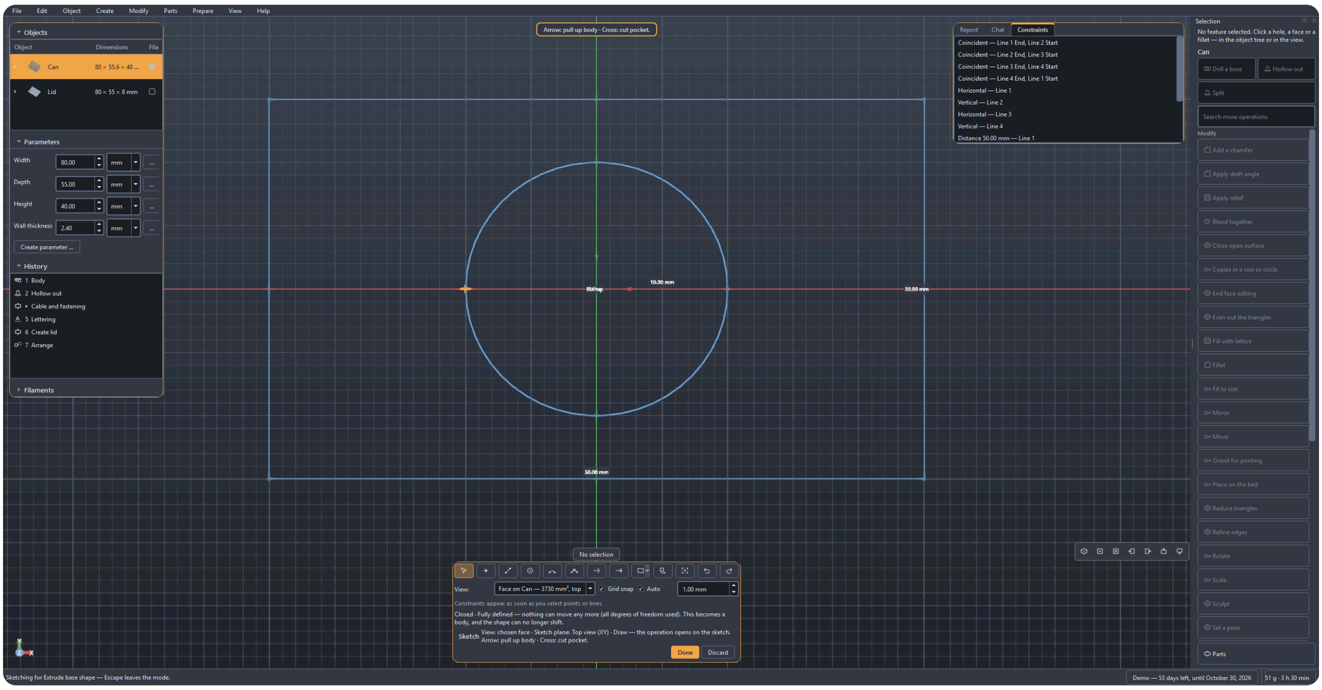
Remove filament clears the assignment for the selected bodies or faces. Other faces keep their filament. Press `Ctrl+Z` to restore the entire assignment.

Draw

Sketching with constraints: lines, arcs, conditions, and what becomes of them.

For the outline no base body gives you. A block with a hole needs no drawing — a base plate with a cut-out that a power supply fits into does.

It starts at the top of the toolbar. *Draw* stands there with its name beside the symbol. It swings the view square onto the drawing plane: you draw in the model, not beside it. What comes out of it you decide at the end — not beforehand. Escape gets you back out, like out of any other tool.



The view stays the view — the drawing lies within it.

The plane first, then the line. *XY* lies flat like the print bed, *XZ* and *YZ* stand upright. With a body in the scene its planar faces are in the list as well — then you draw directly on the top of a part. The sentence next to the choice says how the print layers lie to it: on *XY* parallel to the drawing, on the upright planes across it — and across means that a horizontal line in the picture will be a seam later on. With a picked face its own tilt decides, and the sentence names it: flat, across or at a slant to the layers. Anyone who already has the face in view does not have to recognise it again in this list: right-click on it in the window or in the object tree, *Draw on this face*, done.

The tools and their keys: line `L`, circle `C`, arc `A`, point `P`, curve `S`, trim `T`. The digits `1`, `2` and `3` switch directly between XY, XZ and YZ. `Esc` switches back to selecting and cancels an element you have started. The line runs on as a chain — the end is the start of the next one. A curve collects until you say it is finished: double click, Enter, or click the last point once more.

Finished outlines are ready to hand. Under *Basic shape*: rectangle (`R`), slot, circle, hexagon — and the two that would otherwise be handwork, bolt circle and hole grid. They come in as a drawing with constraints, not as a heap of points, and can be dimensioned afterwards like anything drawn by hand.

Looking, selecting, dragging. Mouse and keyboard drive the view as in the main window — what is written there applies here unchanged. Anyone who rotates stays rotated — the view swings square onto the drawing plane once on entering and not again by itself. *Fit to view* (`Home`) brings everything back into the picture, and on opening it has already happened: an existing drawing fills the area, an empty sheet shows the whole build volume. With *Select*, dragging a point moves that point, and the solver pulls along whatever hangs on it. **Ctrl while clicking collects**, and that is no detail: *Parallel* and *Perpendicular* need two lines, *Symmetric* two points and an axis. The order of the clicks counts — “A parallel to B” is not “B parallel to A”. **A constraint already set is taken back by a second click on the same button**; it stays pressed while it holds. A right-click on a point shows what hangs on it and removes them one at a time. `Del` deletes the selection together with the constraints that hung on it; with the focus on the constraint list instead, `Del` removes the entry there. And `Ctrl+Z` applies here too, to the last stroke on the sheet and not to the last step in the history.

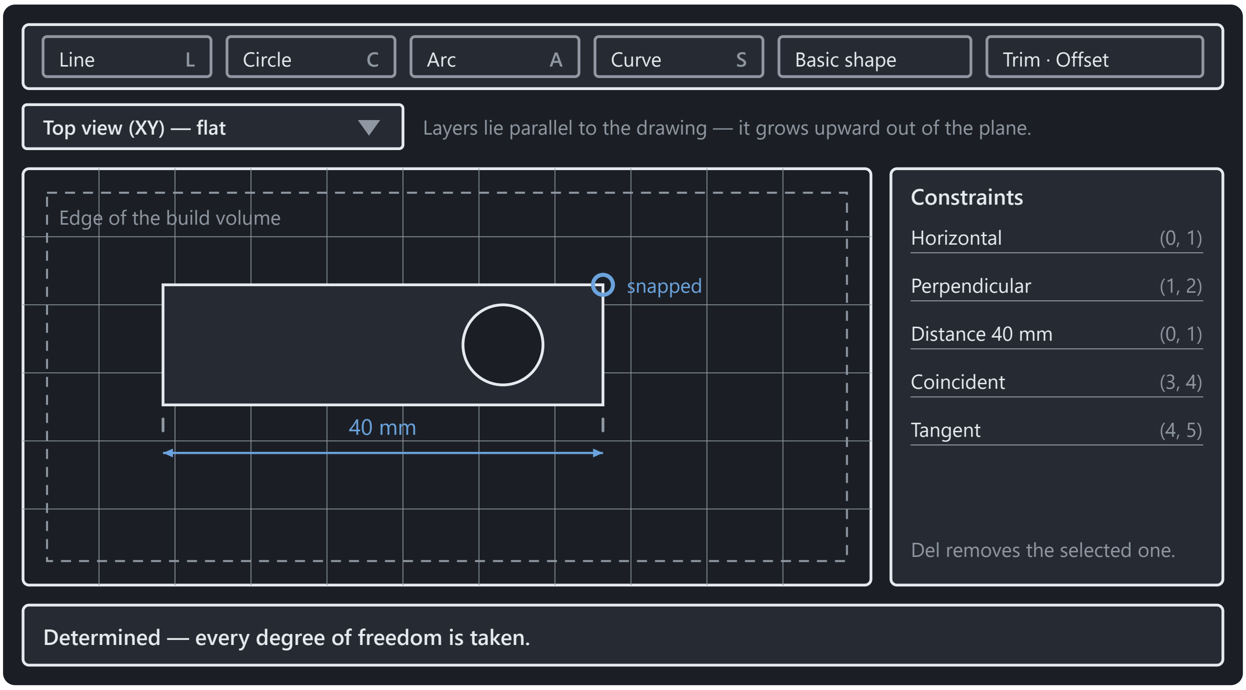
A click near an existing point snaps to it. That is more than convenience: the two points get a *Coincident* constraint and stay together, even when something else moves later. Two points that merely happen to share the same coordinates do not.

And where there is no point, the grid catches. The lines in the background follow the zoom: they step through 1, 2 and 5 and grow coarser as soon as they would sit closer than twenty pixels — a grid whose lines run into each other helps nobody. With *Snap to grid* on, every click falls onto the spacing currently shown, and onto the **nearest** one; truncated, every point would sit offset in the same direction. Existing points take precedence — where one is nearby it wins, because it brings a coincidence with it and the grid only a round number.

The spacing can be pinned. Type one in and it stays, however far you zoom; for a part thought out in two-millimetre steps that is half the construction. Turned all the way down the field reads *Automatic*, and the spacing follows the zoom again.

Dimension while you draw. With line and circle a measure field goes live after the first click, right at the pointer and not in the strip: type a length or a diameter, press Enter — the direction comes from the pointer, the number from the field, and it stays on as a constraint. At a circle, Ø or R sits next to the field: one click swaps diameter for radius, and the choice applies in every circle field of the application. Afterwards the same works with `D` across two selected points, and there an expression may stand: `@width/2` binds the measure to a project parameter instead of repeating it.

Constraints hold the drawing, not the coordinates. Select something, then the button — or the right mouse button on the spot. There are ten: distance, coincident, horizontal, vertical, parallel, perpendicular, tangent, symmetric, fixed, and the reference dimension, which measures along without holding. Whatever does not fit the selection is greyed out. On the right they all stand in a list; whoever runs the pointer over an entry sees the points it holds light up, and `Del` takes it away.

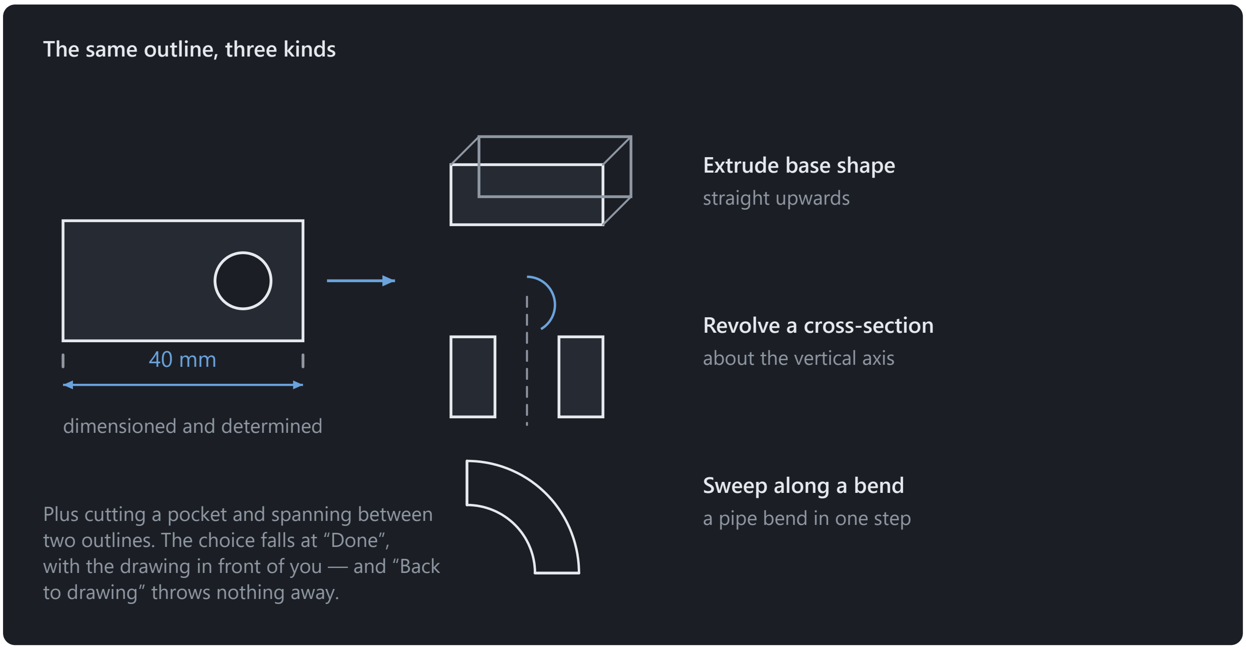


The status line is the real answer: whatever is still free will move on the next drag.

At the bottom stands how firm the drawing is. *Determined* means every number is taken and nothing moves any more. "Four degrees of freedom are still free" means four things are not settled yet — the sketch still works, it is only still movable. If two measures contradict each other, Solidon says which, and the last valid position stays put: an impossible constraint does not destroy what has been drawn.

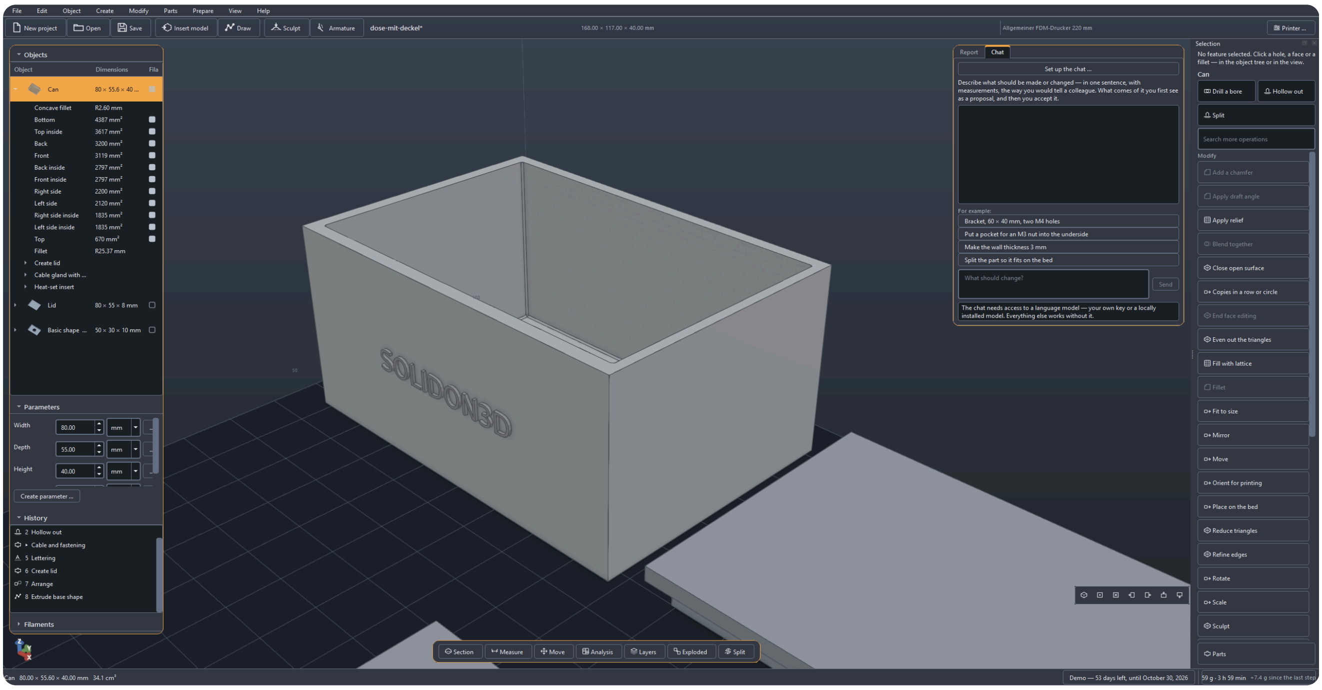
Changing without drawing again: *Trim* cuts away the half you click, *Extend* lets it grow, *Offset* (`O`) lays a contour beside it at a distance — negative goes inwards —, *Mirror* throws the selection about the X or the Y axis. *Guide line* (`X`) turns a line into one that carries constraints but forms no profile: the centre line something hangs on symmetrically should not be extruded along with it. And *Project* brings the edges of the existing bodies on this plane into the drawing — the way to line something up on an imported part instead of measuring it off and typing it in.

The edge of the build volume is drawn in — dashed, with its size written on it, and whoever draws beyond it reads so on that same line: *the sketch sticks out beyond it.* That makes the drawing area the earliest place at which it shows that a part does not fit on the bed — earlier than any report. With a small part the frame lies outside the picture, and that is deliberate: it fits anyway. Once the drawing grows past the plate, the frame comes along into view by itself.



What becomes of the drawing is decided after the drawing.

Finally Solidon asks what it should become. The *Done* button shows the five kinds with a sentence each: extrude it, cut it in as a pocket, revolve it about the vertical axis, sweep it along a bend, or span it between two outlines. *Back to drawing* is there too — it throws nothing away, it opens the same drawing again.



What was drawn is now a step in the history.

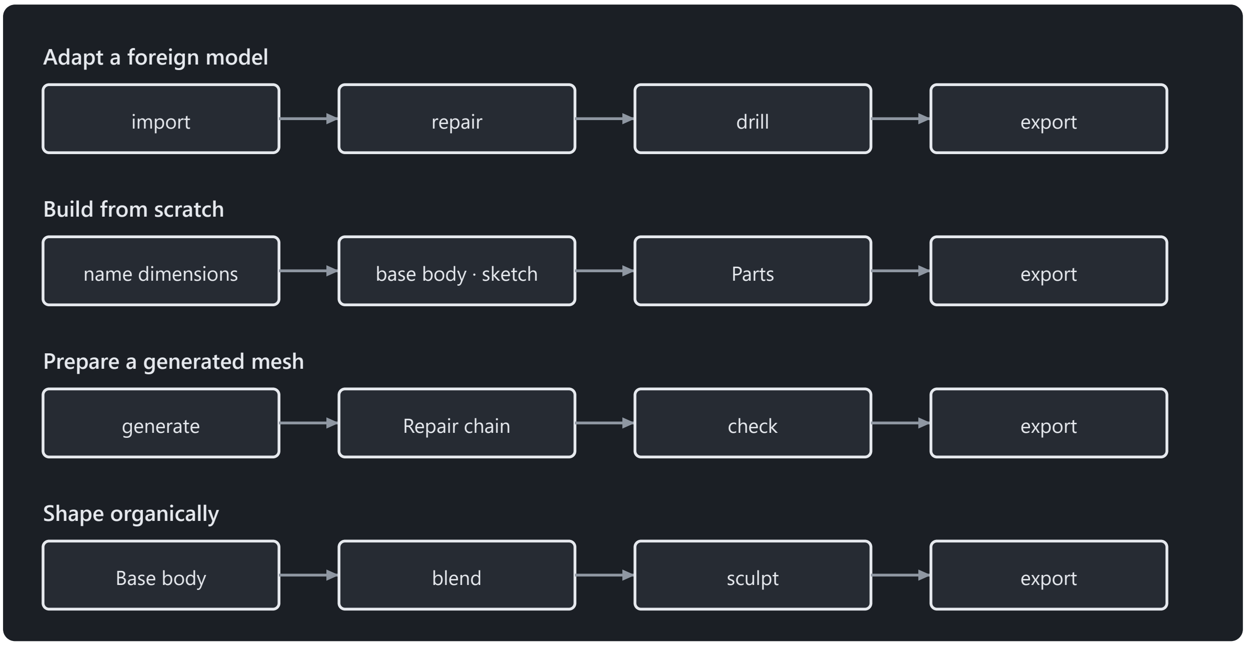
Afterwards the drawing is a value like any other. It stands as a parameter of the operation in the history, and a double click on the step opens it again through *Draw …*. A line that sits two millimetres off gets moved — the operation above it recomputes, and the rest of the project stays as it was.

Drawn curves stay truly round even when enlarged: a circle becomes a round cylinder, not a polygon with many corners.

The four ways

Adapt, construct, generate, sculpt — four ways into the same scene.

Almost every task takes one of four ways. Each has an example project ready — on the start screen, and each opens with a tour: on the right it says, step by step, what to try, and the tour notices by itself when a step is done.



Each way has an example project waiting on the start screen.

Way 1 — adapting a foreign model. Something downloaded that almost fits: import, repair, drill, export. The most common way, and the one where the repair counts.

Way 2 — building from scratch. Something new out of primitives, parts and drawings, with named parameters: width, depth, thickness. Change a number and the part changes. Whatever no primitive gives you is drawn — the *Draw* symbol at the top of the toolbar, described in the chapter before this one.

Way 3 — preparing a model from text or an image. The supplied workflow generates it with TripoSG in a local ComfyUI. What comes back is a surface, not a construction. It passes through the repair chain, and holes are then created as separate steps — not by measuring the mesh.

Way 4 — sculpt a figure. A shape that cannot be dimensioned: roughed out from primitives, blended together and then shaped by hand. Its own chapter is further down; what sets it apart from the other three is that here a gesture counts and not a number.

Sculpt

Sculpting by hand — and why a whole session stays one step.

Some shapes cannot be dimensioned. A handle meant to sit in the hand, a figure, a grown transition — that is what the brush is for.

The way there has three stages, and the first is happily skipped. First a rough shape from primitives, blended together (*Blend together* in the *Modify* menu); then *Even out the triangles*, so that vertices sit everywhere alike; then *Sculpt*. Skip the second stage and you paint on a mesh that cannot follow your brush — the bar says so as soon as the session is open.

The whole session is one step in the history. Not one step per stroke: a figure is several thousand strokes, and a history with four thousand entries is none. Ctrl+Z takes back one stroke during the session, the whole session afterwards.

Six tools. *Add* and *Carve* are the same with opposite sign. *Smooth*, *Inflate* and *Flatten* read what lies before them and therefore cost one extra pass each — the bar shows how many. *Pinch* pulls towards the stroke centre and makes edges.

Order does not matter within a stage. Two strokes over the same spot add up onto the original surface rather than the second sitting on the result of the first. If you do not want that, set *Start afresh* — then the next stroke begins on what is there. The switch applies to one stroke.

Symmetry can be changed afterwards. It sits on the operation, not on the single stroke: a finished session can be mirrored with it. Mirroring is about the object's origin — not its centre of mass, which wanders while sculpting.

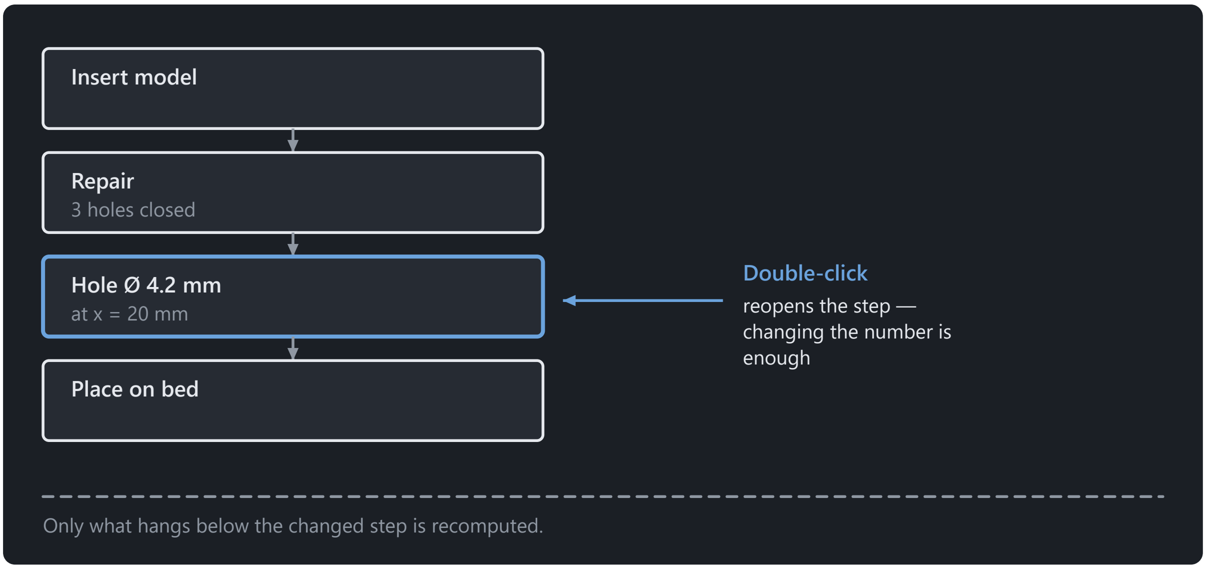
When this tool is not the right one. On a part still being dimensioned: a stroke sits at a place in space, and changing the shape underneath pulls the surface out from under it. For a transition between two bodies *Blend together* is better — three numbers instead of a hundred strokes, and editable at any time. And anyone modelling a portrait bust is better served by a program built for it; six brushes are six brushes.

What this program can do for it and the others cannot: while you sculpt, wall thickness runs along. Spots that are too thin stand as a number in the bar, before the slicer finds them.

The history

Why every step stays editable, and what holds a transaction together.

Every operation sits in the history, and there it stays changeable. That is the difference everything else hangs on: a model here is not a mesh of triangles but the list of steps it came about through.



Every step stays a number — tomorrow as well.

A double-click on a step reopens it — with the values that are in the file. To move a hole two millimetres, change the number; only what hangs below is recomputed. Changing it afterwards is likewise reversible: Ctrl+Z restores the previous value, like with any other step.

Several steps can be selected at once — Ctrl or Shift while clicking. That is exactly the slice a part of your own gets (*Your own parts*), and without a selection the whole stack applies.

Deleting works too, and it is undoable like everything here: the entry sits in the list's context menu and in the bar below it. Because a deleted step hits the ones that build on its result, it is the one place with a confirmation — it names the affected steps and the way back via Ctrl+Z. There are no other safety prompts here.

"Edit this step" also sits on the feature itself, in the context menu in the view and in the object tree: the way back from the result to the step, without hunting for the right row in the history. For that the object tree groups what came from a part under its name — six fillets stay with their pegboard hook this way, instead of sitting in a flat list.

Undo takes back a whole transaction, not half a step. A proposal from the chat is exactly one transaction — an undo takes it back completely.

Two limits are deliberate. Steps taken back fall away as soon as a new one arrives: there are no branches. And a change that alters the *number* of objects while later steps work with them is refused — the new bodies are not the old ones, and an error at the end about a number at the beginning is one nobody can trace.

Clicking sets things up. Select a face in the window and then call an operation, and position and axis are already filled in. The size is not: a countersink takes the head of the screw, not the diameter of the hole it sits on. With a hole that has been clicked, the dialog also says which standard size the measured diameter matches — and where none matches, it asks instead of inventing one.

Parameters and expressions

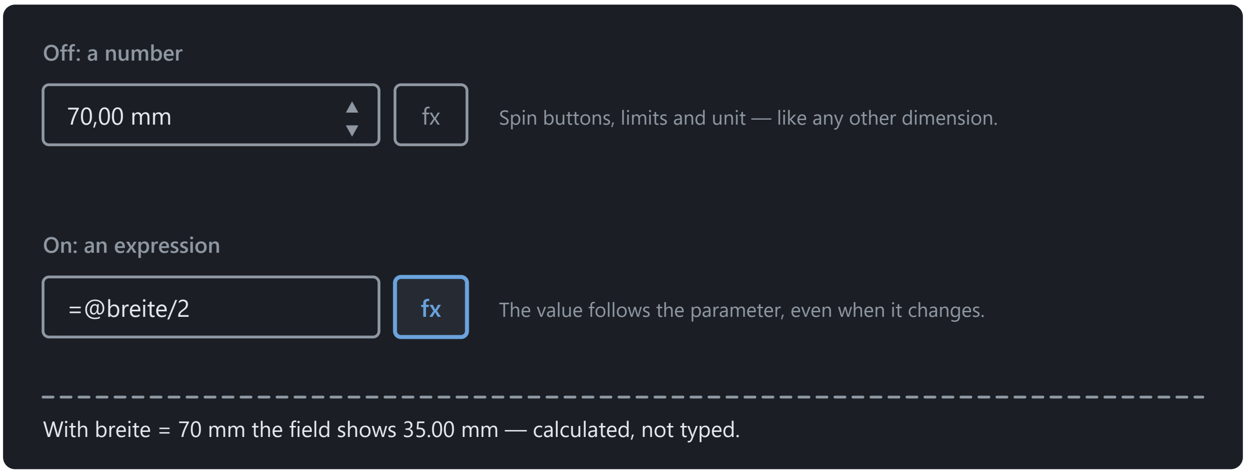
Named dimensions instead of numbers in the model, with expressions between them.

A project has named parameters — `width`, `depth`, `wall` —, and operations may compute with them instead of with fixed numbers.

Why that is more than convenience. A number that sits in seven places is seven numbers: change it and you change six of them and miss the seventh. A parameter is one number in one place. Turn `width`, and everything that depends on it follows in one go — the bore in the middle stays in the middle, because `=@width/2` is written there and not “35”.

This is how you create one. On the left under *Parameters*, click *Create parameter*, enter name and value.

And this is how it gets into a dimension. Next to every number field there is an **fx** button. It switches the field over: off there is a number with spin buttons, on there is an expression. Only then does the field take `=@width/2` — the equals sign turns the field into an expression, the at sign refers to a parameter. Type `@` and the field suggests the names of your parameters, and below it says what is allowed.



The same button sits beside every dimension. Switched on, the field takes an expression instead of a number.

The name in the expression is not always the one in the list. A parameter has the name it was created under and a label that may be translated — the two need not be the same word. The suggestion list shows the right one; whoever uses it never has to know the difference.

Limits, unit and expression can be changed afterwards. Right-click on the parameter, *Change …* — and that is reversible like any other change. An upper limit set too tight otherwise jams a field without a word.

What may appear in an expression: the four basic operations, brackets, `min`, `max`, `abs`, `round` and references to other parameters. `=@width/2 - @wall` is allowed, `=max(@width, 40)` as well.

Nothing else is allowed, and that is deliberate: the expression is computed by an evaluator of its own, not by Python. **A project file is a file and not a program** — it travels between people as an error report, and what is in it must not execute anything. A language model that wrote the expression changes nothing about that.

Circular references are detected and named. If `a` points at `b` and `b` at `a`, Solidon says so instead of computing to the stop.

Everything is computed in millimetres and double precision throughout. Only what is displayed gets rounded — what you read as “40.00 mm” is a number with all its digits in the core.

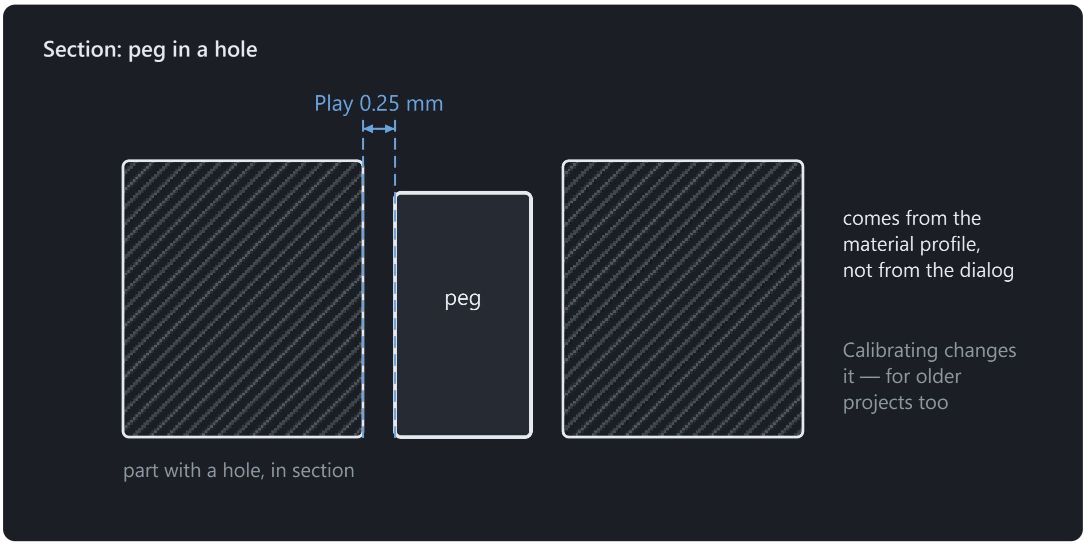
Material, tolerances, fits

Where the clearance a lid needs comes from — and why it is not estimated.

The piece that sets Solidon apart from a slicer.

A printed part is never as large as drawn. Plastic shrinks as it cools, a hole of 5 mm comes out narrower, and the bottom layer is squashed wider than all the others. Anyone who wants to push two parts together has to allow for that — and that is exactly what Solidon takes off your hands.

The clearance is in the material profile, not in the model. To put a pin into a hole you do not enter 0.2 — you say *fit*, and the number comes from the profile of the material being printed.

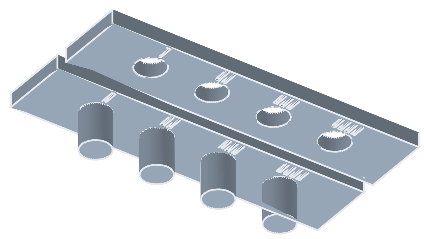


Calibrating also improves parts that were built earlier.

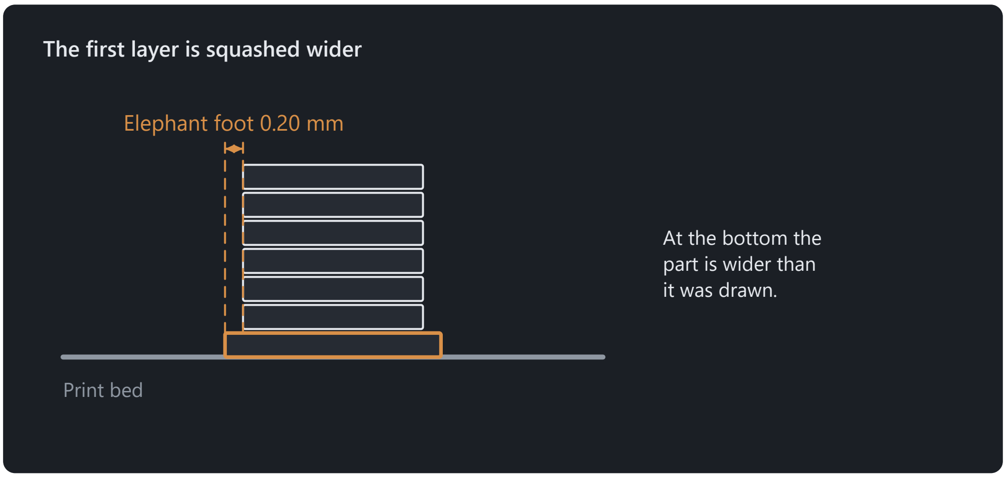
That is why calibrating works backwards. Under *Edit → Calibrate material* the measured value is entered. From then on it applies to every fit — including those in projects that existed before.

Measured is printed, not guessed. The *tolerance ladder* puts pins and holes with stepped clearance on one plate: print once, try, and the value stands. The *wall ladder* and the *overhang fan* do the same for the minimum wall thickness and for the angle at which this printer really needs supports — instead of the rule of thumb of 45 degrees.

The first layer is a special case. It is pressed against the bed and spreads outwards as it goes. For a fit that sits at the bottom of the part, that is the difference between “goes in” and “does not go in”.



Print once, try it, enter the value — every fit is right from then on.



For fits on the first layer, Solidon allows for it.

The test piece cuts a cube out around one spot instead of rebuilding it. What gets printed is the real geometry with the real tolerance: two minutes instead of two hours, and the result holds for the part.

A scene may have several materials. A TPU gasket in a PETG housing shrinks differently and wants more clearance. With *set material* the single body gets its own profile, and tolerances, elephant foot and fit checking compute with it.

The parts

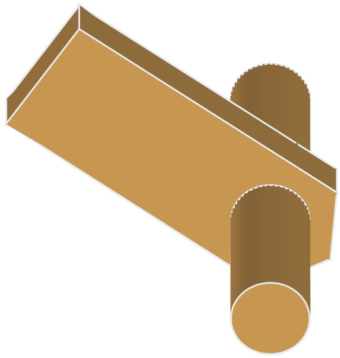
Tested connections from the library instead of self-built geometry.

Sinking a hexagon into a wall deeply enough for an M4 nut to fit while the screw still grips takes half an hour — the first time. The part library does it for you: click a face, choose a part, choose a size. Without a selected body, *Insert* remains locked and explains at the button why — the catalogue can still be browsed, and with no part chosen it says that too.

The dimensions come from a table, not from memory. The width across flats of an M4 hexagon, the depth of a heat-set insert, and the allowable width of a living hinge are looked up rather than estimated.

Nut trap — the pocket for a hex nut inserted during printing. The most common way to give a printed part a load-bearing thread.

Heat-set insert — the stepped bore for a brass insert melted in with a soldering iron. More work than a nut trap, but it can be undone as often as required.

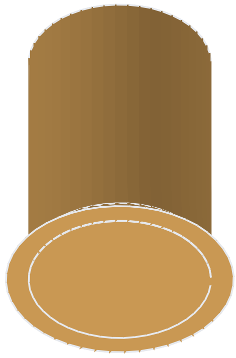


One click instead of half an hour — and the dimension comes from the table.

Snap fit — the springy arm that yields while joining and then latches. A lid that holds without a screw.

Living hinge — a section thin enough to bend instead of breaking. The joint is printed with the part, leaving nothing to assemble.

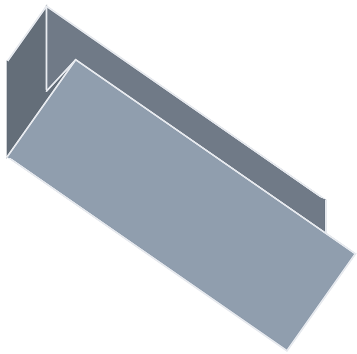
Pegboard hook — one to six hooks directly on the part, matching the grid of a SKÅDIS pegboard; a back plate is optional. Insert it from above and pull down: a springy latch tongue clicks into place and holds the part even when someone pulls on it. To remove it, press the tongue down through the slot. Anyone removing the part frequently can disable the tongue and use the simple lift-off hook instead. It prints lying down so the tongue flexes across its layers.



The thread holds in metal, not in plastic.

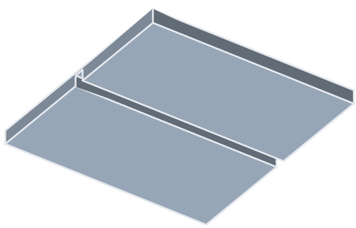
Hinge eye — a tab with a hole. Two of them and a dowel pin form a rotating joint; the living hinge above bends. It is explicitly half a hinge.

Barrel hinge — two lugs around a pin printed with them, separated by a gap the printer leaves open. It comes off the bed already movable: nothing to assemble or insert. The gap is the fit — *Play* sets it and makes the difference between a joint and two fused lugs.



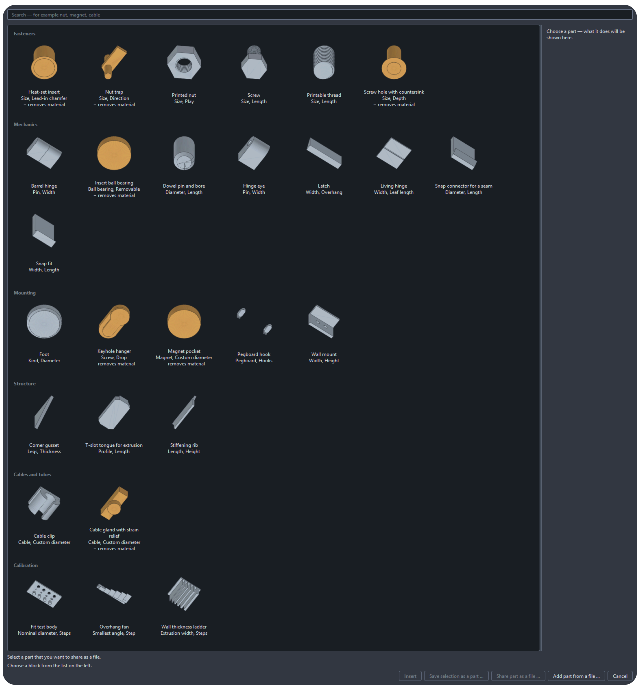
Corner gusset — the triangle in an inside corner that holds two walls at a right angle. A single flexible wall calls for the stiffening rib instead.

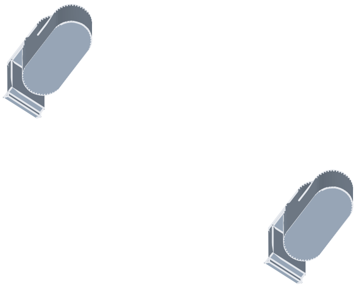
Foot — a printed foot or a pocket for a purchased rubber foot, as selected. The chamfer points down so the elephant foot of the first layer squeezes into empty space.



Cable clip — the bow on a face with an opening narrower than the cable. It guides but does not resist pulling; the cable grommet does that.

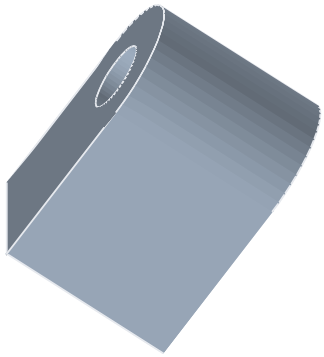
Also included are the screw hole, magnet pocket, cable grommet, rib, keyhole, *Insert ball bearing*, *Screw*, *Printed nut*, printed thread, T-slot tongue, wall mount, snap fit, dowel pin and dowel bore, snap connector for a seam, and the test pieces from the previous chapter. The catalogue sorts them into groups and shows a picture for each — a library invisible to the user does not exist.



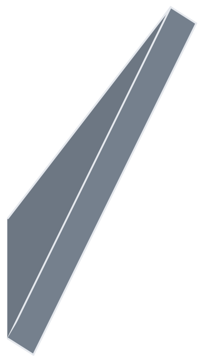


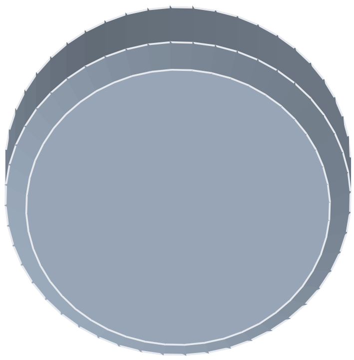
Hooked in from the top, held by the tongue — pressed through the slot to release.

Every part remains as editable as any other operation. It appears in the history, its dimensions can be corrected later, and the places where it attaches retain their names.



Two parts that belong together are created together. A dowel pin without its bore is nothing — and placing the two separately means typing the same dimensions a second time; a 6 mm pin in a 5 mm bore is born right there. *Set counterparts …* in the *Parts* menu turns that into one step: mark the spot on each of the two parts, choose the pair, enter the dimensions once. Solidon places both halves, records the fit between them and puts it all into **one** step — a single Ctrl+Z takes it back together. It knows three pairs: dowel pin and bore, printed screw and nut, heat-set insert and clearance hole.





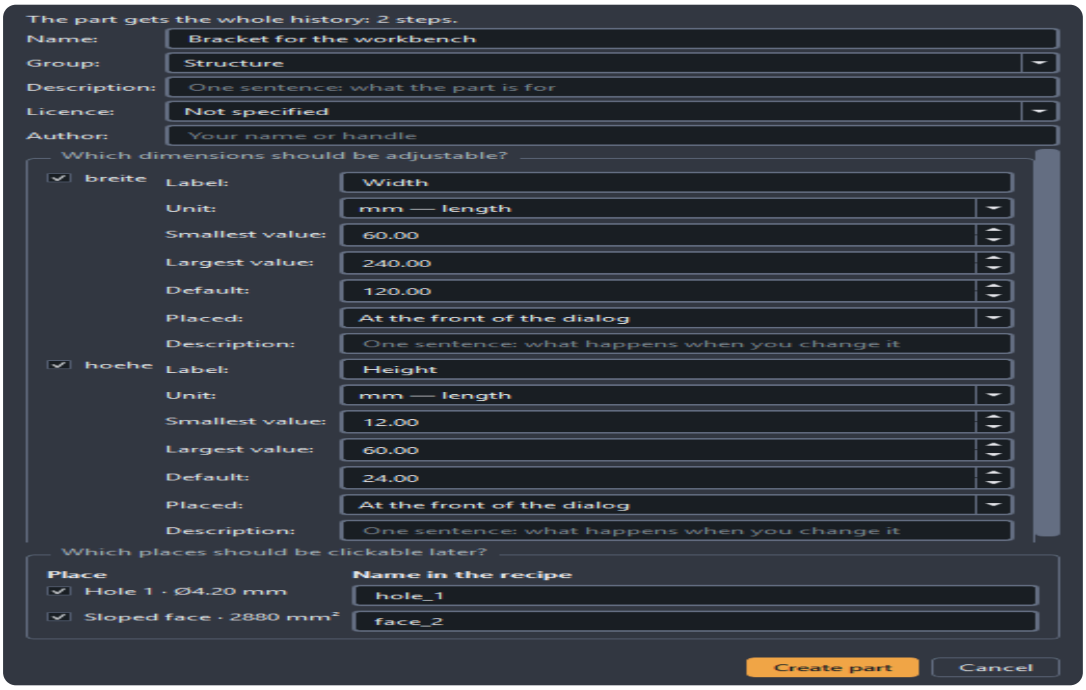
Your own parts

Storing a part you built yourself so that next time it is already there.

The bracket for the workbench took an evening. Next time it should be a menu entry — with an adjustable width, because the next board is thicker.

First choose in the history what should come along. Several steps can be marked there — Ctrl or Shift while clicking, as everywhere —, and exactly these travel into the part. Without a selection the whole stack applies, and that is the common case: whoever has just built their part usually means all of it anyway. The recipe dialog writes into its header what it received, so that this default does not apply silently.

The way starts in the part catalogue (*File → Part catalogue …*, Ctrl+K), not in the parts menu: there you find *Save selection as a part …*. What you built becomes an entry like the nut trap or the snap fit — with a preview image, with fields to adjust, and with places that something can be aligned to later.



What should be adjustable is decided by whoever built the part.

You need project parameters first. That is the one condition it otherwise fails on: exactly what has a name in the project becomes adjustable. Whoever wrote the width of their bracket into the box as a fixed number gets a part that can do exactly one width. Create the values that should change as parameters beforehand and bind the dimensions to them — the chapter *Parameters and expressions* shows how. The button in the catalogue stays locked until that is done, and says next to it what is missing.

The dialog asks five things. A name for the part in the catalogue. A group it is filed under. A description that sits in the catalogue's detail column. For every parameter, whether it should be adjustable — with label, unit, smallest and largest value, default and a sentence about what happens when you change it. And for every detected feature, whether it should be clickable later: a hole another part is aligned to, a face something is placed on. Both are mandatory, not merely offered: without at least one released dimension and at least one released spot, *Create part* stays locked, with the sentence saying what is missing.

Creating it computes, it does not just save. The part is built once at every corner of the range you gave — smallest width with largest height and so on. If no usable body comes out anywhere, that is shown on the entry in the catalogue. A part that falls apart at 8 mm is better known before it is first inserted than after.

It belongs to you and stays on this computer. Nothing is uploaded, nothing is shared. In the catalogue it sits next to the supplied ones with a marker. If you pass on a project that uses it, it travels along as a recipe — as a list of steps and values, not as a program. If your computer already carries a part of the same name, yours wins and the one that travelled gets a name of its own.

Changing means saving again. Save under the same name — the button then reads *Replace part*, and file, catalogue entry and computation are swapped together. And if you no longer have the project it came from — or it reached you as a file — *Open for editing …* in the catalogue brings it back: its steps stand in the history again, and when you save, everything you typed back then is already there. An imported part stays marked as imported. A recipe has a version that follows from its content. If you later open a project that used an older one, it tells you — the old version travels inside the project and sits as its own entry in the catalogue, and you decide which one applies. If nothing changed, nobody says anything either.

Exchange part files

Pass on a part of your own and take in someone else's — through a file, not an account.

Parts change hands exclusively as files. Solidon provides no public library, uploads nothing and needs neither an account nor a network connection for this. You decide how to pass a file on. The file carries the extension `.solidon-part` and is recognisable as a Solidon file by its name and icon.

Exporting starts in the part catalogue (*File → Part catalogue …*, Ctrl+K). Select one of your own parts and press *Share part as a file …*. Solidon places the complete part file, including author and licence, in the location you choose.

Only your own parts can be presented as your own work. Supplied or imported parts retain their origin, author and licence. Saving again does not turn someone else’s work into yours.

State the licence in plain language, not as an abbreviation. The dialog offers three choices named after what they mean: *Public domain — anyone may do anything, without conditions*, *Attribution — my name must be included*, and *Attribution, with adaptations under the same licence*. Choosing the third requires others to share their improvements under the same terms. The name beside it is yours to choose — initials are enough, and nobody verifies it.

The file carries data, never executable source code. It contains the recipe of registered steps and values, together with the information needed for lossless restoration. Required embedded model data may travel locally in the file; Solidon checks its size, structure and integrity before importing it.

Import follows the same path back. The catalogue contains *Add part from a file …*; the dialog accepts a part file. If it passes the checks, the line says “Imported. The part is now in the catalogue.” From then on, you use it like any other part.

Import and export use the same checks. If a check finds something, the reason is stated plainly — which field is missing, which value is not allowed, or which operation is unknown.

An imported part remains permanently in the local catalogue. Its origin stays visible there. With CC BY, every redistribution must name the original author; with CC BY-SA, a derived version must also be shared under the same licence.

When names match, yours wins. If an imported part has the same name as one you already own, yours remains in place and the imported one receives a derived name beside it. An external file does not rewrite your toolbox, even if it tries.

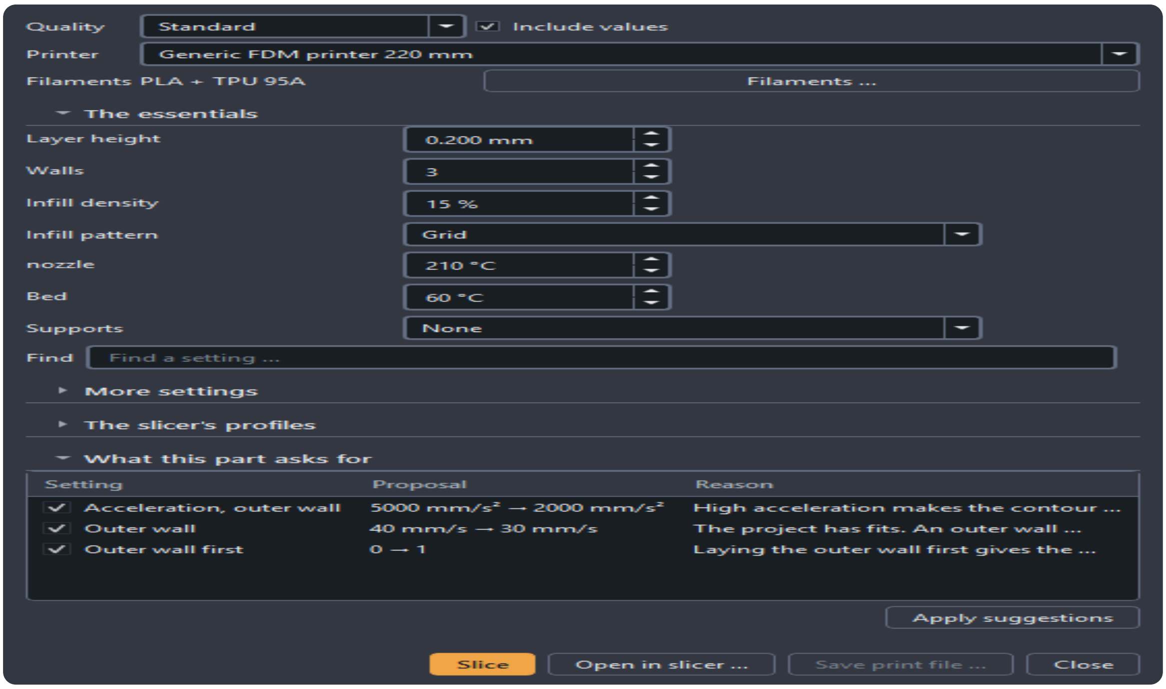
Everything stays offline. Neither the file nor its author, licence or origin is sent to RS Digital.

Printing

From the finished model to the print file, without leaving Solidon.

File → Print settings (Ctrl+P) is the way from the model to the print file. The slicer computes it — but it is driven from here, and as a rule you never get to see it.

The printer is at the top of the dialog and can also be changed directly from the project header. The change applies only to this project; filaments, colours, and their own print settings are preserved.



The slicer computes — but it is driven from here.

Up front are the values you actually change — layer height, infill, supports, adhesion. Every field explains in one sentence what it does, and whatever the geometry itself demands is suggested by the analysis with its reason: supports under the measured overhang, a minimum time per layer where the part tapers to a point.

Which slicer does the computing is set under "The slicer's profiles". If several are installed, a list lets you choose, and the choice is remembered. PrusaSlicer and Cura need no profiles — Solidon describes the machine itself. The Orca family (OrcaSlicer, Bambu Studio, ElegooSlicer) requires a printer and a process profile from its own stock; as long as one is missing, the status line says which.

Slice writes the plate, has the slicer compute it and reads the print file back: print time, material and layers appear in the dialog and report, labelled as coming from G-code. Planned material use remains separate from the Solidon estimate and a weighed remaining quantity. *Save print file* puts the G-code where you want it; with several plates, one file per plate.

Open in slicer is the second way: the plate goes as a file into the slicer's own window, and from there the job is yours — for the last touch in the familiar program, or when a slicer will not accept from outside what its window can do. This way never needs profiles. Whichever of the two ways you used last is remembered by the project: it carries the primary button next time.

If a run fails, it never ends with just "failed". The message names the reason and offers what helps now — choose another slicer, view the slicer's output, check the machine profile, or just export. And the print file is checked after the fact: if it drives beyond the build volume or comes out shorter than the model, that appears as an error in the report before the printer shows it.

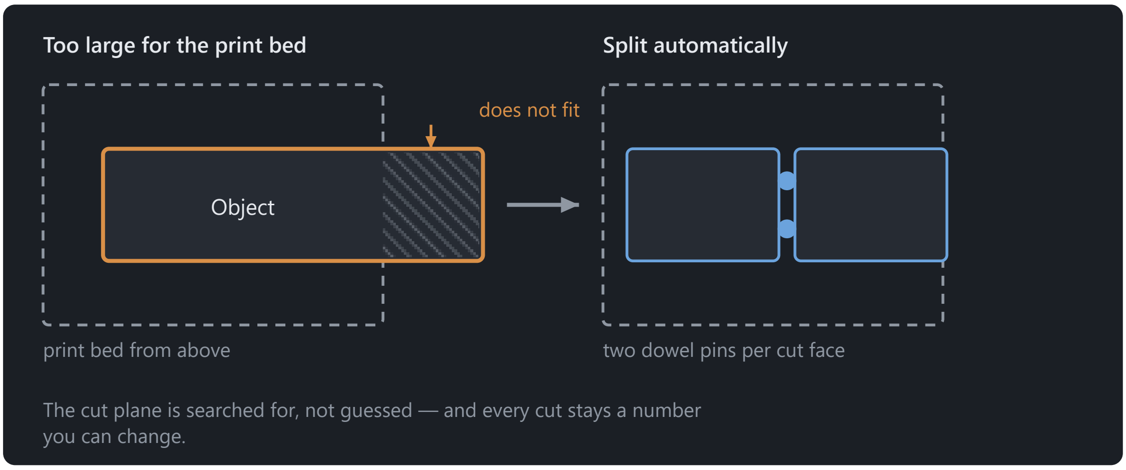
After a successful output, you can **Deduct filament**. Solidon estimates and quantities planned in the G-Code are labelled. Missing individual quantities remain unknown until you enter them or slice the model. A later G-Code figure corrects the print already recorded. *Printed again* explicitly records another print. The filament inventory's consumption history shows every record and lets you reverse it. Inventory settings offer *never*, *ask first* and *record without asking*; the default is to ask.

Onto the bed and out

Arrange, check, export — and what each format carries along.

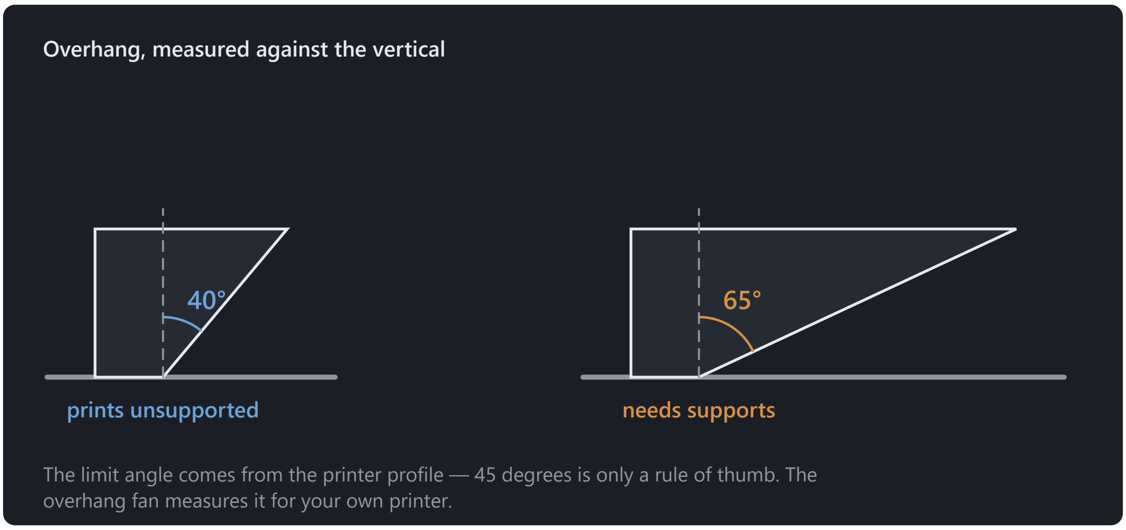
Arrange on the bed lays the objects out side by side and starts a new plate as soon as the current one is full. Whatever does not fit even then is not left out but reported.

Too large for the bed? *Split* cuts where you point; *Split automatically* cuts until every piece fits. The cut plane is searched for rather than guessed, two dowel pins go into every cut face, and each cut stays a number you can change afterwards.

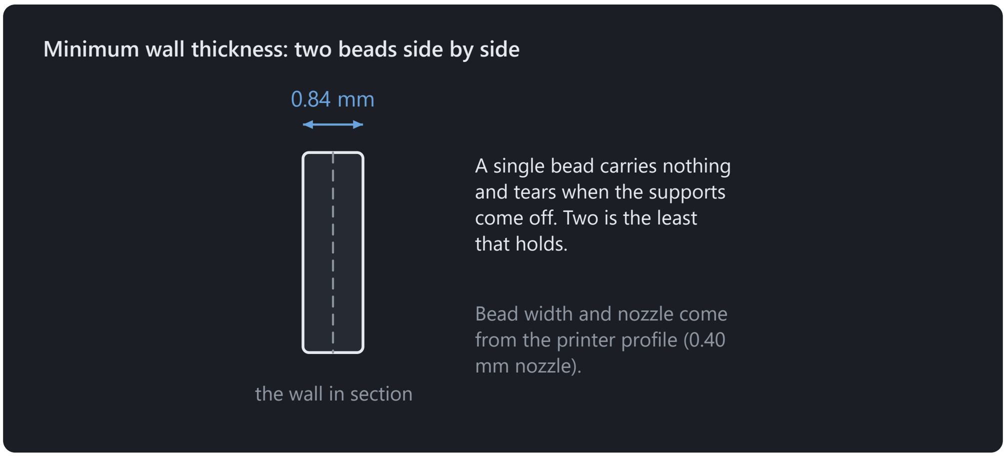


The dowel pins make sure the halves only go together one way round.

Whatever grows outwards at an angle needs supports at some point. How much of an angle depends on the printer and not on a rule of thumb — which is why the overhang fan measures it.



Walls that are too thin do not print. Less than two beads side by side carries nothing and tears as soon as you take the supports off.

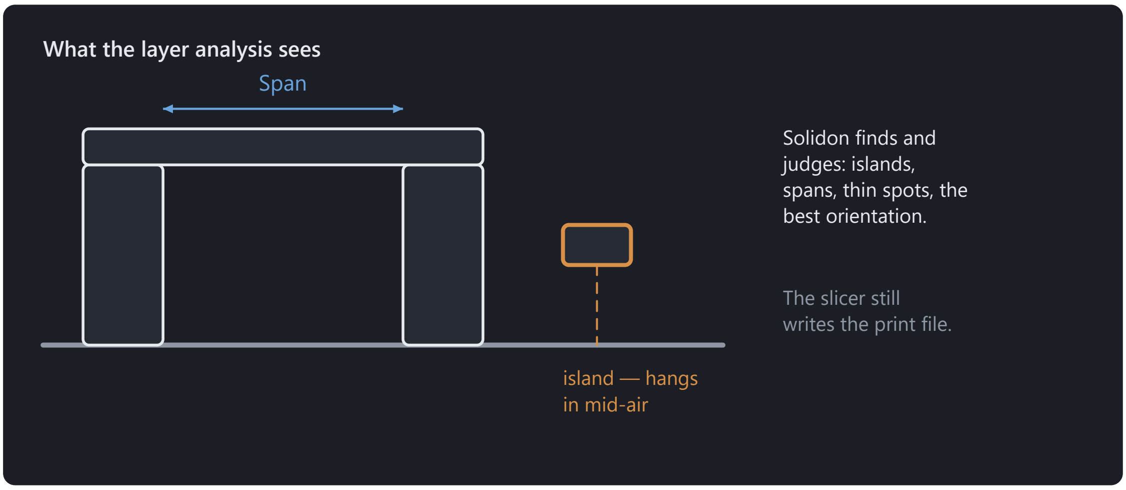


A single bead carries nothing and tears when the supports come off.

Export goes to STL, 3MF or STEP. 3MF is what slicers prefer, and it carries the filament assignments along as colour groups; STL knows no colour and loses them accordingly. STEP preserves faces and edges so other design programs can edit them individually. OBJ and PLY are also available for programs that require either format.

For showing there is GLB. Not for printing — slicers do not read it — but for sending: colours and name travel along, the part arrives true to scale in metres, and the recipient turns it in a browser without installing anything.

The slicer stays outside — yet it is driven from here. The layer analysis finds and judges — islands, spans, thin spots, the best orientation —; the print file is written by the slicer, and how that works without switching programs is in the chapter *Printing*. Where both give numbers, it is always stated which came from where.



Analysed here, sliced as before in the slicer.

When the part does not fit the bed

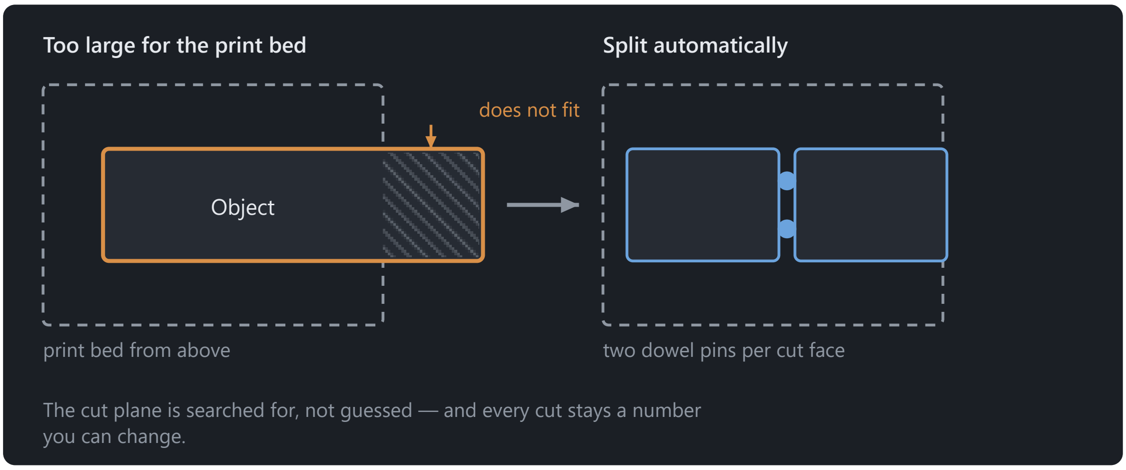
Splitting, pinning and arranging when the build volume is not enough.

A build volume is finite, and the part you need sometimes is not. Then it gets split — and the question is not *whether*, but *where* and *how the halves come back together*.

The shortest way is the *Split* tool at the bottom of the tool strip (Alt+7). Two clicks on the part draw a line, and it is cut between them — straight into the screen. Anyone who can see where the seam belongs no longer has to translate that spot into a coordinate. The *Prepare for plugging together* box is ticked from the start: pins into one half, the matching holes into the other. Beside it stands what holds them:

- **Round** prints cleanest and needs two, or the halves can be twisted against each other.
- **Hexagon** holds against twisting on its own.
- **Dovetail** also holds against being pulled apart across the seam — it is narrower at the back than at the front.
- **Snap** clicks in: a spring arm in one half, a pocket with a catch in the other. It holds without glue, and the halves come apart again. For that it needs room — the seam must give at least 5.4 mm, otherwise the arm would be too short to spring. If it is narrower, round dowels it is, and the report says why.

Prepare → Split automatically takes the search off your hands too. The parting plane is searched for, not guessed: through the same layer analysis as the orientation search, and three things are weighed. A seam made of **one** contour is better than one that falls apart into five thin webs. A prismatic run — the cross section barely changes over a millimetre — glues more cleanly than a slope. And the halves should be of similar size. The number of contours weighs heaviest: a seam in several webs is worse than any imbalance.



The dowel pins make sure the halves only go together one way round.

Two dowel pins go into every cut face. Their diameter comes from the face, their clearance from the calibrated material profile — not from a guessed number. A fit pair is created for every pin, and it is checked on every evaluation. Two pins rather than one, so the halves cannot be twisted against each other.

Every cut is an operation of its own. The parting plane stays a number you can move afterwards, and an undo takes back one cut instead of the whole split.

The “Explode” slider below the view pulls the parts apart for inspection — it only moves the display. What the stack says and what gets exported stays where it is.

If you would rather type the plane than point at it, use *Prepare → Split* and enter axis and position. The same operation, only without the search in front of it; zero pins means: only cut.

The names of the halves say which is which. After pinning, the object tree shows “… A · Pins” and “… B · Holes” — on export the file name is the only clue as to which of the two parts you are holding.

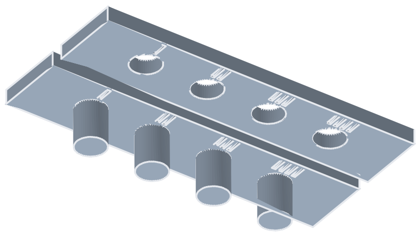
Trying instead of guessing: variants and calibration

Printing several dimensions side by side and adopting the measured one.

The number that decides a fit is the clearance — and it depends on the printer, the material and the nozzle. You guess it once; after that you measure it.

The way there is a test piece. The part catalogue (*File → Part catalogue*, `Ctrl+K`) holds the *Fit test body* under *Calibration*: a row of pins and bores with graded clearance — four steps by default, starting at 0.10 mm and growing by 0.05 mm each, so up to 0.25 mm. Both can be changed in the dialog if that range does not fit. Print it once, try them all, remember the value that slides home snugly.

That value belongs in the material profile, not in the model: *Edit → Calibrate material*. Because every tolerance in the program is a **reference** and not a number, a part made before the calibration computes with it too. A lid from last week fits better after the entry, without anyone touching it.



Print once, try it, enter the value — every fit is right from then on.

The same exists for wall thicknesses (*wall ladder*) and overhangs (*overhang fan*) — the two other numbers that otherwise get estimated.

When a number depends on the part rather than the material, the variant generator helps: *Edit → Generate variants* runs a project parameter through a range and lays the versions out side by side. Five handle diameters on one plate, one print, one decision. The stack stays untouched — the variants are an output, not a change to the project.

The chat

How to talk to the agent, what it sees, and what a proposal costs.

The chat calls the same operations that sit in the menus — it is a second hand on the same tool and not a second way past it.

Two things follow from that. **Every proposal is exactly one transaction**: an undo takes it back completely, never halfway. And **the chat computes no geometry** — it picks operations and values, the computing happens in the program.

The sequence is always the same. You write what you want. The chat answers with a proposal — a list of operations, not yet carried out. The view shows in blue and orange what would be added and what would disappear. Only *Apply* enters it into the history; *Discard* leaves nothing behind.

It was given three rules, and they are not up for negotiation: parts before shapes built by hand, named parameters before typed numbers — and **ask before guessing**. If your request has two readings, a question comes back and not a result.

What it sees is not the raw history but a digest: dimensions, detected features, project parameters, the current selection, the report. A withdrawn contribution counts as discarded — after an undo it no longer argues from a state that no longer exists.

Every transaction says where it came from (§26.4): model, version of the system prompt, version of the rule set. Anyone who wants to know in six months why a bore sits there finds the answer in the history.

It needs a model: either a key of your own (*Edit → Set up the chat*, kept in the system keychain, never in the project file) or a local Ollama. For the tool calls it has to be large enough; below 7B they fail reproducibly.

Everything else in Solidon works without a language model. Without a key the chat is greyed out and says why in one sentence — nothing more happens.

Having a model generated

A mesh from text or an image — and what it takes afterwards to make it printable.

The third route (§2.2): you describe a part or provide an image, and a generator turns it into a mesh. **File → Generate model.**

The calculation does not happen in Solidon but in a local **ComfyUI** on port 8188. If none is running, the entry remains disabled and says why; everything else continues to work. The nodes used are defined in two files beside the program — anyone using a different generator replaces the file, not the source code.

What comes back is a surface, not a construction. This is the most important sentence about this route. A generated mesh has no holes, fits and often no watertightness — it has a shape. Holes and fits are created **afterwards** as separate operations, not by measuring the mesh.

The bundled workflow uses TripoSG. Its source code and model card state the MIT licence; the complete licence chain of the weights and the models it embeds is being verified. Both are fetched only during setup in ComfyUI. Other models have their own terms, which must be checked for the exact version in use before it is used.

The generated result becomes a source, not an operation. A generator is not a function: the same request produces something different after a model change. Its bytes therefore live in the project like a file dragged in, with the usual stack above them — *Load model*, then *Repair*. The request and seed are stored in the source so the file records where its geometry came from.

The repair chain runs without asking, but as a separate step: visible in the history, visible in the report and undoable. Generated meshes are rarely closed, and they consistently need the same treatment — close holes, weld points and correct normals.

The same seed produces the same result as far as the model at the other end allows. It appears in the dialog and is saved with the project.

Setting up additional programs

The three programs Solidon can use — what each one brings, and what is left to do after installing.

None of these is required. Without all three, the whole core workflow still runs: import a model, change it, check it and export it. What is missing are individual functions — and this page tells you which ones.

The list is under **Help → Optional programs**. Each row names the purpose, state and location; where Solidon can install something itself, a button is shown. Installation happens only when someone presses it.

On Windows this goes through **winget**, on macOS through **Homebrew**, and on Linux through **Flatpak** — always without administrator rights. If the package manager is missing, the command to copy appears beside it, together with the manufacturer page.

When something is in an unusual location, *Point at it …* helps: no search procedure can find a portable installation on a second drive, and this specified location takes precedence from then on.

The slicer

For the print file and the cross-check from G-code. All the common slicers are recognised — PrusaSlicer, OrcaSlicer, ElegooSlicer, BambuStudio and Cura. If several are installed, the print settings let you choose which one performs the calculation; the choice is remembered. Solidon also reads its printer profile and, on first start, suggests the printer the slicer used most recently.

Ollama — for chat without a key of your own

There are three steps here, and the first is the smallest. Ollama comes without a model when installed and is not necessarily running — *Edit → Set up the chat* takes care of both:

1. **Is it running?** The dialog says so in one sentence. If it says “installed but not running”, a button starts it. If it runs on another computer, its address belongs in the list of optional programs. 2. **Fetch a model.** The selection lists what is installed, followed by proven models and their sizes. *Fetch the model* downloads it — five to nine gigabytes, with progress and cancellation. A cancelled download continues on the next attempt. 3. **Check tools.** This is the most important step, and it answers two questions. Neither model size nor vendor tells you whether it really calls tools — the test performs a genuine turn. If it says “writes its calls as text”, chat answers but performs no action; a different model will help.

The test also measures **how fast this computer performs**. If Ollama does not use the graphics card, it computes on the processor, which is not merely another speed but another order of magnitude — measured on a computer with Intel Arc graphics at just under eight tokens per second while reading the prompt. A Solidon request is about 20,000 tokens long; on that system, it takes **forty-two minutes** before a response even begins. If the test reports this, it is not a fault but useful information: the local route is not worthwhile on this computer, and a smaller model changes little.

Solidon ends a local request after ten minutes. This is the technical limit of the current transport. A measurement of forty-two minutes therefore means more than a long wait: this model route cannot be used on that computer.

There are other local routes, and Solidon does not set them up: Ollama itself supports AMD graphics cards through **ROCm** on Linux and Windows — which cards, its support list says, and whether it carries on this computer, the tool check tells you. **IPEX-LLM** and **OpenVINO** are separate installations with their own hardware limits and pitfalls. Anyone wanting one of these routes must assess it for their computer; otherwise, use a key for a hosted model. Both choices are valid, and everything except chat works without either.

qwen3:14b has proven itself: in the current test it completed five out of five full instructions correctly, taking 11 to 26 seconds per instruction with a median of 17 on an RTX 4080. The measurement used a graphics card with 16 GB. Smaller models are faster and less accurate; below seven billion parameters, the calls fail reproducibly.

In this check the local model was slower and hit less often than a hosted one — measured on five instructions and one card, not as a rule for every computer. For short instructions it is enough; for long turns a key of your own pays off, and that key then lives in the system's keychain.

ComfyUI — for generating from text or an image

Here too, installation is only half of the setup. ComfyUI needs the nodes addressed by the workflow and the model they load. Solidon places both in it: the ComfyUI row in the optional programs list contains *Set up nodes and model …*.

The short route goes through the generation dialog itself. *File → Generate model* shows how far the generator is ready and distinguishes three states: none is running, one is running without the nodes, or everything is ready. In the middle state, the setup button appears directly beside it.

The dialog finds ComfyUI itself. The **desktop version** from comfy.org records where it was installed. You unpack the portable version yourself; if it is in an unusual location, specify the folder containing `custom_nodes` and `main.py`.

Five steps then run: place the nodes, fetch the specified TripoSG source, fix two places in it, install the missing packages — and **check whether ComfyUI can load the nodes**. If requested, the model follows at about 7.5 GB. If the connection breaks, another run continues where it stopped.

Restart ComfyUI once afterwards. It reads its nodes at startup; without the restart, *Generate model* remains disabled even though everything is in place.

For the route via **text**, an SDXL model is also needed under `models/checkpoints`. The route via an **image** does not need one. The next chapter states which tested model is meant and where it comes from.

The graphics card determines how long it takes. Solidon shows the elapsed time and waits while ComfyUI works. If something stops, the ComfyUI message and the affected step appear in the dialog.

Local AI work runs in sequence. Ollama and ComfyUI often share the same graphics card, so Solidon does not start both at once. After a complete agent proposal, it unloads the Ollama model. After generation — including an error or cancellation — ComfyUI releases the model and cache. Cancelling removes only the job started by Solidon itself. Services on another computer remain untouched.

And what comes with Solidon

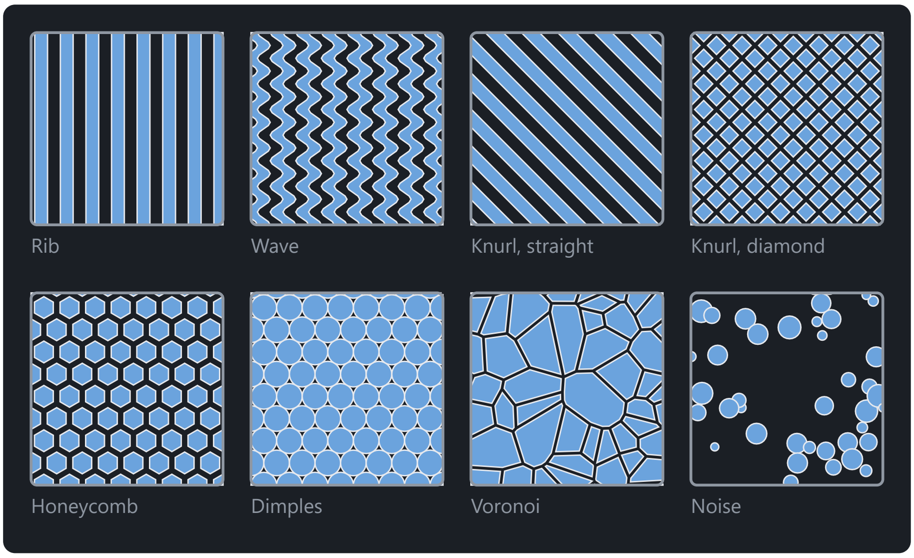
OpenCASCADE keeps faces and edges individually editable; V-HACD shows where a body falls apart by itself. Both are included in the installation package. Anyone running Solidon from source finds them in the same list and installs them from there.

Surfaces and fills

Patterns as real geometry, lattice fills and hollowed bodies.

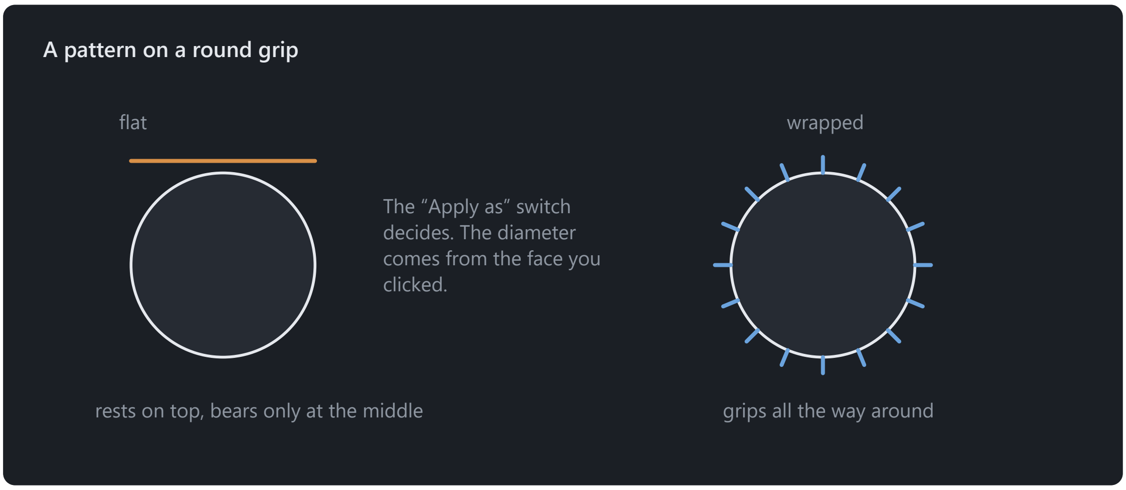
A knurl for the grip, a honeycomb for looks, a lattice instead of solid material — all of it is real geometry, not a texture in the picture. What the slicer gets is what you see.

Eight patterns are on offer — rib, wave, straight and diamond knurl, honeycomb, dimples, Voronoi and noise. Knurling gives grip, honeycomb and rib are ornament, Voronoi and noise hide the layer lines. The dialog shows a preview beside the list, drawn from the very outlines that get printed afterwards.



All eight patterns, drawn from the very outlines that get printed.

Apply texture embosses a pattern onto a face, raised or engraved. Two numbers decide whether it can be printed: the pitch must be wider than two nozzle lines, the depth higher than one layer. Solidon checks both before it computes — an embossing finer than the printer simply disappears, and otherwise you notice that on the finished part.



A knurl belongs around the grip, not as a patch on it.

On a round part the pattern is applied **wrapped**. Laid on flat, the field would meet the cylinder only at its middle and stand in the air at the edges.

Fill with lattice replaces the inside of a body with a structure — gyroid, honeycomb or cubic. This is not the same as the slicer's infill: this one belongs to the model, travels with the file and can be measured. The wall thickness of the structure is checked against the nozzle, just like the texture.

Remote control

Another program operates this installation — local, switchable, reversible.

Another program on the same machine may operate Solidon — over MCP, the same interface Claude Code and similar tools use. It calls exactly the operations that are in the menus.

It is off until you switch it on. The switch is in the settings, together with the port. Solidon then listens on `127.0.0.1` and nowhere else: the interface cannot be reached from another machine, not even on your own network.

Whatever arrives is a transaction like any other. Ctrl+Z in the window undoes it, the history shows it, and the entry says that it came from outside.

One thing never travels over the wire, and that is deliberate: anything that looks like a file path. That would mean a foreign program deciding what gets read on this machine.

In the other program it is entered as a server at `http://127.0.0.1:8787/mcp` .

Which tools exist is written in this manual. Every operation in the chapters that follow is one — with the same parameters, the same limits and the same defaults the dialog shows. There is no separate interface listing for that reason: it would be a second version of the same information, and after two months it would say something else.

Activation

Licence key, device activation and the offline route.

This release is a complete, time-limited demo. Solidon is fully unlocked without a licence key or device activation through 30 October 2026. The status bar shows the remaining days from the beginning. After the cut-off date, this demo no longer starts; your projects remain intact.

The later retail version starts without a trial period. Without a licence key and device activation, all read-only functions remain available there — opening, viewing and measuring models. Only four write actions require activation: changing a model, exporting, handing it to the slicer and using chat. A trial may be offered in a later release.

You enter a key; you do not create an account. Go to *Help → Unlock Solidon …*. After local verification, this computer is activated once — directly with *Activate online* or through request and response files using a second device. Solidon can then remain permanently offline; there are no periodic licence checks. The private device key is stored in the system keychain, while the purchase code and certificate are stored in the settings folder. None of them travels in a project file.

Offline activation takes three steps: open *Activate offline …* and save the request; redeem the file on an internet-connected device at `solidon3d.de/offline-aktivierung.html`; import the downloaded response back into Solidon. There is no long code to type. The response works only on the computer that created the request.

When changing computers, first choose Deactivate this computer. Solidon releases the single device slot and removes the local activation and purchase code. The same key can then be activated on the new computer. If the previous computer is lost or defective, support resets the slot using the order number.

When something goes wrong

The cases that come up most at the start, with what is behind them.

The cases that are most common at the start — what is behind them and what helps.

The file will not open at all. Solidon says on opening what is wrong with it instead of building an empty scene: it is empty, it breaks off mid-sentence, it is not an STL or not a 3MF archive despite its extension, it carries zero triangles, or its coordinates give no extent at all. In the first four cases there is almost always a broken download behind it — then loading it again helps. A server error page arriving instead of the model looks exactly the same. With zero triangles it is worth a look in the originating program: usually nothing was selected when it was exported. Every one of these messages carries the button *Choose another file*, which leads straight to the file picker.

"The model is not closed." Faces are missing from the file, or triangles face the wrong way. With downloaded models that is normal. *Repair* in the report closes small holes and turns faces around. If a whole wall is missing, no repair can guess it — then the file is broken and another source is the shorter way.

A boolean operation fails or eats too much. Boolean operations need clean input bodies. Solidon tries several methods in turn and says afterwards which one carried it. If it says *voxel*, the computation was coarse and accuracy was lost — then repair first and combine again.

Two faces lie exactly on top of each other. The classic case in which a difference removes nothing, or flickering appears. The remedy: let the body being subtracted stick out by a hundredth of a millimetre. The parts do this by themselves.

The part does not fit on the bed. *Split automatically* cuts it until every piece fits and puts dowel pins in the cut faces. If the printer profile holds the wrong build volume, that is the real cause — it is worth checking.

The fit is too tight or too loose. Do not change the model, measure instead: print the tolerance test body, try it and enter the value under *Calibrate material*. It applies to every fit from then on, including those in projects already built.

A step can no longer be changed. If a change would alter the number of objects while later steps work with those objects, the evaluation stops. That is deliberate: the new bodies are not the old ones, and a step that quietly ended up somewhere else would be worse than a message. Take back the later steps, change it, build them again.

Fillet, chamfer or STEP are greyed out. These tools need faces and edges that can be edited individually. Enable *Keep faces and edges editable later* when creating the base shape; the same checkbox is also available in the step dialog. If a later step turned them into fixed triangles, only the order will help: undo the steps from there, apply the tool you want and rebuild the rest. Solidon names the step that is in the way.

The chat does not answer. It needs a language model — your own key or a local Ollama. Small models below about seven billion parameters fail at the tool calls, and they do so reliably. Everything except the chat works without one.

Everything is sluggish. While you work, draft quality is running; the fine computation comes at export. With highly detailed models it helps to simplify early — the triangle count is in the object tree. When a project opens, a loading indicator settles over the view, so that the empty scene does not look like an empty project.

A greyed-out button says why it is grey. Wherever something is locked, the reason stands at the button itself — as a tooltip for the mouse, in the status bar for the glance downwards, and in the description for the screen reader. “Choose a block from the list on the left” is information; a button that only says what it *would* do is not.

As a rule: no error in Solidon stops at the observation. If there is no action beside it that carries on, that is a bug in Solidon and worth a report. The way there is *Help → Send feedback*: screenshot, log and, if you like, the running session go along, the preview shows all of it beforehand, and one button sends it to support. If you would rather hand nothing over, the same dialog stores a folder on your own machine instead.

During the demo Solidon asks by itself. After half an hour of work — only minutes in which you did something are counted — a card settles over the view and asks how it is going. It holds nothing up and waits until you look. *Give feedback* opens the same dialog as *Help → Send feedback*, with three fields in it: how well you are getting on, what worked well, what was missing. No field is required, the preview shows everything beforehand, and nothing goes out without your click. *No thanks* holds for good; if you simply leave the card standing, you see it at most three times.

Glossary

The terms that appear in menus and messages, briefly explained.

The words that turn up in Solidon, in slicers and in printing forums — briefly explained.

Mesh — a body as a shell of triangles. STL and OBJ contain nothing else: no circles, no dimensions, only triangles.

Closed (watertight) — the shell has no hole, inside and outside are unambiguous. Only a closed body can be combined and printed reliably.

Feature — a place in the mesh where Solidon has recognised a known shape: a hole, a peg, a rounding, a face. Ten thousand triangles turn back into something you can click on and change.

Operation — one working step: drilling, filleting, splitting, placing a part. In Solidon geometry comes about this way and no other.

Transaction — everything one undo takes back at once. A proposal from the chat is exactly one, even when it consists of five operations.

Parameter — a named dimension of the project, `width` for instance. Operations may compute with it; when the value changes, everything hanging off it changes.

Part — a ready-made construction element from the library, with dimensions from a table rather than from memory.

Sketch (drawing) — a flat outline of lines, circles and arcs. A body is made from it, and it stays editable afterwards: moving a line means changing the part.

Constraint — a rule that holds the drawing together: two lines parallel, a distance fixed, one point on another. They keep holding when something is dragged.

Degree of freedom — what can still move in a drawing. “Four degrees of freedom still free” means four things are not yet settled. At zero the sketch is fully constrained.

Fully constrained — every number of the drawing is given, nothing shifts any more. Under-constrained is not wrong: the drawing works, it can just still be moved.

Extrude (pull up) — raising an outline perpendicular to its plane until a body comes out of it. The usual way from drawing to part.

Pocket — the same inwards: the outline is cut out of an existing body instead of a new one being added.

Profile — the closed outline a body is made from. Not the same as the material or printer profile, which supplies values.

Curve — a smooth line through several placed points, for shapes that are neither straight nor circular. Other design programs call it a spline.

Guide line — a line that carries constraints but forms no body itself. The centre line something hangs on symmetrically is not meant to be printed.

Fit — how tightly two parts sit inside one another. The clearance needed comes from the material profile.

Clearance — the gap between peg and hole so that the two can be joined. Too little jams, too much wobbles.

Press fit — interference instead of clearance: the part is pressed in and holds by itself.

Tolerance — the amount by which a dimension deliberately departs from the nominal size so that the printed result is right.

Shrinkage — plastic contracts as it cools. That is why a printed part is slightly smaller than drawn.

Elephant foot — the lowest layer is pressed against the bed and spreads outwards. The part is wider at the bottom than intended.

Overhang — a face growing outwards at an angle. Beyond a certain angle the layer below has nothing left to rest on.

Supports — material printed under an overhang and broken off afterwards.

Island — an area of a layer with nothing underneath it. Without a support the printer prints into thin air there.

Bridge (span) — a horizontal run between two points of support. Short bridges succeed without support, long ones sag.

Nozzle — the opening the plastic comes out of, usually 0.4 mm. It limits how fine a detail can be.

Extrusion width — how wide a laid bead becomes. Two beads side by side are the thinnest sensible wall.

Layer height — how thick one layer is, usually 0.2 mm. Smaller means finer and slower.

Slicer — the program that turns the model into the print file. Solidon is not one and replaces none.

G-code — the finished list of instructions for the printer. It comes out of the slicer.

STL — the most widespread format: triangles only, no unit, no colour.

3MF — the modern successor: with unit, colours and several objects in one file. Where the slicer can read it, it is the better choice.

STEP — a format with real faces and edges instead of triangles. What classical CAD programs exchange.

GLB — the format of the viewers: triangles with colours, in a single file that every browser opens. Meant for showing, not for printing.

Report — the list of what stands out about the scene, each entry with an action that fixes it.

Filament — a named spool with its own colour and print settings. For multi-colour printing, it is assigned to a whole part or a recognised face.

Which models Solidon uses

Solidon computes geometry itself and needs nothing for that. Two things come from outside, and each of them needs a model: the chat, and generating from text or an image. Without either, everything else runs — reading in, changing, checking, exporting.

For the chat: a language model

Runs on this machine, through Ollama. It is fetched in *Edit → Set up the chat*: one button starts the service, one loads the model, one checks it. Instead of a local model, your own key for a hosted one also works.

Model	Size	What was measured
qwen3:14b	9.3 GB	Verified twice: five out of five complete instructions were correct in each run; 11 to 26 seconds per instruction (median about 17).
gpt-oss:20b	13.8 GB	Fastest: around 6 seconds per step, hits three of five.
qwen3:8b	5.2 GB	Smaller and faster, hits less often. For short instructions.
qwen3:30b-a3b	18.6 GB	Hits as often as the 14B, but takes twice the time and twice the space.
llama3.1:8b	4.9 GB	Two out of five tool calls — only if the others will not run.
qwen2.5-coder:14b	9.0 GB	Calls no tool at all in this measurement — despite fourteen billion parameters.
mistral-nemo:latest	7.1 GB	Writes about tools instead of calling them. Unusable for Solidon.

The default is in bold. Measured against five requests that each call for one tool call — a model that only writes about it instead of doing it answers in the chat and does nothing. Hence the button *Check tools*: neither size nor provider predicts it.

Below 7 billion parameters the calls fail reproducibly. And the graphics card decides the time: where Ollama does not address it, it computes on the processor, and that is not a different speed but a different order of magnitude.

For generating: three models

They run in ComfyUI, side by side. Solidon sets up two of them itself — in the list of additional programs, ComfyUI's row carries *Set up nodes and model …*. The third one you only need for the way from text.

What for	Model	Size	Where from
A body from an image	TripoSG	7.5 GB	Solidon sets it up
Cutting the object out	BiRefNet	445 MB	Solidon sets it up
An image from text first	SDXL	6.9 GB	yourself, see below

Putting the image model there yourself

Only for the way from text. Anyone bringing a photo or a drawing never needs it — and Solidon does not download seven gigabytes unasked for a way that an existing image gets around.

Tested is **sd_xl_base_1.0.safetensors** from the repository *stabilityai/stable-diffusion-xl-base-1.0* on Hugging Face. The file belongs in *models/checkpoints* inside ComfyUI's folder; restart ComfyUI once afterwards, and *Generate model* will also take a sentence instead of an image.

Another SDXL model works too: Solidon takes what lies in the folder, preferring *juggernaut* and *dreamshaper* over the base model. Models with *refiner*, *inpaint* or *turbo* in the name are not taken — they solve a different job.

How long it takes

Measured on an RTX 4080: from an **image** about fifteen seconds, from **text** about two and a half minutes. The difference is the image model — it first loads seven gigabytes into graphics memory and then computes an image before any body exists at all. After that the same chain follows in both cases: bring to working size, repair, and on very fine meshes reduce the triangles.

Without a suitable graphics card both take many times as long. What breaks off is an error and not slowness — then ComfyUI's own sentence stands in the dialog, together with the step it tore at.

What Solidon judges by

These rules are in front of the agent on every request. They are the reason it takes a screw hole from the table instead of estimating one — and why it asks instead of guessing.

Version: 4

Minimum wall thickness

No wall thinner than two extrusion widths. The value lives in the printer profile and is never written into the operation as a number.

Chamfers instead of overhangs

Beyond 45 degrees of overhang, add a chamfer rather than accepting supports. The layer analysis says where the overhang is.

Tolerances from the material profile

Create fits as a pair and state the tolerance as auto:<material>. A fixed number does not survive a calibration.

Main dimensions as parameters

Every dimension the user might later change becomes a project parameter with a telling name. Stray numbers inside operations are a dead end.

Overlap for boolean operations

Always allow 0.01 mm of overlap for unions and differences. Coincident faces are the most common cause of broken results.

Holes larger than nominal

FDM prints holes tighter than planned. Enlarge the bore by the compensation value from the material profile, not by a guessed number.

First layer

Account for the elephant foot from the material profile whenever the first layer has to hold its dimension.

Parts before primitives

Search the parts library first, construct second. A part brings its features, its limits and its tests along.

Ask before guessing

Use ask_user for ambiguous requests. A question costs one click, a wrong assumption costs a print.

Materials, printers, standard parts

Materials

All dimensions in millimetres. “Clearance” is the gap a moving fit receives, “press fit” the interference of a fixed one. A profile that has been calibrated carries measured values instead of the defaults.

Material	Play	Press fit	Hole allowance	Elephant foot	Shrinkage	calibrated
ABS	0.25	-0.06	0.20	0.15	0.7 %	no
ASA	0.25	-0.06	0.20	0.15	0.6 %	no
PETG	0.25	-0.05	0.20	0.20	0.4 %	no
PETG-CF	0.25	-0.05	0.20	0.15	0.3 %	no
PLA	0.20	-0.05	0.15	0.15	0.2 %	no
TPU 95A	0.35	-0.10	0.30	0.25	0.2 %	no

Printer

The selected printer decides what fits on the bed and when a wall counts as too thin — two extrusion widths are the limit.

Printer	Build volume	nozzle	Layer height	Extrusion width	closed
Allgemeiner FDM-Drucker 220 mm	220 × 220 × 250	0.40	0.20	0.42	no
Anycubic Kobra 2	220 × 220 × 250	0.40	0.20	0.42	no
Bambu Lab A1	256 × 256 × 256	0.40	0.20	0.42	no
Bambu Lab A1 mini	180 × 180 × 180	0.40	0.20	0.42	no
Bambu Lab P1S	256 × 256 × 256	0.40	0.20	0.42	yes
Bambu Lab X1 Carbon	256 × 256 × 256	0.40	0.20	0.42	yes
Creality Ender-3 V3	220 × 220 × 250	0.40	0.20	0.42	no
Creality K1	220 × 220 × 250	0.40	0.20	0.42	yes
Creality K1 Max	300 × 300 × 300	0.40	0.20	0.42	yes
Elegoo Centauri Carbon 2	256 × 256 × 256	0.40	0.20	0.42	yes
Elegoo Neptune 4	225 × 225 × 265	0.40	0.20	0.42	no
Elegoo Neptune 4 Plus	320 × 320 × 385	0.40	0.20	0.42	no
Prusa MINI+	180 × 180 × 180	0.40	0.20	0.45	no
Prusa MK4S	250 × 210 × 220	0.40	0.20	0.45	no
Prusa XL	360 × 360 × 360	0.40	0.20	0.45	no
Sovol SV06	220 × 220 × 250	0.40	0.20	0.42	no

Standard parts

Where the dimensions come from when a part places a screw hole or a nut trap. None of it is estimated.

Size	Nominal	Clearance hole	Tapping hole	Head	Pitch
M2	2.0	2.4	1.60	3.8	0.40
M2.5	2.5	2.9	2.05	4.5	0.45
M3	3.0	3.4	2.50	5.5	0.50
M4	4.0	4.5	3.30	7.0	0.70
M5	5.0	5.5	4.20	8.5	0.80
M6	6.0	6.6	5.00	10.0	1.00
M8	8.0	9.0	6.80	13.0	1.25

Nuts, washers, heat-set inserts, magnets, bearings, extrusion profiles and tubes live in the same table; the parts look them up there.

The remote control tools

Every entry here is the same operation the menu offers, with the same parameters. What comes in becomes a transaction like any other: the history shows it, Ctrl+Z takes it back.

Scene

- **Arrange on the bed** (`arrange_bed`) — Lays all objects out side by side on the build plate.
- **Check overlaps** (`check_collisions`) — Reports overlaps and anything sticking out of the build volume.
- **Check the assembly path** (`check_join_path`) — Checks whether two parts can reach their position — not just whether they fit once there. The first selected part is pushed into the second.
- **Duplicate object** (`duplicate_object`) — Creates further copies of the object. They all stay separate, and the quantity sits in the stack as a number.
- **Remove object** (`delete_object`) — Takes an object out of the scene. The step is in the history, so an undo brings it back.
- **Rename object** (`rename_object`) — Gives an object a different name. The geometry is untouched.

Repair

- **Repair** (`repair`) — Closes holes, removes degenerate triangles and lines the faces up.

Transformation

- **Align to feature** (`align_to_feature`) — Brings a bore axis or a face into line with a second one.
- **Copies in a row or circle** (`pattern`) — Places copies in a row or on a circle — a hole pattern, a ring of screw bosses, a row of clips. One step with a number in it instead of nine identical steps in the stack.
- **Fit to size** (`fit_to_size`) — Scales an object so its longest edge matches the given size. For anything whose size you know but whose factor you would have to work out first.
- **Mirror** (`mirror_object`) — Mirrors an object about an axis. For the counterpart of a part — left and right bracket from the same construction.
- **Move** (`translate_object`) — Moves an object by the given millimetres.
- **Orient for printing** (`orient_for_print`) — Finds the orientation with the least support for each selected body. Every one gets its own — the best lie follows from the geometry of the single part.
- **Place on the bed** (`place_on_bed`) — Puts the underside of the object on the print bed.
- **Rotate** (`rotate_object`) — Rotates an object about an axis. The pivot decides around what: its own centre, the project origin, or the centre of the bed.
- **Scale** (`scale_object`) — Scales an object uniformly or per axis.

Basic shapes

- **Add a box** (`create_box`) — Adds a box, centred on the print bed or standing on a corner.
- **Add a box** (`create_brep_box`) — Creates a cuboid with real edges — chamfers and fillets can be set on them later.
- **Add a cylinder** (`create_brep_cylinder`) — Creates a cylinder with real edges, standing on the print bed — chamfers and fillets can be set on them later.

- **Add a cylinder** (`create_cylinder`) — Adds a cylinder standing on the print bed.
- **Add a sphere** (`create_sphere`) — Adds a sphere sitting on the print bed.
- **Create cone** (`create_cone`) — Creates a cone or truncated cone standing on the print bed.
- **Create ring** (`create_torus`) — Creates a closed circular ring resting on the print bed.
- **Create screw** (`thread_exact`) — A bolt with a true helical external thread and editable faces, supported by STEP export. Enlarged by clearance and subtracted from a body, it becomes the internal thread.

Combine and Subtract

- **Blend together** (`blend_union`) — Joins two bodies with a flowing transition instead of a sharp throat. For handles, struts and anything that must not tear at the inner corner once printed.
- **Intersection** (`intersect_objects`) — Keeps only the area shared by all selected objects. The object selected first keeps its name and material.
- **Subtract** (`subtract_objects`) — Subtracts every subsequently selected object from the first. Select the part to keep first, then everything to remove.
- **Unite** (`union_objects`) — Merges the selected objects into one. The object selected first keeps its name and material; the others become part of it.

Sketch

- **Cut pocket** (`sketch_pocket`) — Cuts a base shape as a pocket into a body — vertically down from the top edge, optionally all the way through. Clicking a face preselects the location. On an exact body, faces and edges are preserved; on an imported mesh, the result is a mesh.
- **Extrude base shape** (`sketch_extrude`) — Raises a base shape straight up into a body. The outline comes from a solved sketch and enters the kernel as an exact curve — a circle really is round.
- **Revolve a cross-section** (`sketch_revolve`) — Revolves a cross-section around the vertical axis: a rectangle becomes a sleeve, a circle a ring. The body stands on the print bed.
- **Span between two outlines** (`sketch_loft`) — Spans a body between the base shape and its scaled copy at height — truncated pyramid, truncated cone, funnel.
- **Sweep along a bend** (`sketch_sweep`) — Sweeps a cross-section along a bend: starting upright, tilting sideways with the bend radius — a pipe elbow in one step.

Shaping

- **Add a chamfer** (`chamfer_edges`) — Chamfers editable edges at 45 degrees.
- **Apply draft angle** (`draft_faces`) — Tilts all vertical, individually editable faces by an angle — for release from a mold or so a stackable container can stack.
- **Fillet** (`fillet_edges`) — Rounds an editable edge as a true curve instead of a sequence of straight segments.
- **Hollow out** (`shell_exact`) — Hollows a body with editable faces to the selected wall thickness and leaves the top open — a box from a cuboid in one step. Clear the option for closed hollowing with a vent.
- **Offset face** (`push_face`) — Grabs a face and moves it along its normal; the neighbouring walls follow. The way to change a wall without hunting for the operation that made it — an imported STEP has none.

Features

- **Change bore** (`resize_hole`) — Changes the diameter of a detected bore.
- **Change feature** (`resize_feature`) — Changes the diameter of a recognised feature: peg, countersink, taper, dome or socket.
- **Countersink** (`countersink_hole`) — Opens the mouth of a bore so a screw head sits flush.

- **Drill a bore** (`drill_brep_hole`) — Drills a round hole and keeps faces and edges individually editable — chamfers, fillets and STEP export remain possible afterwards.
- **Drill a bore** (`drill_hole`) — Drills a round hole, widened by the material tolerance if asked.
- **Duplicate feature** (`duplicate_feature`) — Creates a recognised feature a second time: hole, peg, countersink, taper, dome or socket.
- **Fill a bore** (`plug_hole`) — Fills a bore back in — for instance when a foreign part has one too many.
- **Move feature** (`move_feature`) — Moves a recognised feature to another place: hole, peg, countersink, taper, dome or socket.
- **Remove feature** (`remove_feature`) — Removes a recognised feature: hole, peg, countersink, taper, dome or socket.
- **Turn feature** (`rotate_feature`) — Tilts a recognised feature about its centre: hole, peg, countersink or taper.

Parts

- **Barrel hinge** (`insert_barrel_hinge`) — A hinge that comes off the printer already moving: two leaves around a printed-in pin, with the print gap between them. Nothing to assemble, nothing to insert.
- **Cable clip** (`insert_cable_clip`) — A clamp that sits on a face and holds a cable. The opening is narrower than the cable: you press it in, and it stays.
- **Cable gland with strain relief** (`insert_cable_gland`) — Gland for a round cable with a clamping point behind it. Without it every tug on the cable pulls straight at the solder joint.
- **Corner gusset** (`insert_gusset`) — A triangle in an inside corner: it holds two walls square where they would otherwise spread. The classic answer to a corner that flexes when you grip it.
- **Create lid** (`create_lid`) — Creates a matching lid with a collar for an opening. The cavity is cut out of the body rather than measured again — the clearance comes from the material profile.
- **Create screw lid** (`screw_lid`) — Puts a threaded neck on the opening and creates the matching screw lid. Both threads come from the same pitch, the clearance from the material profile.
- **Dowel pin and bore** (`insert_dowel`) — Pin or bore as a pair, for plugging two parts together. The play comes from the material profile, so a later calibration reaches old projects too.
- **Fit test body** (`create_fit_ladder`) — Pins and bores with staggered play. Print once, try, and the value is settled — it belongs in the material profile afterwards, not in the model.
- **Fit test body** (`insert_fit_ladder`) — Pins and bores with staggered play. Print once, try, and the value is settled — it belongs in the material profile afterwards, not in the model.
- **Foot** (`insert_foot`) — A foot under an enclosure — printed, or as a pocket for a bought rubber one. Four of them keep a device steady and its underside off the table.
- **Heat-set insert** (`insert_heatset_m4`) — Hole for a heat-set insert with a lead-in chamfer. The diameter is deliberately tight: the material is meant to be displaced while pressing.
- **Hinge eye** (`insert_hinge_eye`) — A lug with a hole that turns on a pin. Two of them on two parts, with a dowel between, make a hinge that lasts — unlike the living hinge, which only bends.
- **Insert ball bearing** (`insert_bearing_seat`) — Cuts a matching pocket for a ball bearing. The outside diameter and width come from the table; the fit comes from the selected material.
- **Keyhole hanger** (`insert_keyhole`) — A keyhole shaped recess: the head passes through the round end, the shank holds in the slot.
- **Latch** (`insert_latch`) — A nub for snapping in, with a ramp upwards and a straight holding face downwards — prints without support and holds against pull.
- **Living hinge** (`insert_living_hinge`) — Two leaves joined by a thin spot. The layers run across the bend, otherwise the hinge breaks the first time it is opened.
- **Magnet pocket** (`insert_magnet_pocket`) — Pocket for a round magnet, optionally with a cover layer to print over and a retaining lip at the rim.

- **Nut trap** (`insert_nut_trap`) — Pocket for a hexagon nut, pushed in from the side or laid in from below, optionally with a through screw hole.
- **Overhang fan** (`create_overhang_fan`) — Surfaces from steep to flat. Shows from which angle this printer with this material really needs support — instead of the 45 degree rule of thumb.
- **Overhang fan** (`insert_overhang_fan`) — Surfaces from steep to flat. Shows from which angle this printer with this material really needs support — instead of the 45 degree rule of thumb.
- **Pegboard hook** (`insert_pegboard_hook`) — Hooks for a pegboard, directly on the part. Pushed into the slots from above and pulled down — the nose then grips behind the board, and the springy tongue latches. Print the part lying flat: the tongue in the print plane, so it flexes along its layers instead of tearing across them.
- **Printable thread** (`insert_printed_thread`) — Printable internal thread in a hole or threaded stud on a flat surface — as a helix with a flattened crest. For an open container with a matching screw lid, ‘Create screw lid’ is the shared workflow.
- **Printed nut** (`insert_printed_nut`) — Printable hex nut with matching internal thread.
- **Screw** (`insert_printed_screw`) — Printable screw for a selected hole, with a hex head or an automatically flush countersunk head and matching external thread.
- **Screw hole with countersink** (`insert_screw_hole`) — Through hole for fastening with a metric screw, optionally with a 90 degree countersink and head clearance. Dimensions from the standard parts table.
- **Snap connector for a seam** (`insert_snap_connector`) — Spring arm and pocket as a pair — the halves click together instead of being glued. The arm is a tenth of the length thick; shorter than 8 mm there is none, below that the arm would be thinner than a nozzle lays down as load-bearing.
- **Snap fit** (`insert_snap_fit`) — A springing arm with a hook, for snapping two parts together. The arm is at least ten times as long as it is thick, otherwise it breaks instead of flexing.
- **Stiffening rib** (`insert_rib`) — A rib with a sloped run-out: it stiffens a wall without making it thicker. It stays thinner than the wall it sits on, otherwise it shows on the other side.
- **T-slot tongue for extrusion** (`insert_profile_tongue`) — A T-shaped foot that slides into the slot of an aluminium extrusion from the end and holds there. Neck and head come from the standard parts table. For printing, the tongue is best laid flat along the slot direction — standing upright, the shoulder under the head is an overhang.
- **Wall mount** (`insert_wall_mount`) — A back plate with screw holes and a shelf standing forward. The holes are through holes from the standard parts table.
- **Wall thickness ladder** (`create_wall_ladder`) — Walls from one to several extrusion widths. Shows where the printer still lays down material — the basis for the minimum wall thickness.
- **Wall thickness ladder** (`insert_wall_ladder`) — Walls from one to several extrusion widths. Shows where the printer still lays down material — the basis for the minimum wall thickness.

Split and Adjust

- **Compensate the elephant foot** (`compensate_first_layer`) — Pulls the first layers in by what they spread out while printing. The value comes from the material profile.
- **Create test piece** (`test_piece`) — Cuts a cube out around a spot, to print it and try it out — two minutes instead of two hours.
- **Hollow out** (`hollow_object`) — Hollows an object out and puts vents in. Saves material and time; the wall thickness holds within the raster.
- **Set material** (`set_material`) — Gives this body a material of its own. Tolerances, shrinkage and the elephant foot are then computed with it instead of the project's.
- **Split** (`split_pinned`) — Splits an object at a plane, with dowel pins in the cut face if wanted. The clearance comes from the material profile; zero pins means: only cut.

- **Split along a drawn line** (`split_line`) — Splits an object along a line drawn in the view and, if wanted, sets dowel pins into the cut face.
- **Split into separate parts** (`split_bodies`) — Turns a body made of several unconnected parts into one object each. What does not hang together is not one part — an STL does not know that, the geometry does.

Import

- **Extrude a drawing** (`load_outline`) — Reads an SVG or DXF drawing and gives it a height. Inner contours become holes.
- **Insert model** (`load`) — Reads a model file, converts it to millimetres and cleans it up. A 3MF assembly arrives as several objects rather than as one lump.
- **Load STEP** (`load_step`) — Reads a STEP file with individually editable faces and edges. STEP carries its own unit, so no unit question is needed.

Filament

- **Assign filament** (`assign_slot`) — Assigns a filament to the whole object. A part becomes multi-coloured by uniting bodies with different assignments.
- **Convert texture to filaments** (`slots_from_texture`) — Reduces the texture of an object to the number of filaments loaded and stores the result as filament assignments.
- **Filament on a face** (`paint_slot`) — Colours a recognised face completely in one filament. The boundary of the face comes from recognition — no brush, no radius.
- **Remove filament** (`clear_filament`) — Removes the filament assignment from a body or face. Other faces keep their filament; geometry remains unchanged.

Lettering

- **Lettering as a body** (`create_label`) — Creates lettering as an object of its own — for two-colour printing with two files, and for letters that get glued on.
- **Put text on** (`label_text`) — Puts text raised or engraved on a face. The font is processed as an outline, not as a picture — the edges stay clean at any size.

Surface

- **Apply relief** (`displace_image`) — Turns the brightness of an image into height on the surface. For grain, embossing and crests — everything a sketch is no good for.
- **Apply texture** (`apply_texture`) — Presses a pattern into a face as real geometry — knurling for grip, hexagon or rib for looks. What the slicer gets is what you see.
- **Fill with lattice** (`lattice_fill`) — Fills the cavity of a hollowed body with a lattice as real geometry. It travels in the 3MF and stays the same, whoever slices the part.

Smooth and simplify mesh

- **Close open surface** (`thicken`) — Gives an open surface a wall and turns it into a body. The way out for a mesh that arrives as a surface instead of a volume.
- **End face editing** (`brep_to_mesh`) — Turns individually editable faces into fixed triangles. Individual edges can no longer be chamfered or filleted afterwards; Undo restores the previous state.
- **Even out the triangles** (`remesh_uniform`) — Brings every triangle to roughly the same size — the coarse ones are split, the needlessly fine ones merged. The step before shaping by hand.

- **Reduce triangles** (`decimate_mesh`) — Reduces the triangle count. The largest deviation from the original surface is measured and reported.
- **Refine edges** (`remesh_mesh`) — Splits long edges until the net is even. The shape stays exact.
- **Sculpt** (`sculpt_strokes`) — Adds and removes material with the brush. The whole session is one step in the history and stays editable, however many strokes it holds.
- **Set a pose** (`pose_armature`) — Bends a body around an armature. A pose, not a motion — what gets printed is one state.
- **Smooth** (`smooth_mesh`) — Takes the roughness off a surface without shrinking the body. For generated meshes with stair steps.
- **Subdivide surface** (`subdivide_surface`) — Turns a faceted surface into a curved one: the new points are not placed into the facet but onto the curve it stands for. Sharp edges stay sharp.

Reading, parameters, undo

These tools appear in no menu — they are the information a remote end needs before it changes anything.

- **Undo a transaction** (`undo_transaction`) — Take a transaction back.
- **Create parameter** (`add_parameter`) — Add a main dimension as a project parameter instead of setting it as a number.
- **Change a parameter** (`set_parameter`) — Change the value of an existing project parameter.
- **Create a fit** (`add_fit`) — Add a fit. The tolerance comes from the material profile.
- **Read the report** (`read_report`) — Read the report, optionally only from one severity upwards.
- **Find a part** (`find_part`) — Look for a fitting part before building geometry yourself.
- **Read the digest** (`read_digest`) — Re-read the scene digest — with the features and IDs your previous steps have created.
- **Look up standard part sizes** (`read_standard`) — Look up standard part dimensions: tap hole, clearance hole, wrench size and more — instead of guessing them.
- **Read an analysis** (`read_analysis`) — Read an analysis: printability (overhang, islands, bridges), time and material estimate, settings advice or orientation search. Read-only, the source is always stated.
- **Switch printer and material** (`set_print_target`) — Switch the printer or material of the project. Tolerances are references into the material profile and convert along.

What does not go down the wire

Two operations are blocked, both for the same reason: a foreign program must not decide what gets executed or read on this machine.

- **Question to the user** (`ask_user`)

On top of that, every argument that looks like a file path is refused. Both stay usable in the window; only the way in from outside is closed.

Messages, word for word

What the application says when something does not work — word for word, so it can be looked up. An error never ends in “failed” here: every message comes with at least one way on.

Message	What helps
A value is outside the allowed range.	Correct the input, Cancel
An external program did not answer.	Optional programs …, Try again, Cancel
An unexpected error occurred in the application.	Create an error report, Show details, Cancel
None of the checked orientations fits the printable area. Choose another printer or split the model.	Split the model, Choose a different printer profile, Cancel
Solidon needs a licence key and one-time device activation for this (Help → Unlock Solidon …).	Enter licence key, Buy Solidon, Cancel
That is not unambiguous.	Select, Cancel
The 3D model generation delivered no model.	Cancel
The Solidon installation is damaged.	Open download page, Create an error report, Cancel
The activation file cannot be used.	Correct the input, Activate online, Activate offline …, Cancel
The active licence key cannot be overwritten.	Deactivate this computer, Cancel
The bodies could not be combined by any route.	Repair and try again, Compute more coarsely — dimensions get rounded, Show the places, Cancel
The connection to the activation service did not complete.	Try again, Activate offline …, Create an error report, Cancel
The deactivation of this computer has not yet been confirmed.	Deactivate this computer, Create an error report, Cancel
The device identity could not be stored securely.	Try again, Create an error report, Cancel
The feedback could not be sent.	Send again, Store the report, Send by e-mail yourself, Cancel
The file could not be written.	Try again, Choose another place, Correct the input, Cancel
The geometry did not allow the operation.	Repair and try again, Show the places, Cancel
The input could not be used as given.	Correct the input, Cancel
The language model did not answer.	Optional programs …, Try again, Cancel
The language model took too long.	Optional programs …, Try again, Cancel

Message	What helps
The language model's answer could not be read.	Optional programs …, Try again, Cancel
The model is open in several places.	Repair and try again, Show the places, Cancel
The object does not fit the build volume.	Split the model, Scale down to the build volume, Choose a different printer profile, Cancel
The support map takes too long for this model.	Reduce triangles, Cancel
The tools for editable faces and edges are unavailable.	Optional programs …, Cancel
The unit of the file could not be determined.	Correct the input, Cancel
This computer has not yet been activated for the licence key.	Activate online, Activate offline …, Cancel
This licence key cannot be used.	Correct the input, Buy Solidon
This model is too large for an analysis map.	Reduce triangles, Cancel
This tool needs individually editable faces and edges.	Cancel
Two constraints contradict each other.	Correct the input, Cancel

The line below it in the window names the reason for this particular case — which wall is too thin, which value is out of range, which file was meant.

Scene

Arrange on the bed (`arrange_bed`)

Lays all objects out side by side on the build plate.

Objects: 0 → … · reversible · deterministic · Shortcut `Ctrl+Shift+O`

Parameters	Unit	Default	Range	Meaning
Distance `spacing`	mm	5	0 … 100	Air between the parts. Enough that an elephant foot does not fuse two of them at the edge.
Build plates `plates`		12	1 … 12	What does not fit on one plate goes on the next.
Separate by filament `by_material`		off		Puts parts of different filaments on different plates. Two filaments on one plate cost a change and a purge for every shared layer.

Check overlaps (`check_collisions`)

Reports overlaps and anything sticking out of the build volume.

Objects: 0 → … · reversible · deterministic

Parameters	Unit	Default	Range	Meaning
Minimum clearance `clearance`	mm	0	0 … 50	When two parts count as too close. Zero reports real overlaps only; anything above also reports tight spots.

Check the assembly path (`check_join_path`)

Checks whether two parts can reach their position — not just whether they fit once there. The first selected part is pushed into the second.

When not to: Both parts stand in their final position; what is checked is the path leading there. An open body has no inside and cannot be measured this way.

Objects: 2 → 2 · reversible · deterministic

Parameters	Unit	Default	Range	Meaning
Direction `axis`		x	x, y, z	The axis along which the first part is pushed into the second.
Opposite direction `reverse`		off		Pushes against the axis direction, that is from the other side.
Assembly path `distance`	mm	25	0.1 … 500	How far ahead of the final position to check. As long as the section where the parts engage, plus some run-up.
Steps `steps`		24	2 … 200	How many positions the path is divided into. More finds tighter spots but costs one intersection of both bodies per step.

Duplicate object (`duplicate_object`)

Creates further copies of the object. They all stay separate, and the quantity sits in the stack as a number.

Objects: 1 → … · reversible · deterministic · Shortcut `Ctrl+D`

Parameters	Default	Range	Meaning
Count `count`	2	1 … 100	How many there are afterwards. Two means one copy — the quantity then lives in the stack instead of in the file name.
Name of the copy `name`			Leave empty to derive the name from the original.

Remove object (`delete_object`)

Takes an object out of the scene. The step is in the history, so an undo brings it back.

Objects: 1 → 0 · reversible · deterministic · Shortcut `Del`

Rename object (`rename_object`)

Gives an object a different name. The geometry is untouched.

Objects: 1 → 1 · reversible · deterministic · Shortcut `F2`

Parameters	Default	Meaning
Name `name`	required	New name of the object.

Repair

Repair (`repair`)

Closes holes, removes degenerate triangles and lines the faces up.

Objects: 1 → 1 · reversible · deterministic · Shortcut `Ctrl+Shift+R` · Applies to: Open edge

Parameters	Default	Meaning
Close open places `fill_holes`	on	Closes small holes. It cannot replace a missing wall.
Welding vertices `weld`	on	Merges points that practically coincide. The most common reason a mesh appears to consist of several parts.
Remove degenerate triangles `degenerate`	on	Triangles with no area. They disturb every later computation and carry nothing.
Unifying normals `normals`	on	Settles which side is outside. Without it faces appear dark or vanish.
Delete stray fragments `small_components`	off	Off by default: only what is explicitly asked for gets deleted.
Resolve self-intersections `self_intersections`	off	Recomputes faces that pass through each other. It helps with generated meshes and costs accuracy — hence off until it is needed.

Transformation

Align to feature (`align_to_feature`)

Brings a bore axis or a face into line with a second one.

Objects: 1 → 1 · reversible · deterministic · Applies to: Hole, peg, Face

Parameters	Default	Meaning
Feature `feature`		Bore or face on the moved object. A click in the window enters it.
Target `target`		Feature to align to, written as obj_2:hole_1.
The other way round `flip`	off	Turns the result by 180 degrees when the other side was meant.

Copies in a row or circle (`pattern`)

Places copies in a row or on a circle — a hole pattern, a ring of screw bosses, a row of clips. One step with a number in it instead of nine identical steps in the stack.

Objects: 1 → … · reversible · deterministic

Parameters	Unit	Default	Range	Meaning
Kind `kind`		linear	linear, circular	Linear lines the copies up along a direction, circular puts them around an axis. One operation, two kinds — not two entries.
Count `count`		4	1 … 100	How many copies there are afterwards, the original included. A single one is not a pattern.
Distance `spacing`	mm	20	0 …	Centre to centre, for the linear kind. The circular one counts the angle. Applies when Kind = linear.
Angle `angle`	°	360	-360 … 360	Over which arc the copies are spread. A full 360 degrees closes the ring without two copies landing on each other. Applies when Kind = circular.
Axis `axis`		z	x, y, z	Which axis the ring runs about. It passes through the origin. Applies when Kind = circular.
Direction X `dx`		1		Direction of the linear row. Without one every copy would sit on top of the last. Applies when Kind = linear.
Direction Y `dy`		0		Second axis of the direction — see Direction X. Applies when Kind = linear.
Direction Z `dz`		0		Third axis of the direction — see Direction X. Applies when Kind = linear.

Fit to size (`fit_to_size`)

Scales an object so its longest edge matches the given size. For anything whose size you know but whose factor you would have to work out first.

Objects: 1 → 1 · reversible · deterministic

Parameters	Unit	Default	Range	Meaning
Longest edge `largest`	mm	100	0.1 … 1000	The longest edge grows to this size; the others follow proportionally.
Reference `about`		centre	centre, origin, bed	Which point stays put while scaling.

Mirror (`mirror_object`)

Mirrors an object about an axis. For the counterpart of a part — left and right bracket from the same construction.

Objects: 1 → 1 · reversible · deterministic · Shortcut `Ctrl+M`

Parameters	Default	Range	Meaning
Axis `axis`	x	x, y, z	The axis to mirror about — the other hand of the same part.
Reference point `about`	centre	centre, origin, bed	Where the mirror plane lies: centre of mass, origin or footprint.

Move (`translate_object`)

Moves an object by the given millimetres.

Objects: 1 → 1 · reversible · deterministic · Shortcut `Ctrl+T`

Parameters	Unit	Default	Meaning
Offset X `dx`	mm	0	By how much it moves, not where to. Positive goes right.
Offset Y `dy`	mm	0	Positive goes backwards.
Offset Z `dz`	mm	0	Positive goes up. To set a part down there is *place on bed*.

Orient for printing (`orient_for_print`)

Finds the orientation with the least support for each selected body. Every one gets its own — the best lie follows from the geometry of the single part.

Objects: ≥ 1 → … · reversible · deterministic

Parameters	Unit	Default	Range	Meaning
Search thoroughly `thorough`		on		Checks orientations derived from the geometry and compares the best ones using layer analysis. Off uses a fast face-based heuristic.
Candidates `candidates`		200	8 … 2000	How many convex hull faces are checked as a base. The best orientations are then compared using layer analysis. Applies when Search thoroughly is ticked.
Arrange on the bed afterwards `arrange`		on		A body laid down needs more area than a standing one. Off means each stays where it stood — even when it then sits inside its neighbour.
Distance `spacing`	mm	5	0 … 100	Clearance between the parts when arranging. Applies when Arrange on the bed afterwards is ticked.
Build plates `plates`		12	1 … 12	What does not fit on one plate goes on the next. Applies when Arrange on the bed afterwards is ticked.

Place on the bed (`place_on_bed`)

Puts the underside of the object on the print bed.

Objects: 1 → 1 · reversible · deterministic · Shortcut `Ctrl+Shift+B`

Rotate (`rotate_object`)

Rotates an object about an axis. The pivot decides around what: its own centre, the project origin, or the centre of the bed.

Objects: 1 → 1 · reversible · deterministic · Shortcut `Ctrl+R`

Parameters	Unit	Default	Range	Meaning
Axis `axis`		z	x, y, z	Which axis to turn about. Z spins on the bed, X and Y tip the part.
Angle `angle`	°	90	-360 … 360	Angle counter-clockwise, seen from the tip of the axis.
Pivot `about`		centre	centre, origin, bed, point	Centroid of the object, world origin, footprint — or a named point for several bodies to turn around together.
Pivot X `pivot_x`	mm	0		Applies only when the pivot is “Named point”.
Pivot Y `pivot_y`	mm	0		Applies only when the pivot is “Named point”.
Pivot Z `pivot_z`	mm	0		Applies only when the pivot is “Named point”.

Scale (`scale_object`)

Scales an object uniformly or per axis.

Objects: 1 → 1 · reversible · deterministic

Parameters	Unit	Default	Range	Meaning
Factor `factor`		1	0.001 … 1000	Uniform scaling. Per-axis values are on the back side. When the target size is known, “Fit to size” is the shorter way.
Factor X `fx`		0	0 …	This axis only. Zero means the uniform factor above applies.
Factor Y `fy`		0	0 …	Zero means the uniform factor above applies.
Factor Z `fz`		0	0 …	Zero means the uniform factor above applies. Scaling per axis distorts holes — they turn oval.
Reference point `about`		centre	centre, origin, bed, point	The point that stays put: centroid, origin, footprint — or a named point, so several bodies grow together.
Reference point X `pivot_x`	mm	0		Applies only when the reference point is “Named point”.
Reference point Y `pivot_y`	mm	0		Applies only when the reference point is “Named point”.
Reference point Z `pivot_z`	mm	0		Applies only when the reference point is “Named point”.

Basic shapes

Add a box (`create_box`)

Adds a box, centred on the print bed or standing on a corner.

Objects: 0 → 1 · reversible · deterministic

Parameters	Unit	Default	Range	Meaning
Position X `x`	mm	0		Moves the body's existing reference point along X.
Position Y `y`	mm	0		Another axis of the position — see Position X.
Position Z `z`	mm	0		Another axis of the position — see Position X.
Normal X `nx`		0		Direction of the local Z axis. 0/0/0 keeps the existing upward orientation.
Normal Y `ny`		0		Another axis of the direction — see Normal X.
Normal Z `nz`		0		Another axis of the direction — see Normal X.
Rotation `angle`	°	0	-360 … 360	Turns the body around its own upright axis. The direction says where it points; this angle says how it is oriented while doing so.
Width `width`	mm	40	0.1 … 1000	Extent in X. May be an expression, for instance =@width*2.
Depth `depth`	mm	30	0.1 … 1000	Extent along Y.
Height `height`	mm	10	0.1 … 1000	Extent along Z, that is upwards.
Reference point `anchor`		centre	centre, corner	Centred on the origin or with its corner on it.
Name `name`				What the object is called in the tree. Empty means Solidon picks a name.

Add a box (`create_brep_box`)

Creates a cuboid with real edges — chamfers and fillets can be set on them later.

Objects: 0 → 1 · reversible · deterministic

Parameters	Unit	Default	Range	Meaning
Position X `x`	mm	0		Moves the body's existing reference point along X.
Position Y `y`	mm	0		Another axis of the position — see Position X.
Position Z `z`	mm	0		Another axis of the position — see Position X.
Normal X `nx`		0		Direction of the local Z axis. 0/0/0 keeps the existing upward orientation.
Normal Y `ny`		0		Another axis of the direction — see Normal X.
Normal Z `nz`		0		Another axis of the direction — see Normal X.
Rotation `angle`	°	0	-360 … 360	Turns the body around its own upright axis. The direction says where it points; this angle says how it is oriented while doing so.
Width `width`	mm	40	0.1 … 1000	Extent along X.
Depth `depth`	mm	30	0.1 … 1000	Extent along Y.
Height `height`	mm	20	0.1 … 1000	Extent along Z, that is upwards.
Name `name`				What the object is called in the tree. Empty means Solidon picks a name.

Add a cylinder (`create_brep_cylinder`)

Creates a cylinder with real edges, standing on the print bed — chamfers and fillets can be set on them later.

Objects: 0 → 1 · reversible · deterministic

Parameters	Unit	Default	Range	Meaning
Position X `x`	mm	0		Moves the body's existing reference point along X.
Position Y `y`	mm	0		Another axis of the position — see Position X.
Position Z `z`	mm	0		Another axis of the position — see Position X.
Normal X `nx`		0		Direction of the local Z axis. 0/0/0 keeps the existing upward orientation.
Normal Y `ny`		0		Another axis of the direction — see Normal X.
Normal Z `nz`		0		Another axis of the direction — see Normal X.
Rotation `angle`	°	0	-360 … 360	Turns the body around its own upright axis. The direction says where it points; this angle says how it is oriented while doing so.
Diameter `diameter`	mm	20	0.1 … 1000	Outer diameter. The circle stays truly round even when enlarged.
Height `height`	mm	20	0.1 … 1000	Height upwards, from the standing face.
Name `name`				What the object is called in the tree. Empty means Solidon picks a name.

Add a cylinder (`create_cylinder`)

Adds a cylinder standing on the print bed.

Objects: 0 → 1 · reversible · deterministic

Parameters	Unit	Default	Range	Meaning
Position X `x`	mm	0		Moves the body's existing reference point along X.
Position Y `y`	mm	0		Another axis of the position — see Position X.
Position Z `z`	mm	0		Another axis of the position — see Position X.
Normal X `nx`		0		Direction of the local Z axis. 0/0/0 keeps the existing upward orientation.
Normal Y `ny`		0		Another axis of the direction — see Normal X.
Normal Z `nz`		0		Another axis of the direction — see Normal X.
Rotation `angle`	°	0	-360 … 360	Turns the body around its own upright axis. The direction says where it points; this angle says how it is oriented while doing so.
Diameter `diameter`	mm	20	0.1 … 1000	Outer diameter. The cylinder stands on the print bed.
Height `height`	mm	20	0.1 … 1000	Height upwards, from the standing face.
Segments `segments`		48	8 … 256	More segments means rounder and slower.
Name `name`				What the object is called in the tree. Empty means Solidon picks a name.

Add a sphere (`create_sphere`)

Adds a sphere sitting on the print bed.

Objects: 0 → 1 · reversible · deterministic

Parameters	Unit	Default	Range	Meaning
Position X `x`	mm	0		Moves the body's existing reference point along X.
Position Y `y`	mm	0		Another axis of the position — see Position X.
Position Z `z`	mm	0		Another axis of the position — see Position X.
Normal X `nx`		0		Direction of the local Z axis. 0/0/0 keeps the existing upward orientation.
Normal Y `ny`		0		Another axis of the direction — see Normal X.
Normal Z `nz`		0		Another axis of the direction — see Normal X.
Rotation `angle`	°	0	-360 … 360	Turns the body around its own upright axis. The direction says where it points; this angle says how it is oriented while doing so.
Diameter `diameter`	mm	20	0.1 … 1000	Outer diameter. The sphere rests on the print bed.
Segments `segments`		32	8 … 128	More segments means rounder and slower — from 60 on, the sphere is as round as it gets.
Name `name`				What the object is called in the tree. Empty means Solidon picks a name.

Create cone (`create_cone`)

Creates a cone or truncated cone standing on the print bed.

Objects: 0 → 1 · reversible · deterministic

Parameters	Unit	Default	Range	Meaning
Position X `x`	mm	0		Moves the body's existing reference point along X.
Position Y `y`	mm	0		Another axis of the position — see Position X.
Position Z `z`	mm	0		Another axis of the position — see Position X.
Normal X `nx`		0		Direction of the local Z axis. 0/0/0 keeps the existing upward orientation.
Normal Y `ny`		0		Another axis of the direction — see Normal X.
Normal Z `nz`		0		Another axis of the direction — see Normal X.
Rotation `angle`	°	0	-360 … 360	Turns the body around its own upright axis. The direction says where it points; this angle says how it is oriented while doing so.
Bottom diameter `bottom_diameter`	mm	20	0 … 1000	Diameter at the print bed. Zero makes this end a point.
Top diameter `top_diameter`	mm	10	0 … 1000	Diameter at the top. Zero makes this end a point.
Height `height`	mm	20	0.1 … 1000	Height upwards, from the standing face.
Segments `segments`		48	8 … 256	More segments means rounder and slower.
Name `name`				What the object is called in the tree. Empty means Solidon picks a name.

Create ring (`create_torus`)

Creates a closed circular ring resting on the print bed.

Objects: 0 → 1 · reversible · deterministic

Parameters	Unit	Default	Range	Meaning
Position X `x`	mm	0		Moves the body's existing reference point along X.
Position Y `y`	mm	0		Another axis of the position — see Position X.
Position Z `z`	mm	0		Another axis of the position — see Position X.
Normal X `nx`		0		Direction of the local Z axis. 0/0/0 keeps the existing upward orientation.
Normal Y `ny`		0		Another axis of the direction — see Normal X.
Normal Z `nz`		0		Another axis of the direction — see Normal X.
Rotation `angle`	°	0	-360 … 360	Turns the body around its own upright axis. The direction says where it points; this angle says how it is oriented while doing so.
Outside diameter `outer_diameter`	mm	40	0.1 … 1000	Overall diameter from outer edge to outer edge.
Tube diameter `tube_diameter`	mm	8	0.1 … 500	Diameter of the ring's circular cross-section.
Segments `segments`		48	8 … 128	More segments make the ring and its cross-section rounder but slower to calculate.
Name `name`				What the object is called in the tree. Empty means Solidon picks a name.

Create screw (`thread_exact`)

A bolt with a true helical external thread and editable faces, supported by STEP export. Enlarged by clearance and subtracted from a body, it becomes the internal thread.

Objects: 0 → 1 · reversible · deterministic

Parameters	Unit	Default	Range	Meaning
Position X `x`	mm	0		Moves the body's existing reference point along X.
Position Y `y`	mm	0		Another axis of the position — see Position X.
Position Z `z`	mm	0		Another axis of the position — see Position X.
Normal X `nx`		0		Direction of the local Z axis. 0/0/0 keeps the existing upward orientation.
Normal Y `ny`		0		Another axis of the direction — see Normal X.
Normal Z `nz`		0		Another axis of the direction — see Normal X.
Rotation `angle`	°	0	-360 … 360	Turns the body around its own upright axis. The direction says where it points; this angle says how it is oriented while doing so.
Nominal diameter `diameter`	mm	10	2 … 100	Outer diameter across the thread crests — the dimension M10 refers to.
Pitch `pitch`	mm	1.5	0.25 … 8	Height gained per turn. Printed threads want a coarse pitch — below one millimetre hardly any printer prints cleanly.
Length `length`	mm	12	1 … 500	Thread length from the print bed upwards, at least two turns.
Name `name`				What the object is called in the tree. Empty means Solidon picks a name.

Combine and Subtract

Blend together (`blend_union`)

Joins two bodies with a flowing transition instead of a sharp throat. For handles, struts and anything that must not tear at the inner corner once printed.

When not to: Not on a part whose dimensions matter: the result is built on a grid, and sharp edges of the input bodies come out softer. Where a fit sits, the ordinary union belongs.

Objects: 2 → 1 · reversible · deterministic

Parameters	Unit	Default	Range	Meaning
Transition `radius`	mm	3	0 … 50	How wide the throat between the two bodies fills in. Zero gives the ordinary union. It bridges a gap from about three times its width.
Grid spacing `grid`	mm	1	0.05 … 10	How finely it is computed. Smaller means more accurate and markedly slower — edges below this spacing are lost.

Intersection (`intersect_objects`)

Keeps only the area shared by all selected objects. The object selected first keeps its name and material.

Objects: ≥ 2 → 1 · reversible · with a seed · Shortcut `Ctrl+Shift+X`

Subtract (`subtract_objects`)

Subtracts every subsequently selected object from the first. Select the part to keep first, then everything to remove.

Objects: ≥ 2 → 1 · reversible · with a seed · Shortcut `Ctrl+Shift+A`

Unite (`union_objects`)

Merges the selected objects into one. The object selected first keeps its name and material; the others become part of it.

Objects: ≥ 2 → 1 · reversible · with a seed · Shortcut `Ctrl+Shift+V`

Sketch

Cut pocket (`sketch_pocket`)

Cuts a base shape as a pocket into a body — vertically down from the top edge, optionally all the way through. Clicking a face preselects the location. On an exact body, faces and edges are preserved; on an imported mesh, the result is a mesh.

Objects: 1 → 1 · reversible · deterministic · Applies to: Face

Parameters	Unit	Default	Range	Meaning
Basic shape `shape`		rectangle	rectangle, slot, circle, polygon	Rectangle, slot, circle or polygon — the dimensions sit below.
Length `length`	mm	20	0.1 … 1000	Length in X. For the circle and the polygon this is the diameter, for the slot the overall length across the round ends.
Width `width`	mm	10	0.1 … 1000	Width in Y. Has no effect on the circle and the polygon. Applies when Basic shape = rectangle, slot.
Depth `depth`	mm	5	0.1 … 1000	How deep the pocket cuts down from its top edge. Applies when Through is not ticked.
Through `through`		off		Cuts through the whole height of the body — the depth does not count then.
X `x`	mm	0	-1000 … 1000	Centre of the pocket in the sketch plane, along its x direction. A clicked face fills in the place itself.
Y `y`	mm	0	-1000 … 1000	Centre of the pocket in the sketch plane, along its y direction. A clicked face fills in the place itself.
Top edge `z`	mm	0	-1000 … 1000	Where the pocket starts at the top, measured perpendicular to the sketch plane. Zero means: at the top of the body.
Corners `corners`		6	3 … 64	Number of corners, polygon only. Applies when Basic shape = polygon.
Region `region`		0	0 … 64	For a sketch with several separate outlines: which one. Zero means all of them — they are united into one body.
Sketch `sketch`				A drawn sketch instead of the base shape. Empty means: the base shape above applies.

Extrude base shape (`sketch_extrude`)

Raises a base shape straight up into a body. The outline comes from a solved sketch and enters the kernel as an exact curve — a circle really is round.

Objects: 0 → 1 · reversible · deterministic

Parameters	Unit	Default	Range	Meaning
Basic shape `shape`		rectangle	rectangle, slot, circle, polygon	Rectangle, slot, circle or polygon — the dimensions sit below.
Length `length`	mm	40	0.1 … 1000	Length in X. For the circle and the polygon this is the diameter, for the slot the overall length across the round ends.
Width `width`	mm	20	0.1 … 1000	Width in Y. Has no effect on the circle and the polygon. Applies when Basic shape = rectangle, slot.
Height `height`	mm	10	0.1 … 1000	How high the body is raised, from the print bed upwards.
Corners `corners`		6	3 … 64	Number of corners, polygon only. Applies when Basic shape = polygon.
Region `region`		0	0 … 64	For a sketch with several separate outlines: which one. Zero means all of them — they are united into one body.
Up to face `up_to`				Instead of the height: up to the level of this face. Empty means the height above applies. Clicking a face enters it here.
Name `name`				What the object is called in the tree. Empty means Solidon picks a name.
Sketch `sketch`				A drawn sketch instead of the base shape. Empty means: the base shape above applies.

Revolve a cross-section (`sketch_revolve`)

Revolves a cross-section around the vertical axis: a rectangle becomes a sleeve, a circle a ring. The body stands on the print bed.

Objects: 0 → 1 · reversible · deterministic

Parameters	Unit	Default	Range	Meaning
Basic shape `shape`		rectangle	rectangle, slot, circle, polygon	Rectangle, slot, circle or polygon — the dimensions sit below.
Length `length`	mm	5	0.1 … 1000	Extent of the cross-section away from the axis. For the circle and the polygon this is the diameter.
Width `width`	mm	8	0.1 … 1000	Height of the cross-section along the axis. No effect on the circle and the polygon. Applies when Basic shape = rectangle, slot.
Distance to the axis `offset`	mm	10	0 … 1000	Distance from the axis to the inner edge of the cross-section. Zero makes the body solid to the centre.
Angle `angle`	°	360	1 … 360	How far around the axis it turns. 360 closes the body.
Name `name`				What the object is called in the tree. Empty means Solidon picks a name.
Corners `corners`		6	3 … 64	Number of corners, polygon only. Applies when Basic shape = polygon.
Sketch `sketch`				A drawn sketch as the cross-section, used as drawn: x is the distance from the axis, y the height — the distance above does not apply then.

Span between two outlines (`sketch_loft`)

Spans a body between the base shape and its scaled copy at height — truncated pyramid, truncated cone, funnel.

Objects: 0 → 1 · reversible · deterministic

Parameters	Unit	Default	Range	Meaning
Basic shape `shape`		rectangle	rectangle, slot, circle, polygon	Rectangle, slot, circle or polygon — the dimensions sit below.
Length `length`	mm	40	0.1 … 1000	Length in X. For the circle and the polygon this is the diameter, for the slot the overall length across the round ends.
Width `width`	mm	20	0.1 … 1000	Width in Y. Has no effect on the circle and the polygon. Applies when Basic shape = rectangle, slot.
Height `height`	mm	20	0.1 … 1000	Distance between the lower and the upper outline.
Upper outline `top`		scaled	scaled, drawn	Where the upper outline comes from: computed from the lower one, or drawn separately. A truncated cone needs one number; a transition from round to square needs two outlines.
Taper `top_scale`		0.5	0.05 … 2	Size of the upper outline relative to the lower one. 0.5 halves it — a truncated pyramid or cone; above 1 it widens towards the top. Applies when Upper outline = scaled.
Upper drawing `top_sketch`				The second outline, drawn freely — on the same plane as the lower one, and spanned the given height above it. Applies when Upper outline = drawn.
Name `name`				What the object is called in the tree. Empty means Solidon picks a name.
Corners `corners`		6	3 … 64	Number of corners, polygon only. Applies when Basic shape = polygon.
Sketch `sketch`				A drawn sketch instead of the base shape. Empty means: the base shape above applies.

Sweep along a bend (`sketch_sweep`)

Sweeps a cross-section along a bend: starting upright, tilting sideways with the bend radius — a pipe elbow in one step.

Objects: 0 → 1 · reversible · deterministic

Parameters	Unit	Default	Range	Meaning
Basic shape `shape`		circle	rectangle, slot, circle, polygon	Rectangle, slot, circle or polygon — the dimensions sit below.
Length `length`	mm	10	0.1 … 1000	Length in X. For the circle and the polygon this is the diameter, for the slot the overall length across the round ends.
Width `width`	mm	10	0.1 … 1000	Width in Y. Has no effect on the circle and the polygon. Applies when Basic shape = rectangle, slot.
Path `along`		arc	arc, drawn	What the cross-section runs along: an arc from two numbers, or a drawn path. A pipe bend needs no drawing; a duct around two corners does.
Bend radius `bend_radius`	mm	20	0.1 … 1000	Radius of the path the cross-section follows. It must be larger than half the cross-section, or the inside kinks. Applies when Path = arc.
Bend angle `bend_angle`	°	90	1 … 180	How far the bend goes — 90 degrees is a right-angle pipe elbow. Applies when Path = arc.
Drawn path `path_sketch`				The course the cross-section follows — drawn open, on the front or side view. Its start moves to the origin: a path describes a course, not a place. Applies when Path = drawn.
Name `name`				What the object is called in the tree. Empty means Solidon picks a name.
Corners `corners`		6	3 … 64	Number of corners, polygon only. Applies when Basic shape = polygon.
Sketch `sketch`				A drawn sketch instead of the base shape. Empty means: the base shape above applies.

Shaping

Add a chamfer (`chamfer_edges`)

Chamfers editable edges at 45 degrees.

Objects: 1 → 1 · reversible · deterministic

Parameters	Unit	Default	Range	Meaning
Width `distance`	mm	1	0.01 … 100	How far the chamfer takes the edge back, on each of the two faces.
Edges `edges`		vertical	all, vertical, horizontal, top, bottom, named	Which edges are meant — upright, level, top, bottom, all, or picked one by one.
Individual edges `edge_keys`				The edges picked one by one. They hang on their place on the body, not on a number — a step before may change something else without the fillet wandering off. Applies when Edges = named.

Apply draft angle (`draft_faces`)

Tilts all vertical, individually editable faces by an angle — for release from a mold or so a stackable container can stack.

Objects: 1 → 1 · reversible · deterministic

Parameters	Unit	Default	Range	Meaning
Angle `angle`	°	2	0.1 … 30	By how many degrees the vertical faces are tilted. The base keeps its size; the body narrows towards the top.

Fillet (`fillet_edges`)

Rounds an editable edge as a true curve instead of a sequence of straight segments.

Objects: 1 → 1 · reversible · deterministic

Parameters	Unit	Default	Range	Meaning
Radius `radius`	mm	2	0.01 … 100	Radius of the fillet. Larger than the thinnest adjoining material will not work — the kernel then has no room left.
Edges `edges`		vertical	all, vertical, horizontal, top, bottom, named	Which edges are meant — upright, level, top, bottom, all, or picked one by one.
Individual edges `edge_keys`				The edges picked one by one. They hang on their place on the body, not on a number — a step before may change something else without the fillet wandering off. Applies when Edges = named.

Hollow out (`shell_exact`)

Hollows a body with editable faces to the selected wall thickness and leaves the top open — a box from a cuboid in one step. Clear the option for closed hollowing with a vent.

When not to: Not on parts that carry loads — a thin shell breaks differently from a solid body.

Objects: 1 → 1 · reversible · deterministic

Parameters	Unit	Default	Range	Meaning
Wall thickness `wall`	mm	2	0.2 … 50	How thick the remaining wall becomes. More than half the body is impossible — nothing would be left to hollow.

Offset face (`push_face`)

Grabs a face and moves it along its normal; the neighbouring walls follow. The way to change a wall without hunting for the operation that made it — an imported STEP has none.

Objects: 1 → 1 · reversible · deterministic · Applies to: Face

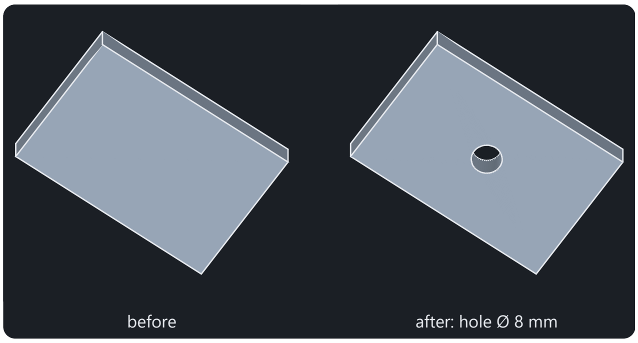
Parameters	Unit	Default	Meaning
Distance `distance`	mm	2	How far the face travels. Positive outwards, negative inwards — the same tool for both.
Direction X `nx`		0	Which faces move: those whose normal points this way. A clicked face fills the direction in by itself.
Direction Y `ny`		0	Second axis of the direction — see Direction X.
Direction Z `nz`		1	Third axis of the direction. Up by default.

Features

Change bore (`resize_hole`)

Changes the diameter of a detected bore.

Objects: 1 → 1 · reversible · with a seed · Applies to: Hole



Both pictures are computed by the same operation that sits in the menu.

Parameters	Unit	Default	Range	Meaning
Diameter `diameter`	mm	5	0.2 …	New finished diameter of the detected bore. Clicking it first shows its measured size here.
Hole `at_feature`		required		The detected bore whose diameter is being changed. Click the bore to enter it.
Account for the material tolerance `compensate`		off		Increases the selected finished dimension by the value from the material profile. When off, the measured dimension remains unchanged.

Change feature (`resize_feature`)

Changes the diameter of a recognised feature: peg, countersink, taper, dome or socket.

Objects: 1 → 1 · reversible · with a seed · Applies to: peg, Cone, sphere

Parameters	Unit	Default	Range	Meaning
Feature `at_feature`		required		The recognised feature whose dimension changes.
Diameter `diameter`	mm	8	0.5 …	The new diameter. When you click it, its measured one appears here.

Countersink (`countersink_hole`)

Opens the mouth of a bore so a screw head sits flush.

Objects: 1 → 1 · reversible · deterministic · Applies to: Hole, Cone

Parameters	Unit	Default	Range	Meaning
Head diameter `diameter`	mm	8.4	0.5 … 100	Diameter of the screw head, not of the hole below it. A clicked hole fills in the head of the matching screw — never its own measured size.
Angle `angle`	°	90	30 … 170	Full head angle — 90 degrees for metric countersunk screws.
Position X `x`	mm	0		Centre of the hole in the object's coordinates. A clicked face fills in all three values itself.
Position Y `y`	mm	0		Second axis of the position — see Position X.
Position Z `z`	mm	0		Third axis of the position — see Position X.
Axis `axis`		z	x, y, z	The direction to drill in. Z goes straight down from above, X and Y through a side wall.
Reference point `anchor`		mouth	mouth, centre	What the position means: the mouth of the bore being countersunk, or the spot itself. A clicked bore reports its centre — countersinking there creates a cavity instead of a chamfer.

Drill a bore (`drill_brep_hole`)

Drills a round hole and keeps faces and edges individually editable — chamfers, fillets and STEP export remain possible afterwards.

Objects: 1 → 1 · reversible · deterministic · Applies to: Face

Parameters	Unit	Default	Range	Meaning
Diameter `diameter`	mm	5	0.2 … 200	Nominal diameter of the hole. For a screw there is *screw hole* among the parts — its dimensions come from the standards table.
Position X `x`	mm	0		Centre of the hole in the object's coordinates. A clicked face fills in all three values itself.
Position Y `y`	mm	0		Second axis of the position — see Position X.
Position Z `z`	mm	0		Third axis of the position — see Position X.
Axis `axis`		z	x, y, z	The direction to drill in. Z goes straight down from above, X and Y through a side wall.
Normal X `nx`		0		Free direction from the selected surface. Zero in all three fields uses the axis.
Normal Y `ny`		0		Another axis of the direction — see Normal X.
Normal Z `nz`		0		Another axis of the direction — see Normal X.
Depth `depth`	mm	0	0 …	Zero drills right through the part.
Widening diameter `widening_diameter`	mm	0	0 …	Zero keeps the bore straight. A larger diameter creates space above its opening.
Widening depth `widening_depth`	mm	0	0 …	Depth of the wider straight section, measured from the clicked surface.
Transition angle `transition_angle`	°	90	5 … 180	Full angle between the bore and widened section. 180 degrees creates a flat shoulder.
Reference point `anchor`		mouth	mouth, centre	What the position means: the opening where the bore starts, or its centre. A widened section starts at the opening.

Parameters	Unit	Default	Range	Meaning
Account for the material tolerance `compensate`		on		Widens the bore by the value from the material profile.

Drill a bore (`drill_hole`)

Drills a round hole, widened by the material tolerance if asked.

Objects: 1 → 1 · reversible · with a seed · Shortcut `Ctrl+B` · Applies to: Face

Parameters	Unit	Default	Range	Meaning
Diameter `diameter`	mm	5	0.2 … 200	Nominal diameter of the hole. For a screw there is *screw hole* among the parts — its dimensions come from the standards table.
Position X `x`	mm	0		Centre of the hole in the object's coordinates. A clicked face fills in all three values itself.
Position Y `y`	mm	0		Second axis of the position — see Position X.
Position Z `z`	mm	0		Third axis of the position — see Position X.
Axis `axis`		z	x, y, z	The direction to drill in. Z goes straight down from above, X and Y through a side wall.
Normal X `nx`		0		Free direction from the selected surface. Zero in all three fields uses the axis.
Normal Y `ny`		0		Another axis of the direction — see Normal X.
Normal Z `nz`		0		Another axis of the direction — see Normal X.
Depth `depth`	mm	0	0 …	Zero drills right through the part.
Widening diameter `widening_diameter`	mm	0	0 …	Zero keeps the bore straight. A larger diameter creates space above its opening.
Widening depth `widening_depth`	mm	0	0 …	Depth of the wider straight section, measured from the clicked surface.
Transition angle `transition_angle`	°	90	5 … 180	Full angle between the bore and widened section. 180 degrees creates a flat shoulder.
Reference point `anchor`		mouth	mouth, centre	What the position means: the opening where the bore starts, or its centre. A widened section starts at the opening.

Parameters	Unit	Default	Range	Meaning
Account for the material tolerance `compensate`		on		Widens the bore by the value from the material profile.

Duplicate feature (`duplicate_feature`)

Creates a recognised feature a second time: hole, peg, countersink, taper, dome or socket.

Objects: 1 → 1 · reversible · with a seed · Applies to: Hole, peg, Cone, sphere

Parameters	Unit	Default	Range	Meaning
Normal X `nx`		0		Free direction at the new position. Zero in all three fields preserves the previous direction.
Normal Y `ny`		0		Another axis of the direction — see Normal X.
Normal Z `nz`		0		Another axis of the direction — see Normal X.
Feature `at_feature`		required		The recognised feature that is created a second time. Clicking it in the object tree or in the view selects it.
X `x`	mm	0	-1000 … 1000	The centre of the copy. On clicking, the old one appears here, offset by one diameter.
Y `y`	mm	0	-1000 … 1000	The centre of the copy. On clicking, the old one appears here, offset by one diameter.
Z `z`	mm	0	-1000 … 1000	The centre of the copy. On clicking, the old one appears here, offset by one diameter.

Fill a bore (`plug_hole`)

Fills a bore back in — for instance when a foreign part has one too many.

Objects: 1 → 1 · reversible · deterministic · Applies to: Hole

Parameters	Unit	Default	Range	Meaning
Hole `at_feature`				The recognised bore that is filled in. Clicking it enters it here; without it the values under “More” apply.
Diameter `diameter`	mm	5	0.2 … 200	Diameter of the hole being closed — slightly more does no harm.
Position X `x`	mm	0		Centre of the hole in the object's coordinates. A clicked face fills in all three values itself.
Position Y `y`	mm	0		Second axis of the position — see Position X.
Position Z `z`	mm	0		Third axis of the position — see Position X.
Axis `axis`		z	x, y, z	The direction to drill in. Z goes straight down from above, X and Y through a side wall.
Depth `depth`	mm	0	0 … 1000	Zero fills through the whole part.
Reference point `anchor`		mouth	mouth, centre	What the position means: the mouth where the plug starts, or its centre. For a through plug it makes no difference.
Account for the material tolerance `compensate`		on		Fills as far as *Drill hole* cuts with the same setting — otherwise the gap the bore was widened by stays open around the plug.

Move feature (`move_feature`)

Moves a recognised feature to another place: hole, peg, countersink, taper, dome or socket.

Objects: 1 → 1 · reversible · with a seed · Applies to: Hole, peg, Cone, sphere

Parameters	Unit	Default	Range	Meaning
Normal X `nx`		0		Free direction at the new position. Zero in all three fields preserves the previous direction.
Normal Y `ny`		0		Another axis of the direction — see Normal X.
Normal Z `nz`		0		Another axis of the direction — see Normal X.
Feature `at_feature`		required		The recognised feature to move. Click it in the object tree or in the view to select it.
X `x`	mm	0	-1000 … 1000	The new centre of the feature. When you click it, its current one appears here.
Y `y`	mm	0	-1000 … 1000	The new centre of the feature. When you click it, its current one appears here.
Z `z`	mm	0	-1000 … 1000	The new centre of the feature. When you click it, its current one appears here.

Remove feature (`remove_feature`)

Removes a recognised feature: hole, peg, countersink, taper, dome or socket.

Objects: 1 → 1 · reversible · with a seed · Applies to: Hole, peg, Cone, sphere

Parameters	Default	Range	Meaning
Feature `at_feature`	required		The recognised feature to remove.
Connected sections `sections`	ask	ask, chain, single	What happens to the remaining sections of a cavity — a countersink above the selected hole, for instance. With “Ask first” you decide as soon as the case arises.

Turn feature (`rotate_feature`)

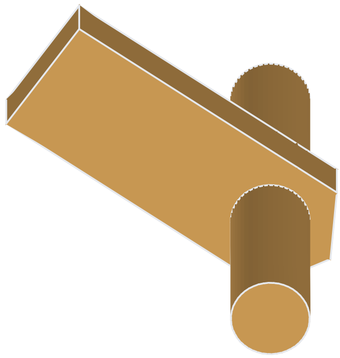
Tilts a recognised feature about its centre: hole, peg, countersink or taper.

Objects: 1 → 1 · reversible · with a seed · Applies to: Hole, peg, Cone

Parameters	Unit	Default	Range	Meaning
Feature `at_feature`		required		The recognised feature to turn.
Axis `axis`		x	x, y, z	Which axis to turn about. The turn happens about the centre of the feature.
Angle `angle`	°	90	-360 … 360	How far to turn, in degrees.

Parts

All parts share the same placement values. They are listed here once and are therefore missing from the tables below:



One click instead of half an hour — and the dimension comes from the table.

Parameters	Unit	Default	Range	Meaning
Position X `x`	mm	0		Where the part sits, measured in the object's coordinates. A clicked face fills the value in itself.
Position Y `y`	mm	0		Second axis of the position — see Position X.
Position Z `z`	mm	0		Height above the base of the object.
Normal direction X `nx`		0		Outward direction at the selected surface. Three zeros use the axis.
Normal direction Y `ny`		0		Second component of the surface direction — see Normal direction X.
Normal direction Z `nz`		0		Third component of the surface direction — see Normal direction X.
Axis `axis`		z	x, y, z	The direction the part points in — as long as no feature is chosen. A clicked face sets it itself.
Rotation `angle`	°	0	-360 … 360	Turns the part about its own axis. It matters for anything that is not round — a nut trap has to line up with the wall the nut slides in through.
At feature `at_feature`				Name of a recognised feature, for example hole_1. Its place then counts, and the position above is taken as an offset.
At several features `at_features`		()		Several recognised features at once. The part is placed at each of them — one step in the history, one Ctrl+Z. Empty means the single feature above applies.

Barrel hinge (`insert_barrel_hinge`)

A hinge that comes off the printer already moving: two leaves around a printed-in pin, with the print gap between them. Nothing to assemble, nothing to insert.

When not to: The gap decides: too tight welds during printing, too wide rattles. It comes from the calibrated material — change the material and print a test piece before printing twenty hinges.

Objects: 1 → 1 · reversible · deterministic · Applies to: Face

Parameters	Unit	Default	Range	Meaning
Pin `pin`	mm	4	2 … 20	Diameter of the axle. It is printed along and not inserted.
Width `width`	mm	24	8 … 120	Total width across both leaves, measured along the axle.
Reach `reach`	mm	12	4 … 60	How far each leaf reaches from the axle — the lever of the joint.
Wall thickness `wall`	mm	2.5	1 … 15	Material around the pin. Too thin tears at the first pull.
Play `play`	mm	0	0 … 2	Zero means: the value from the calibrated material profile.

Cable clip (`insert_cable_clip`)

A clamp that sits on a face and holds a cable. The opening is narrower than the cable: you press it in, and it stays.

When not to: Not for cables under tension — the cable gland with strain relief is there for that. A clip guides, it does not anchor.

Objects: 1 → 1 · reversible · deterministic · Applies to: Face

Parameters	Unit	Default	Range	Meaning
Cable `size`		cable-5	ptfe-4x2, ptfe-6x3, cable-5, cable-7	What the clip should hold — cable or tube from the table.
Custom diameter `diameter`	mm	0	0 … 100	Only fill this in if the matching size is not listed. Zero uses the selected size.
Width `width`	mm	8	2 … 40	How wide the clamp is, measured along the cable.
Wall thickness `wall`	mm	2	0.8 … 10	Thickness of the clamp. Too thin and it springs open, too thick and it will not spring at all.
Narrowing `grip`	mm	0	0 … 5	How much narrower the opening is than the cable, per side — that is what makes the clip hold. Zero means a fifth of the diameter.
Play `play`	mm	0	0 … 2	Zero means: the value from the calibrated material profile.

Cable gland with strain relief (`insert_cable_gland`)

Gland for a round cable with a clamping point behind it. Without it every tug on the cable pulls straight at the solder joint.

Objects: 1 → 1 · reversible · deterministic · Applies to: Face

Parameters	Unit	Default	Range	Meaning
Cable `size`		cable-5	ptfe-4x2, ptfe-6x3, cable-5, cable-7	Outer diameter of the cable — the number printed on the jacket.
Custom diameter `diameter`	mm	0	0 … 100	Only fill this in if the matching size is not listed. Zero uses the selected size.
Wall thickness `wall`	mm	3	1 … 30	Thickness of the wall the gland passes through.
Play `play`	mm	0	0 … 2	Zero means: the value from the calibrated material profile.
Strain relief `strain_relief`		on		Two ribs clamping the cable so nothing pulls at the solder joint.
Clamping gap `relief_gap`	mm	0	0 … 20	Zero means: four fifths of the cable diameter.

Corner gusset (`insert_gusset`)

A triangle in an inside corner: it holds two walls square where they would otherwise spread. The classic answer to a corner that flexes when you grip it.

When not to: Not for a corner something has to pass through — it fills it diagonally. For a wall that is too soft on its own, use the stiffening rib.

Objects: 1 → 1 · reversible · deterministic · Applies to: Face

Parameters	Unit	Default	Range	Meaning
Legs `legs`	mm	12	2 … 100	How far the gusset reaches along both walls.
Thickness `thickness`	mm	0	0 … 20	How thick the gusset is, measured along the edge. Zero means as thick as the rib would be — two thirds of the wall.
Wall thickness `wall`	mm	2	0.4 … 20	The wall it sits against — it sets the default for the thickness.

Create lid (`create_lid`)

Creates a matching lid with a collar for an opening. The cavity is cut out of the body rather than measured again — the clearance comes from the material profile.

Objects: 1 → 2 · reversible · deterministic · Applies to: Face

Parameters	Unit	Default	Range	Meaning
Lid thickness `thickness`	mm	2.4	0.4 … 50	How thick the lid plate becomes — the collar below it comes from the cavity.
Collar depth `collar`	mm	4	0 … 100	How far the collar reaches into the opening. Zero means a flat lid without one.
Play `clearance`	mm	0	0 … 2	Zero means the value from the material profile.
Name `name`				What the object is called in the tree. Empty means Solidon picks a name.

Create screw lid (`screw_lid`)

Puts a threaded neck on the opening and creates the matching screw lid. Both threads come from the same pitch, the clearance from the material profile.

Objects: 1 → 2 · reversible · deterministic · Applies to: Face

Parameters	Unit	Default	Range	Meaning
Thread height `height`	mm	8	2 … 100	How tall the threaded neck becomes. Two turns hold, three sit tight.
Pitch `pitch`	mm	3	1 … 10	Coarse, so the printer resolves it and half a turn closes it.
Lid thickness `thickness`	mm	2.4	0.8 … 50	Thickness of the lid plate above the thread.
Wall thickness of the lid `wall`	mm	2.4	0.8 … 20	Thickness of the rim carrying the thread. Too thin splits along the layers when screwed on.
Neck diameter `neck`	mm	0	0 … 400	Zero takes the narrower side of the opening.
Play `clearance`	mm	0	0 … 2	Zero means the value from the material profile.

Dowel pin and bore (`insert_dowel`)

Pin or bore as a pair, for plugging two parts together. The play comes from the material profile, so a later calibration reaches old projects too.

Objects: 1 → 1 · reversible · deterministic · Applies to: Face

Parameters	Unit	Default	Range	Meaning
Diameter `diameter`	mm	4	1 … 30	Nominal diameter. Pin and hole carry the same value — the clearance between them comes from the material profile.
Length `length`	mm	8	1 … 100	How far the pin stands out, or how deep the hole goes.
Kind `kind`		pin	pin, bore	The pin itself or the bore for it.
Shape `shape`		round	round, hex, dovetail	The cross-section. Round is the simplest and needs two of them against twisting; hexagon and dovetail hold on their own, the dovetail also against being pulled apart.
Chamfer `chamfer`	mm	0.6	0 … 3	Makes joining easier and keeps the first layer to size.
Play `play`	mm	0	0 … 1	Zero means: the value from the calibrated material profile.

Fit test body (`create_fit_ladder`)

Pins and bores with staggered play. Print once, try, and the value is settled — it belongs in the material profile afterwards, not in the model.

Objects: 0 → 1 · reversible · deterministic

Parameters	Unit	Default	Range	Meaning
Nominal diameter `diameter`	mm	6	2 … 30	Diameter of pin and hole. Best the one the part will really use — clearance does not behave the same across all sizes.
Steps `steps`		4	2 … 8	How many pairs get printed. Four is usually enough to bracket the value.
Smallest play `first`	mm	0.1	0 … 1	Where the ladder starts. The first step may well be too tight.
Step `step`	mm	0.05	0.01 … 0.5	By how much the clearance grows from step to step.
Height `height`	mm	8	2 … 40	Height of the pins. Taller takes longer to print but mates more honestly.

Fit test body (`insert_fit_ladder`)

Pins and bores with staggered play. Print once, try, and the value is settled — it belongs in the material profile afterwards, not in the model.

Objects: 1 → 1 · reversible · deterministic

Parameters	Unit	Default	Range	Meaning
Nominal diameter `diameter`	mm	6	2 … 30	Diameter of pin and hole. Best the one the part will really use — clearance does not behave the same across all sizes.
Steps `steps`		4	2 … 8	How many pairs get printed. Four is usually enough to bracket the value.
Smallest play `first`	mm	0.1	0 … 1	Where the ladder starts. The first step may well be too tight.
Step `step`	mm	0.05	0.01 … 0.5	By how much the clearance grows from step to step.
Height `height`	mm	8	2 … 40	Height of the pins. Taller takes longer to print but mates more honestly.

Foot (`insert_foot`)

A foot under an enclosure — printed, or as a pocket for a bought rubber one. Four of them keep a device steady and its underside off the table.

When not to: Not for parts meant to rest flat on the surface — a foot lifts them off. And not meant to be screwed down: it has no hole, and one drilled into it would leave the screw standing on the table.

Objects: 1 → 1 · reversible · deterministic · Applies to: Face

Parameters	Unit	Default	Range	Meaning
Kind `kind`		foot	foot, pocket	A printed foot, or the pocket for a bought rubber one.
Diameter `diameter`	mm	10	3 … 60	How wide the foot stands. Wider tips later.
Height `height`	mm	3	0.6 … 30	How high it lifts — for the pocket: how deep the rubber foot sinks in.
Chamfer `chamfer`	mm	0	0 … 10	Bevel at the edge. For the foot, zero means a fifth of the height — enough that the first layer does not stand out as a ridge. For the pocket it is a lead-in chamfer.
Play `play`	mm	0	0 … 2	Zero means: the value from the calibrated material profile.

Heat-set insert (`insert_heatset_m4`)

Hole for a heat-set insert with a lead-in chamfer. The diameter is deliberately tight: the material is meant to be displaced while pressing.

When not to: Not without a soldering iron: the insert is pressed in warm, and the diameter is kept tight for that. Pushed in cold it splits the wall — without a soldering iron a nut trap carries about as much, and a plain screw hole is enough where the screw may pass through the part.

Objects: 1 → 1 · reversible · deterministic · Applies to: Hole, Face

Parameters	Unit	Default	Range	Meaning
Size `size`		M3	M2, M2.5, M3, M3S, M4, M4S, M5, M5S, M6	Thread of the insert. The bore diameter for it comes from the standards table.
Lead-in chamfer `lead_in`		on		A chamfer at the rim so the insert goes in straight.
Extra depth `extra_depth`	mm	0.5	0 … 5	Room under the insert for displaced material.

Hinge eye (`insert_hinge_eye`)

A lug with a hole that turns on a pin. Two of them on two parts, with a dowel between, make a hinge that lasts — unlike the living hinge, which only bends.

When not to: Half a hinge: the second eye belongs on the counterpart, and the pin comes from the library (dowel) or from the shop.

Objects: 1 → 1 · reversible · deterministic · Applies to: Face

Parameters	Unit	Default	Range	Meaning
Pin `pin`	mm	3	1 … 20	Diameter of the pin that goes through — a dowel or a screw.
Width `width`	mm	8	2 … 60	How wide the eye is, measured along the axis of rotation.
Distance `reach`	mm	8	1 … 60	How far the axis stands off the face — that is the pivot.
Wall thickness `wall`	mm	2	0.8 … 15	Material all around the hole. Too thin tears open on the first pull.
Play `play`	mm	0	0 … 2	Zero means: the value from the calibrated material profile.

Insert ball bearing (`insert_bearing_seat`)

Cuts a matching pocket for a ball bearing. The outside diameter and width come from the table; the fit comes from the selected material.

When not to: For a through shaft, first make a hole and place the bearing seat on it.

Objects: 1 → 1 · reversible · deterministic · Applies to: Hole, Face

Parameters	Unit	Default	Range	Meaning
Ball bearing `size`		608	608, 623, 624, 625, 626, 6800, 6000, 6001	The number is printed on the bearing. Next to it, Solidon shows the bore, outside diameter and width.
Removable `removable`		off		On means the bearing can be replaced. Off means it is pressed firmly into place.
Extra depth `extra_depth`	mm	0	0 … 5	Extra space behind the bearing.
Play `play`	mm	0	0 … 2	Zero means: the value from the calibrated material profile. Applies when Removable is ticked.
Oversize `grip`	mm	0	0 … 2	Zero means: the value from the calibrated material profile. Applies when Removable is not ticked.

Keyhole hanger (`insert_keyhole`)

A keyhole shaped recess: the head passes through the round end, the shank holds in the slot.

Objects: 1 → 1 · reversible · deterministic · Applies to: Face

Parameters	Unit	Default	Range	Meaning
Screw `size`		M4	M2, M2.5, M3, M4, M5, M6, M8	The screw in the wall. Its head passes through the round end, its shank holds in the slot.
Drop `drop`	mm	8	2 … 60	How far the part drops after it is hung up.
Depth `depth`	mm	4	1 … 40	How deep the recess goes into the material.
Head depth `head_room`	mm	2.5	0.5 … 20	How deep the screw head sinks in.
Play `play`	mm	0	0 … 2	Zero means: the value from the calibrated material profile.

Latch (`insert_latch`)

A nub for snapping in, with a ramp upwards and a straight holding face downwards — prints without support and holds against pull.

Objects: 1 → 1 · reversible · deterministic · Applies to: Face

Parameters	Unit	Default	Range	Meaning
Width `width`	mm	6	2 … 60	Width of the latch along the edge.
Overhang `depth`	mm	1	0.4 … 6	How far the latch protrudes. That is the travel the counterpart has to spring over.
Height `height`	mm	3	1 … 30	Height of the latch. The lead-in slope is on top, the flat holding face below.
As a recess `negative`		off		The other side: the same shape, but with play and to be subtracted.
Play `play`	mm	0	0 … 1	Zero means: the value from the calibrated material profile.

Living hinge (`insert_living_hinge`)

Two leaves joined by a thin spot. The layers run across the bend, otherwise the hinge breaks the first time it is opened.

When not to: Not for a part printed upright: the thin spot only holds while the layers run across the bend. If the hinge stands vertically on the bed, it breaks the first time it opens.

Objects: 1 → 1 · reversible · deterministic · Applies to: Face

Parameters	Unit	Default	Range	Meaning
Width `width`	mm	30	5 … 300	Length of the hinge axis — that is how wide the lid hanging on it becomes.
Leaf length `leaf`	mm	15	3 … 200	How far each of the two leaves reaches away from the hinge.
Material thickness `thickness`	mm	2	0.8 … 10	Thickness of the two leaves — not of the thin section between them.
Hinge thickness `film`	mm	0.4	0.2 … 1.2	Thinnest spot. Below 0.3 mm PLA tears; PETG takes more.
Hinge width `gap`	mm	1.5	0.5 … 8	Width of the thin section. Too narrow folds along one line and breaks, too wide lets the lid wobble.

Magnet pocket (`insert_magnet_pocket`)

Pocket for a round magnet, optionally with a cover layer to print over and a retaining lip at the rim.

Objects: 1 → 1 · reversible · deterministic · Applies to: Face

Parameters	Unit	Default	Range	Meaning
Magnet `size`		8x3	6x3, 8x3, 10x3, 12x2, 20x3	Diameter by height of the round magnet, as it is sold.
Custom diameter `diameter`	mm	0	0 … 100	Only fill this in if the matching size is not listed. Zero uses the selected size.
Custom height `height`	mm	0	0 … 30	Only fill this in if the matching size is not listed. Zero uses the selected size.
Play `play`	mm	0	0 … 1	Zero means: the value from the calibrated material profile.
Cover layer `cover`	mm	0	0 … 5	Material above the magnet. Zero leaves the pocket open.
Retaining lip `press_lip`		on		A tenth of a millimetre narrower at the rim so the magnet cannot fall out.
Oversize `grip`	mm	0	0 … 1	Zero means: the value from the calibrated material profile.

Nut trap (`insert_nut_trap`)

Pocket for a hexagon nut, pushed in from the side or laid in from below, optionally with a through screw hole.

When not to: Only with a matching hexagon nut: the pocket is built to the width across flats from the standard-parts table, and any other nut wobbles in it or will not go in. And it has to stay reachable — slid in from the side it needs a free flank, laid in from below an opening that no later step builds over. Where no nut is at hand, a printed thread carries light loads; for load-bearing joints a heat-set insert is the right choice.

Objects: 1 → 1 · reversible · deterministic · Applies to: Hole, Face

Parameters	Unit	Default	Range	Meaning
Size `size`		M3	M2, M2.5, M3, M4, M5, M6, M8	Thread of the nut. Width across flats and height come from the standards table.
Direction `direction`		side	side, bottom	Pushed in from the side or laid in from below.
Slide-in length `slide`	mm	12	0 … 100	How far the slot reaches outwards. Zero means: the pocket only.
Play `play`	mm	0	0 … 1	Zero means: the value from the calibrated material profile.
Cut the screw hole as well `screw_hole`		on		Also cuts the through-hole for the screw across the part.

Overhang fan (`create_overhang_fan`)

Surfaces from steep to flat. Shows from which angle this printer with this material really needs support — instead of the 45 degree rule of thumb.

Objects: 0 → 1 · reversible · deterministic

Parameters	Unit	Default	Range	Meaning
Smallest angle `first`	°	20	5 … 80	The steepest face, measured against vertical. Smaller means steeper.
Step `step`	°	10	2 … 30	By how many degrees each face gets shallower than the one before.
Steps `steps`		6	2 … 10	How many angles get tested.
Width per step `width`	mm	8	2 …	Width of a single face. Narrower saves time, wider shows more.
Overhang length `length`	mm	15	3 …	How far each face reaches out unsupported. Too short and the printer forgives everything.

Overhang fan (`insert_overhang_fan`)

Surfaces from steep to flat. Shows from which angle this printer with this material really needs support — instead of the 45 degree rule of thumb.

Objects: 1 → 1 · reversible · deterministic

Parameters	Unit	Default	Range	Meaning
Smallest angle `first`	°	20	5 … 80	The steepest face, measured against vertical. Smaller means steeper.
Step `step`	°	10	2 … 30	By how many degrees each face gets shallower than the one before.
Steps `steps`		6	2 … 10	How many angles get tested.
Width per step `width`	mm	8	2 …	Width of a single face. Narrower saves time, wider shows more.
Overhang length `length`	mm	15	3 …	How far each face reaches out unsupported. Too short and the printer forgives everything.

Pegboard hook (`insert_pegboard_hook`)

Hooks for a pegboard, directly on the part. Pushed into the slots from above and pulled down — the nose then grips behind the board, and the springy tongue latches. Print the part lying flat: the tongue in the print plane, so it flexes along its layers instead of tearing across them.

When not to: To take the part off, the latch tongue must be pressed down through the slot — without a tool that only works standing in front of the board. If a part comes off often, switch the tongue off; the hook then releases the same way it was hung. Slot dimensions and grid are checked against a dimensioned drawing; the 5 mm board thickness was measured on 28 August 2026 — the depth of the nose and tongue follows from it.

Objects: 1 → 1 · reversible · deterministic · Applies to: Face

Parameters	Unit	Default	Range	Meaning
Pegboard `system`		skadis	skadis	Which board the part goes on. The dimensions come from the table.
Hooks `count`		2	1 … 6	How many hooks. Two side by side keep a part from twisting; one is enough for something light.
Grid steps `steps`		1	1 … 4	How many holes lie between two hooks. One means every hole, two means every second. Wide parts tip between two hooks only forty millimetres apart — further out they carry more steadily.
Stacked `upright`		off		Places the hooks vertically instead of side by side — for narrow parts that would otherwise stick out past the grid.
Latch tongue `latch`		on		A springy tongue on the peg that latches behind the board: the part stays hung even if someone lifts it while taking things off. To release, press it down through the slot. Switched off, the hook is the plain form that lifts off the board.
Back plate `plate`	mm	0	0 … 10	Zero means none. The part the hooks sit on already joins them — a plate in between would be material nobody needs. If you want one anyway, it sits **inside** the part, not on top of it.
Play `play`	mm	0	0 … 1.5	Zero means: the value from the calibrated material profile.
Lip depth `lip`	mm	0	0 … 6	How far the nose reaches behind the pegboard. Zero means two thirds of the pegboard thickness.

Printable thread (`insert_printed_thread`)

Printable internal thread in a hole or threaded stud on a flat surface — as a helix with a flattened crest. For an open container with a matching screw lid, ‘Create screw lid’ is the shared workflow.

When not to: Not where a metal screw has to bite: the crest is flattened so a printer can resolve it at all — a standard counterpart will not grip cleanly. For load-bearing joints a heat-set insert is the right choice, and where no soldering iron is at hand, a nut trap.

Objects: 1 → 1 · reversible · deterministic · Applies to: Hole, Face

Parameters	Unit	Default	Range	Meaning
Size `size`		M6	M2, M2.5, M3, M4, M5, M6, M8	Nominal diameter and pitch. The profile is flattened to be printable, not an ISO profile — a printer would not resolve that anyway.
Length `length`	mm	12	2 … 200	Length of the thread, not of the bolt.
Internal thread (off = external thread) `internal`		off		Enable for an internal thread in a hole; disable for a threaded stud on a flat outer surface. For a container with a matching screw lid, use 'Create screw lid'.
Play `play`	mm	0	0 … 1	Zero means: the value from the calibrated material profile.

Printed nut (`insert_printed_nut`)

Printable hex nut with matching internal thread.

When not to: Print it together with the printed screw in the same material: clearance comes from the material profile. For high loads or frequent removal, metal screws with a nut trap or heat-set insert are more reliable.

Objects: 1 → 1 · reversible · deterministic · Applies to: Face

Parameters	Unit	Default	Range	Meaning
Size `size`		M5	M2, M2.5, M3, M4, M5, M6, M8	Nominal diameter and pitch of the matching printed thread.
Play `play`	mm	0	0 … 1	Zero means: the value from the calibrated material profile.

Screw (`insert_printed_screw`)

Printable screw for a selected hole, with a hex head or an automatically flush countersunk head and matching external thread.

When not to: Print it together with the printed nut in the same material: clearance comes from the material profile. For high loads or frequent removal, metal screws with a nut trap or heat-set insert are more reliable.

Objects: 1 → 1 · reversible · deterministic · Applies to: Hole

Parameters	Unit	Default	Range	Meaning
Size `size`		M5	M2, M2.5, M3, M4, M5, M6, M8	Nominal diameter and pitch of the matching printed thread.
Length `length`	mm	12	2 … 200	Length of the thread below the screw head.
Countersunk head `countersunk`		off		Creates a 90-degree countersunk head and countersinks the selected hole to match in the same undoable step.
Play `play`	mm	0	0 … 1	Zero means: the value from the calibrated material profile.

Screw hole with countersink (`insert_screw_hole`)

Through hole for fastening with a metric screw, optionally with a 90 degree countersink and head clearance. Dimensions from the standard parts table.

Objects: 1 → 1 · reversible · deterministic · Applies to: Face

Parameters	Unit	Default	Range	Meaning
Size `size`		M3	M2, M2.5, M3, M4, M5, M6, M8	Thread size of the screw. Every dimension comes from the standards table.
Depth `depth`	mm	10	1 … 200	How deep to drill. More than the wall thickness gives a through-hole.
Countersunk head `countersink`		on		A 90-degree countersink for a flat head. Off for a socket-head cap screw.
Recess washer `washer`		off		Cuts a matching seat for the washer. Applies when Countersunk head is not ticked.
Play `play`	mm	0	0 … 2	Zero means: the value from the calibrated material profile. Applies when Recess washer is ticked. Applies when Countersunk head is not ticked.
Head depth `head_room`	mm	0	0 … 50	How deeply the screw head and washer sink into the part.

Snap connector for a seam (`insert_snap_connector`)

Spring arm and pocket as a pair — the halves click together instead of being glued. The arm is a tenth of the length thick; shorter than 8 mm there is none, below that the arm would be thinner than a nozzle lays down as load-bearing.

When not to: Not for short seams: the arm is a tenth of its length thick, and below eight millimetres a nozzle no longer lays it down strongly enough. For small parts a latch nub holds better.

Objects: 1 → 1 · reversible · deterministic · Applies to: Face

Parameters	Unit	Default	Range	Meaning
Diameter `diameter`	mm	6	4 … 30	The circle the connector fits inside — the same measure as for the dowel, so the seam planner reckons alike for both.
Length `length`	mm	9	8 … 100	How far the arm sticks out, or how deep the pocket goes. It also sets the arm thickness: a tenth of it.
Kind `kind`		pin	pin, bore	The spring arm itself, or the pocket with the catch for it.
Play `play`	mm	0	0 … 1	Zero means: the value from the calibrated material profile.

Snap fit (`insert_snap_fit`)

A springing arm with a hook, for snapping two parts together. The arm is at least ten times as long as it is thick, otherwise it breaks instead of flexing.

Objects: 1 → 1 · reversible · deterministic · Applies to: Face

Parameters	Unit	Default	Range	Meaning
Width `width`	mm	8	2 … 60	Width of the arm. Wider takes more load and flexes less far.
Length `length`	mm	16	4 … 120	Length of the arm. At least ten times its thickness — shorter and it breaks instead of flexing.
Arm thickness `thickness`	mm	1.6	0.6 … 8	Thickness of the arm. It governs the spring force more than anything else.
Hook overhang `hook`	mm	1.2	0.2 … 6	How far the hook protrudes — that is, how far it travels when snapping in.
Lead angle `lead_angle`	°	35	10 … 60	Flatter means easier to snap in, steeper means it holds harder.

Stiffening rib (`insert_rib`)

A rib with a sloped run-out: it stiffens a wall without making it thicker. It stays thinner than the wall it sits on, otherwise it shows on the other side.

Objects: 1 → 1 · reversible · deterministic · Applies to: Face

Parameters	Unit	Default	Range	Meaning
Length `length`	mm	20	2 … 300	How far the rib runs along the wall.
Height `height`	mm	10	1 … 200	How far it stands off the wall. That is what buys the stiffness.
Wall thickness `wall`	mm	2	0.4 … 20	The wall the rib sits on — it sets the rib thickness.
Rib thickness `thickness`	mm	0	0 … 20	Zero means: two thirds of the wall thickness, otherwise it shows through.
Run-out `fillet`	mm	2	0 … 20	A sloped run-out at the foot instead of a sharp edge.

T-slot tongue for extrusion (`insert_profile_tongue`)

A T-shaped foot that slides into the slot of an aluminium extrusion from the end and holds there. Neck and head come from the standard parts table. For printing, the tongue is best laid flat along the slot direction — standing upright, the shoulder under the head is an overhang.

Objects: 1 → 1 · reversible · deterministic · Applies to: Face

Parameters	Unit	Default	Range	Meaning
Profile `size`		2020	2020, 3030, 4040	The slot size of the rail — on the usual extrusions it is the number in the name.
Length `length`	mm	20	6 … 200	How far the tongue runs along the slot. Longer holds more.
Lead-in `lead_in`	mm	1.5	0 … 6	Over this length the ends taper to the width of the neck, so the tongue slides in instead of jamming on the first edge.
Play `play`	mm	0	0 … 1	Zero means: the value from the calibrated material profile.
Head height `head`	mm	0	0 … 8	Zero means: tall enough that the head fills the chamber without touching the bottom of the slot. More than the chamber depth does not fit.

Wall mount (`insert_wall_mount`)

A back plate with screw holes and a shelf standing forward. The holes are through holes from the standard parts table.

Objects: 1 → 1 · reversible · deterministic · Applies to: Face

Parameters	Unit	Default	Range	Meaning
Width `width`	mm	30	8 … 300	Width of the back plate that goes against the wall.
Height `height`	mm	25	8 … 300	Height of the back plate. The shelf stands forward from it.
Wall thickness `thickness`	mm	3	1 … 20	Thickness of back plate and shelf. Below two millimetres the mount bends.
Screw `size`		M4	M2, M2.5, M3, M4, M5, M6, M8	What the holes are for. They are clearance holes from the standards table.
Holes `holes`		2	1 … 6	Spread across the width.
Shelf `lip`	mm	12	0 … 100	A shelf standing forward for the device to sit on.

Wall thickness ladder (`create_wall_ladder`)

Walls from one to several extrusion widths. Shows where the printer still lays down material — the basis for the minimum wall thickness.

Objects: 0 → 1 · reversible · deterministic

Parameters	Unit	Default	Range	Meaning
Extrusion width `extrusion`	mm	0.42	0.2 … 1.2	How wide this printer lays a bead — usually a little more than the nozzle. Each step of the ladder is one bead thicker than the last.
Steps `steps`		6	2 … 10	How many walls stand side by side, each one bead thicker than the last.
Height `height`	mm	15	3 … 60	Height of the walls. Tall enough that a wall too thin actually falls over.
Length `length`	mm	25	5 … 120	Length of each wall.

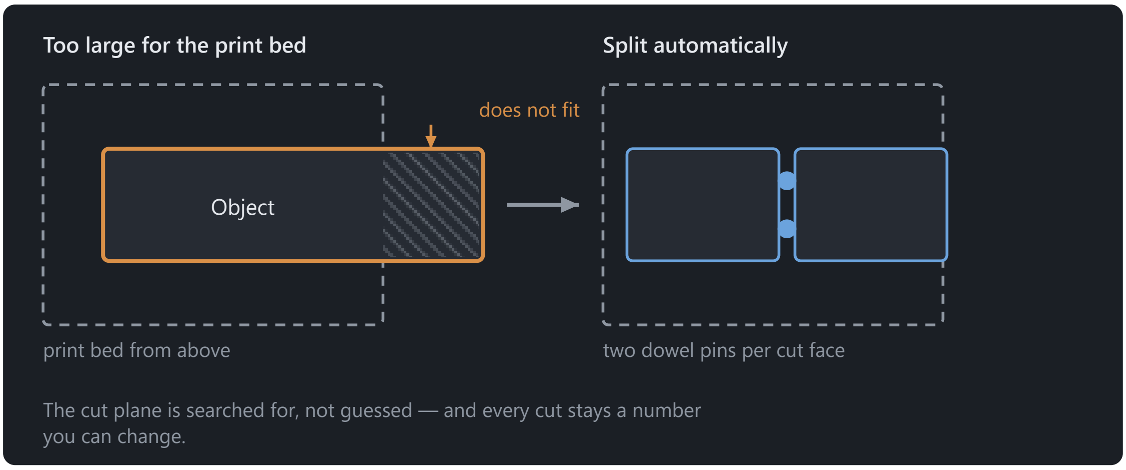
Wall thickness ladder (`insert_wall_ladder`)

Walls from one to several extrusion widths. Shows where the printer still lays down material — the basis for the minimum wall thickness.

Objects: 1 → 1 · reversible · deterministic

Parameters	Unit	Default	Range	Meaning
Extrusion width `extrusion`	mm	0.42	0.2 … 1.2	How wide this printer lays a bead — usually a little more than the nozzle. Each step of the ladder is one bead thicker than the last.
Steps `steps`		6	2 … 10	How many walls stand side by side, each one bead thicker than the last.
Height `height`	mm	15	3 … 60	Height of the walls. Tall enough that a wall too thin actually falls over.
Length `length`	mm	25	5 … 120	Length of each wall.

Split and Adjust



The dowel pins make sure the halves only go together one way round.

Compensate the elephant foot (`compensate_first_layer`)

Pulls the first layers in by what they spread out while printing. The value comes from the material profile.

Objects: 1 → 1 · reversible · deterministic

Parameters	Unit	Default	Range	Meaning
Height `height`	mm	0.6	0.1 … 5	Over how much height it is pulled in — about the first three layers.
Amount `amount`	mm	0	0 … 2	Zero means: the value from the calibrated material profile.

Create test piece (`test_piece`)

Cuts a cube out around a spot, to print it and try it out — two minutes instead of two hours.

Objects: 1 → 1 · reversible · deterministic · Applies to: Hole, peg, Face

Parameters	Unit	Default	Range	Meaning
Edge length `size`	mm	20	2 … 200	How big the cut-out gets. Big enough that the fit has material around it.
Position X `x`	mm	0		Centre of the cut-out — the spot in question.
Position Y `y`	mm	0		Second axis of the position — see Position X.
Position Z `z`	mm	0		Third axis of the position — see Position X.
Place on the bed `on_bed`		on		Lays the test piece down flat, so it prints without supports.

Hollow out (`hollow_object`)

Hollows an object out and puts vents in. Saves material and time; the wall thickness holds within the raster.

When not to: Not without a vent if the slicer will add supports: the cavity fills with material nobody can get out again. And not on parts that carry loads — a thin shell breaks differently from a solid body.

Objects: 1 → 1 · reversible · deterministic · Shortcut `Ctrl+H`

Parameters	Unit	Default	Range	Meaning
Wall thickness `wall`	mm	2	0.2 … 50	What stays. Two extrusion widths are the minimum.
Open the top `open_top`		off		Removes the ceiling above the cavity. The hollow body becomes a tin, and *Create lid* finds the opening it needs.
Vents `vents`		1	0 … 6	Zero means a closed cavity — in FDM printing that pushes the ceiling up. An open tin needs none.
Vent diameter `vent_diameter`	mm	4	1 … 20	Width of the openings. Wide enough for unused material to come out.

Set material (`set_material`)

Gives this body a material of its own. Tolerances, shrinkage and the elephant foot are then computed with it instead of the project's.

Objects: 1 → 1 · reversible · deterministic

Parameters	Meaning
Material `material`	Which material this part is made of. Empty means: the project's.

Split (`split_pinned`)

Splits an object at a plane, with dowel pins in the cut face if wanted. The clearance comes from the material profile; zero pins means: only cut.

When not to: Not on parts whose cut face stays visible: the seam lies on a plane and looks like it afterwards. Where it would be in the way, change the orientation or pick a place where an edge already runs.

Objects: 1 → 2 · reversible · deterministic

Parameters	Unit	Default	Range	Meaning
Axis `axis`		z	x, y, z	Which axis the cut is perpendicular to. Z makes a horizontal cut.
Position `position`	mm	0		Where the cutting plane lies, measured on that axis. The number stays editable: a double-click on the step moves the cut.
Dowel pins `pins`		2	0 … 6	Zero means: cut only. Two hold the halves against twisting.
Pin shape `shape`		round	round, hex, dovetail, snap	What holds the halves together. Round prints cleanest and needs two against twisting; hexagon and dovetail hold on their own, the dovetail also against being pulled apart. The snap clicks in and holds without glue — it needs a seam of at least 5.4 mm.
Glue recommended `glue_hint`		off		Split automatically turns this on when neither a dovetail nor a snap fits the seam.
Pin diameter `diameter`	mm	0	0 … 8	Zero means: derive it from the cut face.
Play `play`	mm	0	0 … 1	Zero means: the value from the calibrated material profile.

Split along a drawn line (`split_line`)

Splits an object along a line drawn in the view and, if wanted, sets dowel pins into the cut face.

When not to: The cut is a plane, not a curve: the line sets where and at what angle the split runs, and the plane goes straight through the part from there. To follow a curve, split twice.

Objects: 1 → 2 · reversible · deterministic

Parameters	Unit	Default	Range	Meaning
Position `position`	mm	0		How far the split plane sits from the origin, measured along the split direction. The drawn line fills the number in; changing it later moves the cut without turning it.
Dowel pins `pins`		2	0 … 6	Pins on one half, bores on the other — they hold the parts aligned while gluing. Zero means: only split.
Split direction X `normal_x`		0		Direction the plane faces. The drawn line fills it in; set by hand, the three numbers form an arrow perpendicular to the cut face.
Split direction Y `normal_y`		0		Second axis of the split direction — see Split direction X.
Split direction Z `normal_z`		1		Third axis of the split direction — see Split direction X.
Pin shape `shape`		round	round, hex, dovetail, snap	What holds the halves together. Round prints cleanest and needs two against twisting; hexagon and dovetail hold on their own, the dovetail also against being pulled apart. The snap clicks in and holds without glue — it needs a seam of at least 5.4 mm.
Pin diameter `diameter`	mm	0	0 … 8	Zero means: derive it from the cut face.
Play `play`	mm	0	0 … 1	Zero means: the value from the calibrated material profile.

Split into separate parts (`split_bodies`)

Turns a body made of several unconnected parts into one object each. What does not hang together is not one part — an STL does not know that, the geometry does.

When not to: Only what does not touch is separated. Two parts meeting at a face are one to the geometry — there Split along a plane is the way.

Objects: 1 → … · reversible · deterministic

Parameters	Default	Range	Meaning
Count `count`	2	2 … 64	How many objects it is split into. The number belongs here and not in the result, because the stack hands out the identifiers of its outputs before anything is computed. How many parts the body actually has is in the check report — and this operation names the number when it does not fit.
Keep slivers `keep_tiny`	off		Keep fragments below one per cent of the largest part as well. Scans often bring single loose triangles — rarely something you want as its own model.

Import

Extrude a drawing (`load_outline`)

Reads an SVG or DXF drawing and gives it a height. Inner contours become holes.

Objects: 0 → 1 · reversible · deterministic

Parameters	Unit	Default	Range	Meaning
Source `source`		required		The embedded SVG or DXF file in the project.
Height `height`	mm	3	0.1 … 500	How far the drawing gets pulled up.
Width `width`	mm	0	0 … 1000	Scale to this width. Zero takes the numbers in the file as millimetres.
Name `name`				What the object is called in the tree. Empty takes the file name.

Insert model (`load`)

Reads a model file, converts it to millimetres and cleans it up. A 3MF assembly arrives as several objects rather than as one lump.

Objects: 0 → … · reversible · deterministic

Parameters	Default	Range	Meaning
Source `source`	required		The embedded or linked file in the project.
Unit `unit`	auto	auto, mm, cm, in, m	STL carries no unit. Automatic means: guess, and ask when in doubt.
Name `name`			Leave empty to use the file name.
Place on the bed `place_on_bed`	off		Puts the underside of the model on the print bed.
Centre on the bed `centre`	off		Moves the model to the middle of the print bed. A model from a CAD program often has its origin in a corner and would otherwise sit far off to one side.
Welding vertices `weld`	on		Merges points that practically coincide — the most common reason an STL appears to be made of separate pieces.
Remove degenerate triangles `remove_degenerate`	on		Triangles with no area. They disturb every later computation and carry nothing.
Unifying normals `unify_normals`	on		Settles which side is outside. Without it faces appear dark or missing.

Load STEP (`load_step`)

Reads a STEP file with individually editable faces and edges. STEP carries its own unit, so no unit question is needed.

Objects: 0 → 1 · reversible · deterministic

Parameters	Default	Meaning
Source `source`	required	The embedded STEP file in the project.
Name `name`		Leave empty to use the file name.

Filament

Assign filament (`assign_slot`)

Assigns a filament to the whole object. A part becomes multi-coloured by uniting bodies with different assignments.

Objects: 1 → 1 · reversible · deterministic

Parameters	Default	Range	Meaning
Filament `slot`	0	0 … 7	The filament assigned to the whole object.
Name `name`			What the filament is called. Empty uses “Filament” and the number.
Colour `colour`			Display colour as #RRGGBB. For the view only — what prints is what is loaded.
Type `material_type`			The filament material type, such as PLA or PETG.
Slicer profile `slicer_profile`			Manufacturer profile from which the slicer takes this spool's print values.

Convert texture to filaments (`slots_from_texture`)

Reduces the texture of an object to the number of filaments loaded and stores the result as filament assignments.

Objects: 1 → 1 · reversible · with a seed

Parameters	Default	Range	Meaning
Filaments `filaments`	4	1 … 8	How many colours to reduce to — as many as are loaded.

Filament on a face (`paint_slot`)

Colours a recognised face completely in one filament. The boundary of the face comes from recognition — no brush, no radius.

Objects: 1 → 1 · reversible · deterministic · Applies to: Face

Parameters	Default	Range	Meaning
Filament `slot`	1	0 … 7	Which filament the face gets. In the 3MF it stays assigned to the same colour-change position.
Face `at_feature`	required		The recognised face that is coloured completely — set by clicking the feature.
Colour `colour`			Display colour as #RRGGBB. For the view only — what prints is what is loaded.
Name `name`			Name of the filament. It appears in the 3MF at the colour change.
Type `material_type`			The filament material type, such as PLA or PETG.
Slicer profile `slicer_profile`			Manufacturer profile from which the slicer takes this spool's print values.
Replace filament completely `replace_filament`	off		Uses the selected spool's complete details. Unknown material values remain unknown.

Remove filament (`clear_filament`)

Removes the filament assignment from a body or face. Other faces keep their filament; geometry remains unchanged.

Objects: 1 → 1 · reversible · deterministic · Applies to: Face

Parameters	Default	Meaning
Faces `at_features`	()	The faces whose filament is removed. Leave empty to remove it from the whole body.
Face `at_feature`		The face whose filament is removed. Leave empty to remove it from the whole body.

Lettering

Lettering as a body (`create_label`)

Creates lettering as an object of its own — for two-colour printing with two files, and for letters that get glued on.

Objects: 0 → 1 · reversible · deterministic

Parameters	Unit	Default	Range	Meaning
Text `text`		required		What the body should say.
Font size `size`	mm	8	3 … 200	Height of the capitals. Below three millimetres the print loses the shape.
Thickness `depth`	mm	0.6	0.1 … 50	How thick the letters become. A few tenths are enough for gluing on.
Font `font`		DejaVu Sans	DejaVu Sans, DejaVu Serif, DejaVu Sans Mono	DejaVu ships with Solidon, so a project looks the same on every machine.
Position X `x`	mm	0		Where the text sits. A clicked face fills in place and direction itself.
Position Y `y`	mm	0		Another axis of the position — see Position X.
Position Z `z`	mm	0		Another axis of the position — see Position X.
Normal X `nx`		0		The direction the text faces. A clicked face supplies it; by hand, 0/0/1 is up.
Normal Y `ny`		0		Another axis of the direction — see Normal X.
Normal Z `nz`		0		Another axis of the direction — see Normal X.
Rotation `angle`	°	0		Rotates the text in its plane around the selected point.
Name `name`				What the object is called in the tree. Empty means Solidon picks a name.

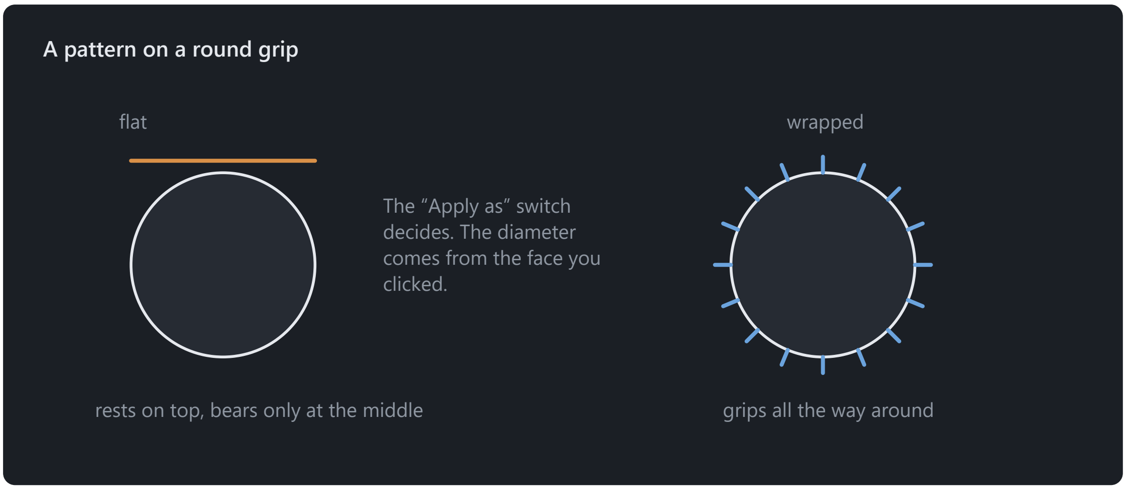
Put text on (`label_text`)

Puts text raised or engraved on a face. The font is processed as an outline, not as a picture — the edges stay clean at any size.

Objects: 1 → 1 · reversible · deterministic · Applies to: Face

Parameters	Unit	Default	Range	Meaning
Text `text`		required		What it should say.
Font size `size`	mm	8	3 … 200	Height of the capitals. Below three millimetres the print loses the shape.
Depth `depth`	mm	0.6	0.1 … 10	How far raised or how deep engraved.
Kind `mode`		raised	raised, engraved	Raised prints better, engraved survives sanding.
Filament `slot`		0	0 … 7	Puts the lettering into a slot of its own — the 3MF export turns that into the colour change, without a second file.
Font `font`		DejaVu Sans	DejaVu Sans, DejaVu Serif, DejaVu Sans Mono	DejaVu ships with Solidon, so a project looks the same on every machine.
Position X `x`	mm	0		Where the text sits. A clicked face fills in place and direction itself.
Position Y `y`	mm	0		Another axis of the position — see Position X.
Position Z `z`	mm	0		Another axis of the position — see Position X.
Normal X `nx`		0		The direction the text faces. A clicked face supplies it; by hand, 0/0/1 is up.
Normal Y `ny`		0		Another axis of the direction — see Normal X.
Normal Z `nz`		1		Another axis of the direction — see Normal X.
Rotation `angle`	°	0	-360 … 360	Turns the text within the face it sits on.

Surface



A knurl belongs around the grip, not as a patch on it.

Apply relief (`displace_image`)

Turns the brightness of an image into height on the surface. For grain, embossing and crests — everything a sketch is no good for.

When not to: Not on a coarse mesh: a relief only shows where there are vertices, and below one vertex per two pixels nothing of the image survives. Even out the mesh first.

Objects: 1 → 1 · reversible · deterministic

Parameters	Unit	Default	Range	Meaning
Image `source`		required		The greyscale image whose brightness becomes height. Light stands high.
Height `strength`	mm	1	0 … 50	How tall the relief becomes — the distance between black and white.
Apply as `projection`		planar	planar, cylindrical, spherical, face	How the rectangular image gets onto the body: from above, wrapped around the axis or over the sphere.
Face `at_feature`				Which detected face the image is laid onto. Only for “Onto a face” — the other kinds need none. Applies when Apply as = face.
Neutral level `middle`		0	0 … 1	Which grey value leaves the surface alone. Zero only adds; a half raises and lowers — the difference between a stamp and an embossing.
Soften `smooth`		0	0 … 20	How often the heights are averaged over their neighbours. Takes hard edges out of an image drawn too sharply for printing.

Apply texture (`apply_texture`)

Presses a pattern into a face as real geometry — knurling for grip, hexagon or rib for looks. What the slicer gets is what you see.

Objects: 1 → 1 · reversible · with a seed · Applies to: Face

Parameters	Unit	Default	Range	Meaning
Pattern `pattern`		knurl_diamond	rib, wave, knurl_straight, knurl_diamond, hexagon, dimple, voronoi, noise	Which pattern is applied. Knurling gives grip, hexagon and rib are decoration, Voronoi and noise hide the layer lines.
Width `width`	mm	40	0.5 …	How wide the pattern field becomes.
Height `height`	mm	30	0.5 …	How tall the pattern field becomes. It sits centred on the chosen spot.
Pitch `pitch`	mm	2	0.1 …	Distance from land to land. The lands are half as wide; below the nozzle size the pattern does not print, and the operation says so instead of trying.
Depth `depth`	mm	0.6	0.02 …	How far the pattern stands out or cuts in. Shallower than one layer it disappears when printed.
Kind `mode`		raised	raised, engraved	Raised puts the pattern onto the face, engraved cuts it in. An engraved knurl grips differently from a raised one — which is better is for the hand to decide.
Apply as `wrap`		flat	flat, cylinder	Flat onto a plane, or wrapped around a cylinder. A knurl belongs around the grip, not as a patch on it.

Parameters	Unit	Default	Range	Meaning
Diameter `wrap_diameter`	mm	0	0 …	The diameter the pattern runs around. Only when wrapping. Clicking a cylindrical face enters it here. Applies when Apply as = cylinder.
Rotation `angle`	°	0	-360 … 360	Turns the pattern within the face. Applies when Apply as = flat.
Position X `x`	mm	0		Centre of the field. A clicked face fills this in by itself.
Position Y `y`	mm	0		Second axis of the position — see Position X.
Position Z `z`	mm	0		Height of the face the pattern goes on.
Direction X `nx`		0		Normal of the face. A clicked face brings it along.
Direction Y `ny`		0		Second axis of the direction — see Direction X.
Direction Z `nz`		1		Third axis of the direction. Up by default.

Fill with lattice (`lattice_fill`)

Fills the cavity of a hollowed body with a lattice as real geometry. It travels in the 3MF and stays the same, whoever slices the part.

When not to: Not for parts that have to be watertight: a lattice has cavities, and water finds them. For load-bearing parts, increase the wall thickness first — a fill does not replace a wall.

Objects: 1 → 1 · reversible · deterministic

Parameters	Unit	Default	Range	Meaning
Structure `structure`		gyroid	gyroid, honeycomb, cubic	Gyroid carries in every direction and traps nothing, honeycomb stiffens a plate, the cubic grid is the stiffest and the heaviest.
Cell size `cell`	mm	8	1 …	From cell to cell. Smaller means more struts and more material.
Strut thickness `wall`	mm	1	0.1 …	How thick the struts become. Below two extrusion widths the machine will not print them — then the operation says so instead of trying.

Smooth and simplify mesh

Close open surface (`thicken`)

Gives an open surface a wall and turns it into a body. The way out for a mesh that arrives as a surface instead of a volume.

Objects: 1 → 1 · reversible · deterministic

Parameters	Unit	Default	Range	Meaning
Wall thickness `thickness`	mm	2	0.1 … 50	How thick the wall becomes. Below two extrusion widths it is fragile — what that means for this material is in the report.

End face editing (`brep_to_mesh`)

Turns individually editable faces into fixed triangles. Individual edges can no longer be chamfered or filleted afterwards; Undo restores the previous state.

Objects: 1 → 1 · reversible · deterministic

Parameters	Unit	Default	Range	Meaning
Fineness `deflection`	mm	0.05	0.001 … 1	How far the triangles may deviate from the real surface.

Even out the triangles (`remesh_uniform`)

Brings every triangle to roughly the same size — the coarse ones are split, the needlessly fine ones merged. The step before shaping by hand.

When not to: Not for refining alone: if you only want more triangles and none to disappear anywhere, use “Refine edges” — it splits and never merges.

Objects: 1 → 1 · reversible · deterministic

Parameters	Unit	Default	Range	Meaning
Edge length `edge`	mm	1	0.05 … 50	Roughly how long the edges are everywhere afterwards. Shorter means finer and larger — and finer is only needed where shaping happens later.
Permitted deviation `deviation`	mm	0	0 … 1	How far the surface may travel for it. Zero leaves the shape untouched; only above that do the needlessly fine spots disappear as well.

Reduce triangles (`decimate_mesh`)

Reduces the triangle count. The largest deviation from the original surface is measured and reported.

When not to: Not on a part that is still being dimensioned: decimating shifts faces, and a hole placed afterwards sits on a different surface than planned. Construct first, decimate last.

Objects: 1 → 1 · reversible · deterministic

Parameters	Default	Range	Meaning
Triangles `triangles`	50000	500 … 5e+06	Target count. Fewer means faster and less accurate — the report says how much.

Refine edges (`remesh_mesh`)

Splits long edges until the net is even. The shape stays exact.

Objects: 1 → 1 · reversible · deterministic

Parameters	Unit	Default	Range	Meaning
Edge length `edge`	mm	1	0.05 … 50	Every longer edge is split. Shorter means more even and larger.

Sculpt (`sculpt_strokes`)

Adds and removes material with the brush. The whole session is one step in the history and stays editable, however many strokes it holds.

When not to: Not on a part still being dimensioned: a stroke sits at a place in space, and changing the shape underneath moves the surface out from under it. Construct first, sculpt last.

Objects: 1 → 1 · reversible · deterministic

Parameters	Default	Range	Meaning
Strokes `strokes`			The brush strokes gathered in this session. They are painted, not typed — this field only shows what came of it.
Baked `baked`			A fixed state of this session. Once set, the result comes from there and no longer from the strokes — they remain as a record but no longer take effect.
Symmetry `symmetry`	none	none, x, y, z, xy, xz, yz, xyz	Which planes every stroke is additionally mirrored across. Changeable afterwards — a finished session can be made symmetric this way.

Set a pose (`pose_armature`)

Bends a body around an armature. A pose, not a motion — what gets printed is one state.

When not to: Not for strong bends: the skin pinches at bent joints, and an outstretched arm is an overhang. What the print makes of it, the overhang map says.

Objects: 1 → 1 · reversible · deterministic

Parameters	Meaning
Armature `armature`	The bones the body hangs on. They are placed, not typed — this field only shows what came of it.
Pose `pose`	Three angles per bone. An angle may be a project parameter.

Smooth (`smooth_mesh`)

Takes the roughness off a surface without shrinking the body. For generated meshes with stair steps.

Objects: 1 → 1 · reversible · deterministic

Parameters	Default	Range	Meaning
Passes `iterations`	5	1 … 50	More passes means smoother. Edges go with them.

Subdivide surface (`subdivide_surface`)

Turns a faceted surface into a curved one: the new points are not placed into the facet but onto the curve it stands for. Sharp edges stay sharp.

When not to: Not a replacement for a coarse original: what comes out is an interpolation of the facets at hand, not a recovered construction. Where a dimension matters, the curve belongs in the sketch.

Objects: 1 → 1 · reversible · deterministic

Parameters	Unit	Default	Range	Meaning
Edge length `edge`	mm	1	0.05 … 50	How fine the new surface becomes. Shorter means rounder and dearer.
Edge angle `angle`	°	52.5	0 … 180	From which angle an edge stays sharp. Lower rounds off more; above ninety even a box loses its corners.